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Additional Information

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An Open-Source Deep Learning Pipeline for automated Brain Organoid segmentation

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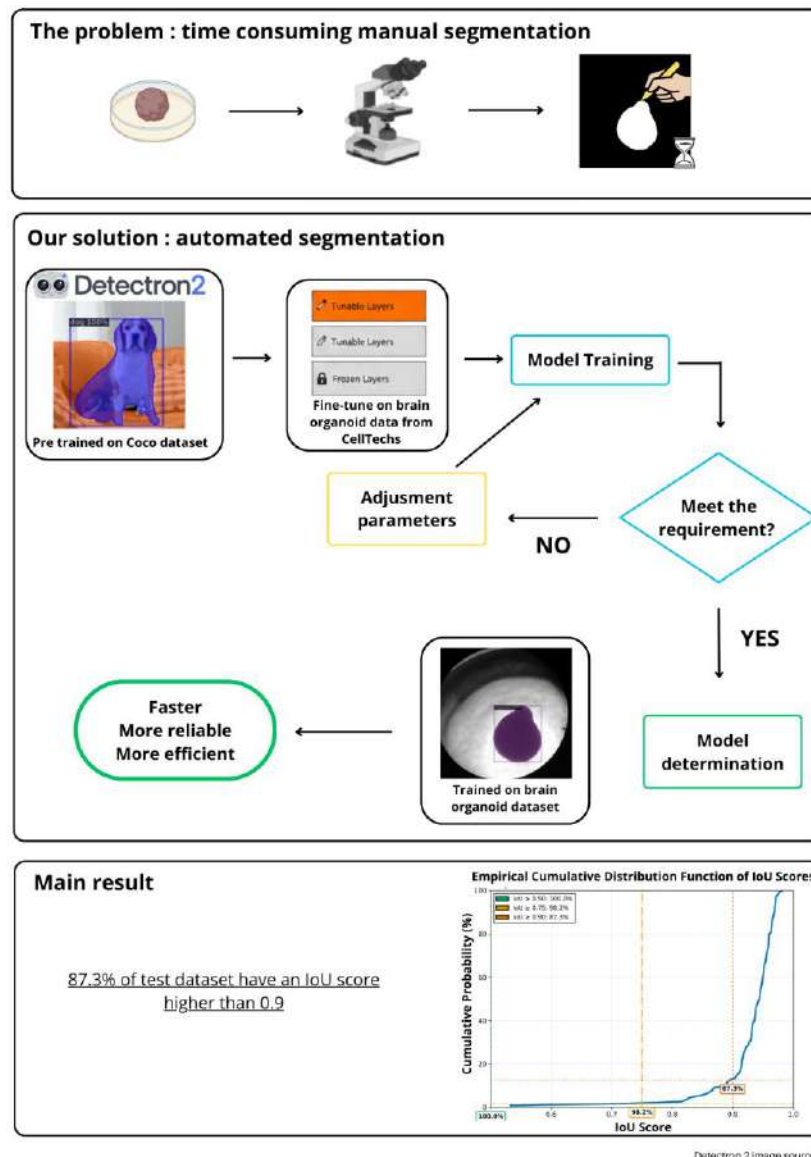
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VISUAL ABSTRACT



An Open-Source Deep Learning Pipeline for automated Brain Organoid segmentation

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Abstract – Organoids are increasingly established as relevant 3D models for mimicking human tissue complexity. However, their characterization remains a significant bottleneck: manual analysis is prone to error, while automated high-content screening systems are often cost-prohibitive or lack the necessary robustness. This study demonstrates that the democratization of open-source Artificial Intelligence (AI) now places robust, automated analysis within the reach of standard laboratories.

We present an accessible deep learning pipeline based on Detectron2, an open-source model developed by Meta, optimized for the segmentation of cerebral organoids. By fine-tuning pre-trained models, we show that high-performance segmentation is achievable without expensive local hardware. We leveraged cloud-based GPU resources (Google Colab) to overcome computational barriers, proving that advanced image analysis no longer requires specialized infrastructure.

The model achieves a promising Intersection over Union (IoU) score, validating the feasibility of this low-cost approach. To facilitate immediate adoption by the research community, we have released our inference code and a ready-to-use testing environment on GitHub. This tool aims to streamline organoid monitoring, making high-throughput analysis accessible to all biomedical researchers.

Index Terms – Automated Quantification, Deep Learning, Instance Segmentation, Open-source, Brain Organoids.

INTRODUCTION

Organoids are three-dimensional models grown *in vitro* from pluripotent stem cells (embryonic or induced pluripotent stem cells) or adult stem cells. They are capable of faithfully reproducing the structure and functional properties of human

tissues or organs. Their self-organizing nature gives them the ability to grow and organize themselves in 3D autonomously through cellular interactions¹. This three-dimensional organization allows them to approximate the cellular architecture and microenvironment observed *in vivo*, addressing many of the limitations of classical two-dimensional models such as the lack of nutrient gradients, mechanical cues, and diverse cell-to-cell contacts essential for realistic physiological modeling².

Due to their structural and functional proximity to living tissues, organoids have become essential models across various biomedical fields, including the study of embryonic development, carcinogenesis, and pathogen response. These models offer new perspectives for understanding complex pathologies and developing innovative therapeutic strategies. Among these models, cerebral organoids are particularly valuable for investigating human neurodevelopment, modeling neuropsychiatric and neurodegenerative disorders, and studying brain-specific responses to environmental or pharmacological stimuli. Consequently, the ability to obtain accurate measurements of parameters such as cell proliferation, viability, and differentiation is critical for evaluating the growth and stability of organoids as biological models³. Monitoring these characteristics ensures the reliability of experimental results. However, despite their considerable potential, monitoring these key characteristics remains a major challenge today. The morphological complexity and high variability of organoids make it difficult to observe these parameters continuously and accurately.

Historically, growth monitoring has relied on imaging and morphological analysis techniques. Brightfield Imaging is typically used to visualize organoid morphology, with subsequent morphometric analysis providing quantitative data on size and shape. While imaging is indispensable, its analysis remains time-consuming and heavily reliant on manual processing in tools like Fiji (ImageJ). This reliance on

manual labor makes it increasingly inadequate for handling the expanding volumes of generated data. These constraints increase the risk of human error, which can compromise the reproducibility and reliability of the results obtained. Standardizing procedures would limit manual manipulations, thus reducing handling time, the risk of experimental errors, and producing more reliable data that is comparable between laboratories.

To overcome the limitations of traditional approaches, advanced imaging and analysis solutions have been developed through the use of AI, a field whose rapid maturation over the past decade has revolutionized many areas of science and engineering. Although AI is a broad field, it generally refers to systems designed to perform tasks associated with human intelligence, such as perception, reasoning, decision making, or learning. The canonical reference by Russell & Norvig⁴ defines AI as the study of agents that perceive their environment and take actions to maximize a cumulative reward, categorizing approaches into search, logical reasoning, and machine learning.

Within this framework, machine learning is the subfield that focuses on building models capable of generalizing from data. Jordan and Mitchell (2015) provide a lucid panorama of the evolution of machine learning methods, highlighting the shift from classical, feature-engineering-based approaches, such as support vector machines or random forests, to end-to-end statistical learning paradigms based on large datasets and flexible models⁵.

These techniques have allowed for more precise and sometimes automated monitoring in biological research. A further leap occurs when one moves from shallow learning to deep learning, in which neural networks comprising multiple layers of nonlinear transformations learn hierarchical representations. The classic review by Y. LeCun, Y. Bengio, G. Hinton⁶ popularized this paradigm, showing how deep convolutional neural networks (CNNs) can automatically extract increasingly abstract features from raw pixel data, enabling breakthroughs in image classification, object detection, and segmentation. The full theoretical foundations, including backpropagation, optimization, and convolutional architectures, are given in the reference text of Goodfellow, Bengio & Courville⁷. In the specific domain of imaging, convolutional neural networks have become the workhorse model. The original LeNet⁶ demonstrated how convolutional filters and weight sharing can learn to recognize digits, while the resurgence in 2012 with AlexNet exploited large-scale labeled datasets and GPU training. More recent advances, such as ResNet⁸ introduced residual connections to allow much deeper networks to train without suffering vanishing gradients.

These convolutional architectures form the typical backbone of computer vision systems today. Reviews⁹ synthesize their principles and applications. However, many biological

imaging tasks, and in particular the segmentation and quantification of organoids, pose challenges that go beyond simple classification. Researchers must identify and delineate individual instances (organoids), which are often overlapping or touching, and measure their morphological features.

This is the domain of instance segmentation, which represents a more complex challenge than mere classification where a single label is assigned to the entire image. It also goes beyond semantic segmentation, which labels each pixel by its respective class but does not distinguish between individual instances. In the context of instance segmentation, every object must be localized through methods such as a bounding box (BBox) and then masked so that its precise per-pixel boundary is predicted. This category of AI technology has already been applied to organoid research for tasks such as morphological phenotype classification, growth dynamics monitoring, and the detection of structural aberrations during drug treatments³. Commercial platforms, including IN Carta (Molecular Devices), provide automated high-content image analysis solutions for organoid cultures. Similarly, companies such as Orakl market proprietary end-to-end systems integrating advanced image analysis pipelines with data management and predictive modeling capabilities.

While these platforms represent significant progress toward standardized and automated quantitative organoid phenotyping, they are typically closed-source and commercially licensed. Their implementation often requires specialized hardware, costly infrastructure, and substantial computational expertise, limiting accessibility for many academic laboratories, particularly those operating in early-stage or low-resource settings³. Moreover, given the rapid evolution of AI technologies, maintaining up-to-date tools and technical knowledge remains a considerable challenge. These limitations underscore the need for accessible, open-source, and computationally affordable solutions that can be readily implemented in standard biological research environments.

To address this need, Detectron2 was selected as the core framework. This open-source framework, developed by Facebook AI Research (FAIR), is an established and versatile platform for object detection and instance segmentation¹⁰. For automated organoid detection and segmentation, Detectron2 offers an optimal balance between performance, accessibility, and flexibility, while its modular design and extensive documentation ensure the interpretability and reproducibility required for research environments. The availability of pre-trained models, tutorials, and community resources facilitated the adaptation process and ensured that the resulting workflow could be reproduced or extended by other researchers^{11, 12}. To bypass the hardware limitations often encountered in biological laboratories, the training and deployment of the robust model were executed utilizing accessible cloud-based GPU platforms, specifically Google Colab.

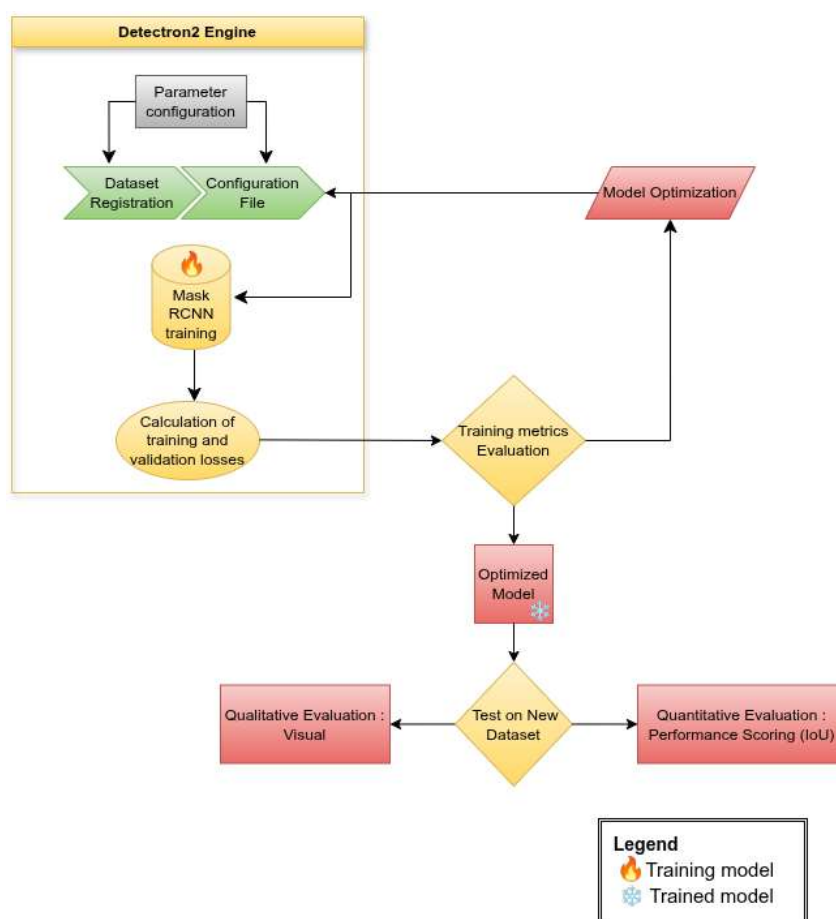


Figure 1: Schematic overview of the Mask R-CNN training, optimization, and evaluation pipeline within Detectron2. The workflow illustrates the iterative process ranging from dataset registration and parameter configuration to the training phase (indicated by the fire icon) and the calculation of losses. The pipeline includes a model optimization loop based on training metrics evaluation, which updates the configuration file. Once optimized, the final frozen model (indicated by the snowflake icon) is tested on a new dataset for both qualitative visual evaluation and quantitative performance scoring (IoU).

The objective of this article is to show how an existing deep-learning framework can be adapted and fine-tuned to address a specific biological imaging challenge. To facilitate community adoption and ensure reproducibility, the inference code and a dedicated testing environment have been made publicly available on GitHub:

[\[https://github.com/CodeByte341/detectron2-organoid-finetuning\]](https://github.com/CodeByte341/detectron2-organoid-finetuning).

MATERIAL AND METHOD

A schematic overview of the complete experimental pipeline is presented in Figure 1, illustrating the sequential workflow from dataset registration and model configuration through training, iterative optimization, and final performance evaluation. The following sections detail each step of this pipeline, encompassing image acquisition and dataset composition, data labeling and preparation, model

architecture, training strategy, and both qualitative and quantitative evaluation frameworks.

IMAGE ACQUISITION AND DATASET COMPOSITION

The dataset utilized in this study was provided by CellTechs (Villejuif, France), a laboratory specializing in stem cell and organoid research. The images were obtained from cerebral organoid culture experiments conducted under controlled conditions to monitor growth and morphogenesis. Image acquisition was performed using a Leica optical microscope equipped with a 40×/0.45 objective. As no dedicated scientific camera was available, images were captured using a smartphone positioned at the microscope eyepiece. This acquisition setup reflects common laboratory conditions in resource-limited environments. The images were subsequently exported and saved in .tif format.

To ensure the development of a robust and generalizable segmentation model, the dataset was constructed to represent a high degree of heterogeneity. It includes images of large morphological diversity and varying optical quality, ranging from high-contrast samples to those exhibiting experimental artifacts such as air bubbles, cellular debris, and morphological disintegration. This diversity was intentional to simulate real-world experimental variability.

DATA LABELING AND PREPARATION

The segmentation framework relies on a supervised learning approach, with ground truth data generated through manual annotation using FIJI (ImageJ version 2.14.0) software. The annotation task was performed by a cohort of engineering students (n=6). To ensure biological relevance, the labeling process was guided by technical discussions and morphological criteria established in collaboration with the CellTechs laboratory.

A semi-automated workflow was employed: initial binarization was performed via automatic thresholding using a custom FIJI Macro code to isolate primary structures, followed by manual refinement. On average, the manual annotation of each image required 2 to 4 minutes. To maximize the precision of the segmentation and minimize false positives, a conservative labeling strategy was adopted. Any structures perceived as ambiguous such as air bubbles,

debris, or fragments exhibiting signs of advanced necrosis were intentionally left unlabeled in the ground truth masks.

The resulting binary masks were converted into the COCO JSON format to ensure compatibility with the Detectron2 API. This project was defined as a single-class instance segmentation task (class: 'organoid'); all unannotated regions were treated as background. The final dataset comprised 1095 labeled images. These were randomly partitioned into three stratified subsets: 80% (875 images) allocated for model training, 10% (110 images) reserved for validation and 10% (110 images) for final performance testing.

IMAGE PRE-PROCESSING

Prior to ingestion into the neural network, images were resized from their native resolution (3120×4160 pixels) to a standardized dimension of 519×693 pixels. This uniform resizing was applied to both images and corresponding masks to reduce computational overhead and accommodate GPU memory constraints while preserving the spatial aspect ratio. No additional photometric pre-processing such as intensity normalization or cropping was applied, ensuring the model was trained on raw biological data features.

COMPUTATIONAL INFRASTRUCTURE AND DATA MANAGEMENT

Model development and training were executed on the Google Colab Pro cloud platform using a Python-based environment. Preliminary tests and code optimization were conducted on NVIDIA Tesla T4 GPUs (16 GB VRAM). The final, intensive model training phases were performed using

NVIDIA A100 GPUs (40 GB VRAM) to handle the computational load efficiently¹³.

To optimize data loading times within the cloud environment, labeled images were stored in a shared Google Drive folder and transferred to the execution environment as ZIP archives before being uncompressed for processing. This integration streamlined data management and overcame local hardware storage limitations.

MODEL ARCHITECTURE

The organoid segmentation task was addressed using the Mask R-CNN architecture, implemented within the Detectron2 framework (Meta AI). This architecture was selected for its ability to simultaneously perform object detection (localization) and instance segmentation (pixel-wise masking), which is essential for distinguishing individual organoids in potentially overlapping cultures¹⁰.

The model backbone consisted of a ResNet-50 deep residual network coupled with a Feature Pyramid Network (FPN). The ResNet-50 component extracts hierarchical visual features, while the FPN aggregates these features at multiple scales, enhancing the detection of organoids of varying sizes. The Region Proposal Network (RPN) subsequently generates candidate regions of interest (RoIs), which are aligned spatially using the RoIAlign operation to correct for quantization errors¹⁴ (Figure 2).

The architecture comprises two parallel output heads:

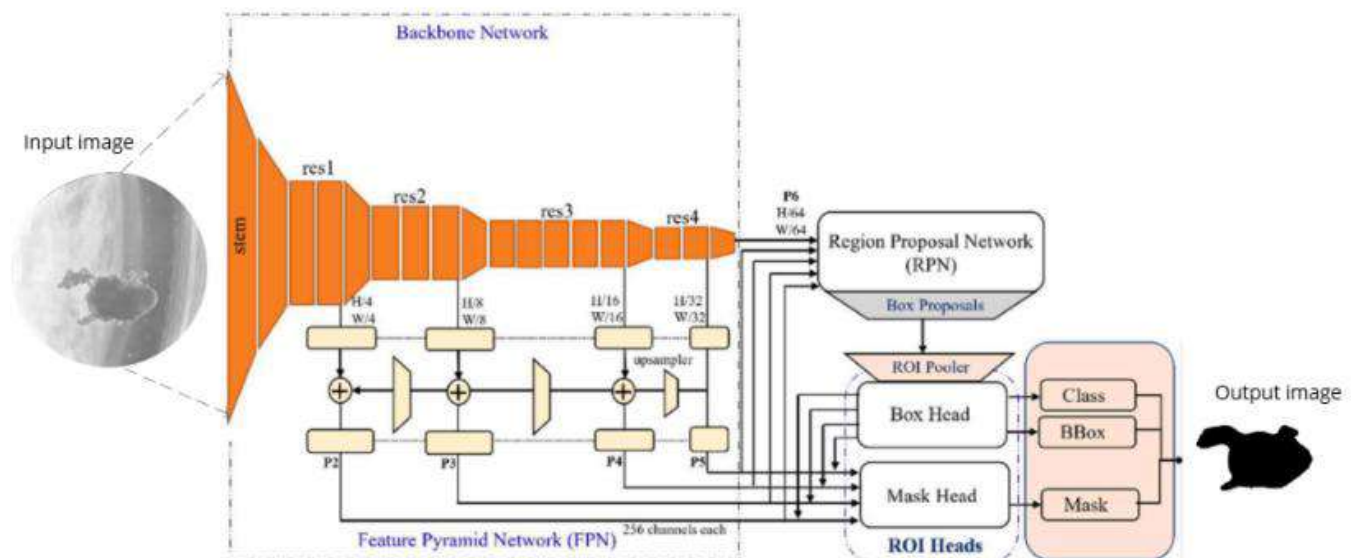


Figure 2: Architecture of the Detectron2 model based on Mask R-CNN (Rani et al., 2024). The input image is processed by a convolutional backbone (ResNet) to extract multi-scale feature maps, which are combined using a Feature Pyramid Network (FPN). A Region Proposal Network (RPN) generates candidate regions (ROIs), which are then refined by ROI heads to predict the object class, bounding box (BBox), and pixel-wise segmentation mask.

- **Box Head:** Responsible for BBox regression and object classification.
- **Mask Head:** A fully convolutional branch that predicts the binary segmentation mask for each detected instance.

To accelerate convergence and leverage generalized visual feature extraction (e.g., edge and texture detection), the model was initialized using weights pre-trained on the COCO (Common Objects in Context) dataset via transfer learning¹².

TRAINING STRATEGY AND PARAMETER OPTIMIZATION

The model was fine-tuned using the DefaultTrainer provided by Detectron2, with hyperparameters systematically optimized to balance convergence speed and generalization capability on the limited dataset. The optimization process focused on adjusting critical parameters, specifically the learning rate, batch size, and iteration count, to minimize loss without inducing overfitting^{10, 13}.

The final optimized configuration utilized Stochastic Gradient Descent (SGD) with a base learning rate of 0.00025 and a total batch size of 8 images per iteration. Training was conducted over 1,000 iterations, with the ROI Heads configured to sample 256 regions per image. The same learning rate strategy was employed throughout the training phase to ensure stability given the limited number of iterations.

Model optimization was guided by the rigorous monitoring of specific training metrics:

- **Loss Dynamics:** The total loss and its decomposed components localization, classification, BBox regression and mask segmentation losses were tracked to verify the simultaneous minimization of detection and segmentation errors.
- **Accuracy Metrics:** The evolution of classification accuracy (Fast R-CNN) and mask accuracy (Mask R-CNN) were monitored alongside the mask prediction accuracy to ensure the model correctly distinguished organoids from the background.
- **Error Rate Analysis:** To fine-tune the decision boundaries, we analyzed the False Positive (over-detection) and False Negative (missed detection) rates throughout the training process, aiming to minimize hallucinated artifacts while maximizing object retrieval.

No extensive manual data augmentation was applied. The training pipeline utilized only the standard Detectron2 transformations: ResizeShortestEdge (which resizes images while maintaining the aspect ratio using bilinear interpolation) and RandomFlip (horizontal flipping with a probability of 0.5) to ensure basic invariance.

INFERENCE AND POST-PROCESSING

For inference on new data, a confidence threshold of 0.7 was applied to filter out low-probability detections. Non-Maximum Suppression (NMS) was automatically handled by the framework to eliminate duplicate predictions for the same object. Furthermore, based on the experimental design where only a single organoid is present per culture well, a post-processing constraint was implemented to retain only the single instance with the highest confidence score per image, thereby preventing over-segmentation.

QUALITATIVE AND QUANTITATIVE PERFORMANCE EVALUATION

Model performance was evaluated through a dual-framework approach following the general workflow (Figure 1). Qualitatively, the model's morphological fidelity was assessed by stratifying results into three complexity levels: Level 1 (typical morphology), Level 2 (intermediate complexity with artifacts), and Level 3 (high-complexity edge cases). This allowed for a systematic validation of the model's ability to handle biological heterogeneity. Quantitatively, segmentation accuracy was benchmarked using the IoU metric (Figure 3). To ensure the reliability and reproducibility of these results, the IoU scores obtained from the validation dataset (n = 110) were then subjected to the rigorous statistical characterization described below.

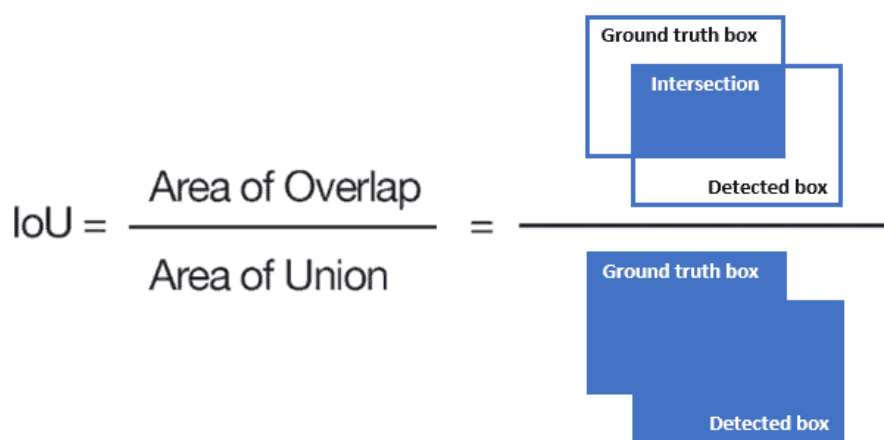


Figure 3: Schematic representation of the Intersection over Union (IoU) metric. The diagram illustrates the IoU formula, defined as the ratio of the overlapping area to the combined union area of the ground truth and detected bounding boxes, serving as a measure of detection accuracy.

STATISTICAL ANALYSIS

All statistical analyses were conducted using Python (version 3.10/3.11) with the NumPy, SciPy, and statsmodels libraries.

Descriptive statistics were computed to characterize the central tendency and dispersion of IoU values, including the mean (μ), median, standard deviation (σ), variance, range, and quartiles (Q1, Q2, Q3). The coefficient of variation (CV) was calculated to quantify the relative variability of the performance metric. Additional percentiles (10th, 90th, 95th, and 99th) were extracted to provide a comprehensive distribution profile.

The normality of the IoU distribution was assessed using the Shapiro-Wilk test ($\alpha = 0.05$). Statistical moments, specifically skewness and kurtosis, were computed to characterize the asymmetry and tailedness of the distribution. Outliers were identified using the interquartile range (IQR) method, with boundaries defined as $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$.

A 95% confidence interval (CI) for the mean IoU was calculated using the Student's t-distribution to estimate the precision of the average performance. The proportion of images achieving clinically relevant segmentation quality was quantified by computing the percentage of predictions exceeding critical IoU thresholds (0.50, 0.75, and 0.90), following established benchmarks in instance segmentation literature.

To identify the most probable performance range, the Highest Density Region (HDR) was computed using kernel density estimation (KDE) with Gaussian kernels. The HDR interval was defined to contain 50% of the distribution's probability mass, providing insight into the model's typical operational range.

Empirical cumulative distribution function (ECDF) analysis was performed to visualize the progression of IoU values across the dataset and to extract additional percentile-based metrics for model characterization.

RESULTS

TRAINING DYNAMICS AND CONVERGENCE OF THE DETECTRON2 MODEL

The performance of the analysis protocol was quantified by studying the training dynamics over 1,000 iterations. The evolution of the global error was monitored by detailing the total loss and its individual components (Figure 4A). The total loss (black dashed line) decreases from 1.62 to 0.16 during the process. This reduction is driven by the classification loss (blue) and the mask loss (green), both of which follow a similar downward trajectory until stabilizing after iteration 500. Spatial localization precision was evaluated via the BBox regression loss (yellow), which drops from approximately 0.30 to stabilize around 0.06. Finally, the

efficiency of preliminary detection is confirmed by the RPN losses (orange and purple), which remain stable at approximately 0.002 and 0.003 respectively from the first 100 iterations.

The model's predictive efficiency was subsequently measured through the evolution of accuracy rates (Figure 4B). The global classification accuracy (blue) increases very rapidly, exceeding 0.90 within the first 50 iterations and reaching a plateau of 0.99 starting from iteration 800. Segmentation quality is confirmed by the mask accuracy (orange) which, after a very slight initial dip, stabilizes at 0.97 as of iteration 300.

System reliability was further examined through an analysis of error rates (Figure 4C). The model's sensitivity is demonstrated by a constant and low false negative rate (blue), oscillating between 0.015 and 0.025. Specificity was evaluated by the false positive rate (orange), which shows a transient increase to 0.94 around iteration 20 before gradually decreasing to stabilize at 0.05 from iteration 500.

Finally, the control of training parameters was verified by the evolution of the learning rate (Figure 4D). The latter follows a linear increasing progression throughout the training, starting at a value near 0 to reach its maximum of 2.50×10^{-4} at iteration 1,000.

QUALITATIVE EVALUATION OF INSTANCE SEGMENTATION

The performance of the Detectron2-based Mask R-CNN model was evaluated qualitatively by comparing its automated predictions against manual annotations across a representative sample of data. These examples were selected to cover a spectrum of morphological scenarios, categorized into three levels of complexity (Figure 5).

Level 1 - Typical Morphology: For organoids exhibiting typical morphology and well-defined contours, the model-generated masks demonstrate a high degree of fidelity to the structures of interest. As illustrated in (Figure 5.1), the predicted segmentation boundaries align precisely with the manual annotations, accurately capturing the edges of the organoid without significant deviation.

Level 2 - Intermediate Complexity: When applied to more challenging cases of organoids involving irregular shapes, partially overlapping instances, or complex positioning within the imaging frame, the model mask successfully identifies individual organoids and maintains the integrity of their global surface area. While some minor discrepancies are visible at the edges, the instance masks remain coherent with the global morphology and surface area of the organoids (Figure 5.2). Specifically, in the first example, the organoid is surrounded by bubbles of similar size; the model detects the organoid with a high confidence score (98%) but also identifies two bubbles as organoids with lower confidence (74-78%). However, the final mask correctly isolates only the

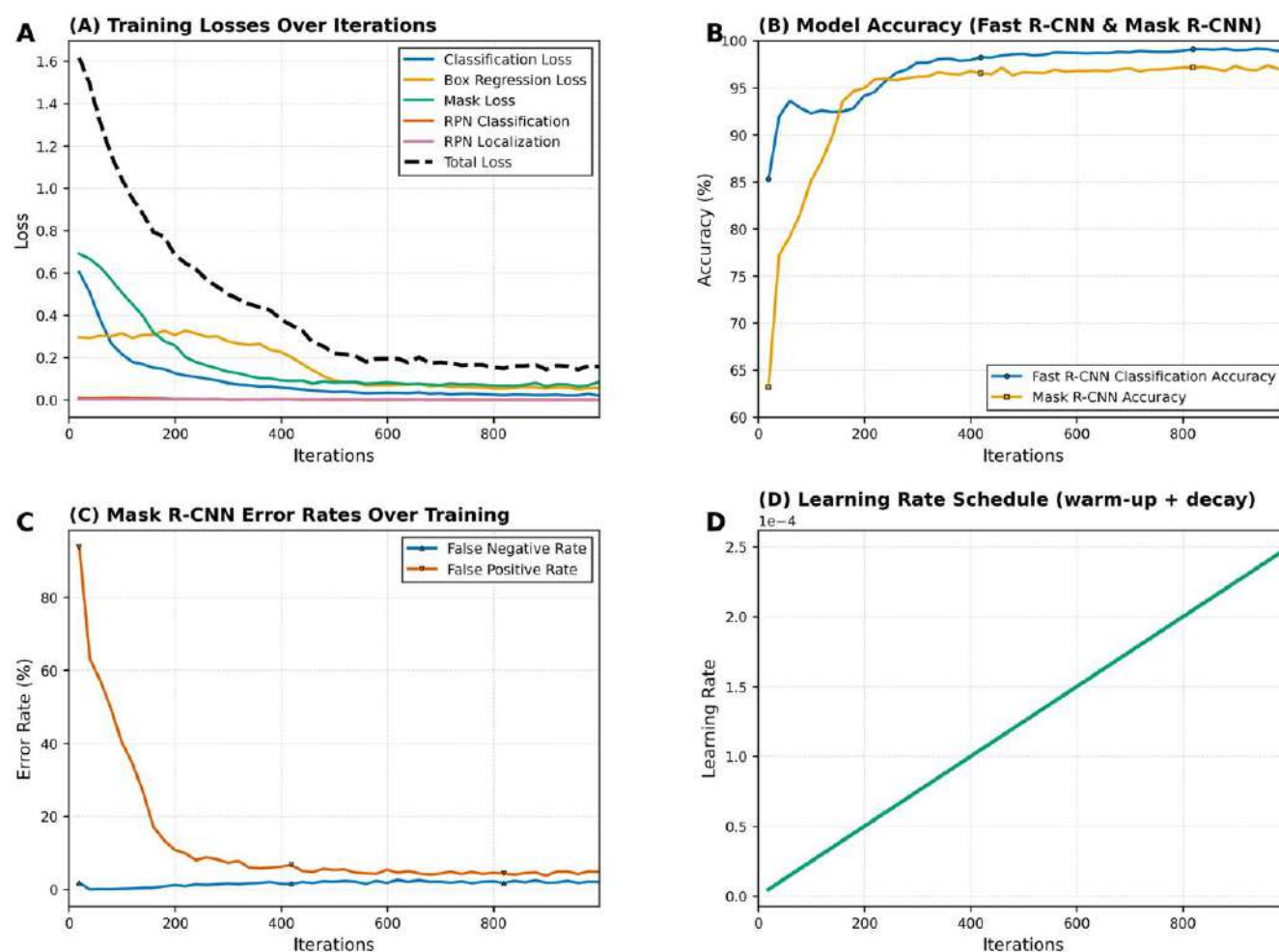


Figure 4: Training dynamics of the fine-tuned Detectron2 model for organoid instance segmentation. (A) Evolution of the total loss and its main components over training iterations: classification loss (blue), bounding box regression loss (yellow), mask segmentation loss (green), RPN classification loss (orange), RPN localization loss (purple), and total loss (black dashed line). The x-axis indicates the iteration number, and the y-axis the corresponding loss value. (B) Evolution of accuracy metrics for both Fast R-CNN (classification) and Mask R-CNN (segmentation) heads. The fast_rcnn/cls accuracy (blue) indicates overall object classification accuracy, while mask_rcnn/accuracy (yellow) represents mask prediction accuracy for instance segmentation. The x-axis denotes training iterations, and the y-axis the accuracy percentage. (C) Evolution of the Mask R-CNN error rates. The orange curve (mask_rcnn/false_positive) represents the false positive rate, i.e., regions predicted as containing an object when they do not. The blue curve (mask_rcnn/false_negative) corresponds to the false negative rate, i.e., regions where the model did not detect an object that was present. (D) Variation in learning rate over the model's training iterations. The x-axis represents the number of iterations, while the y-axis shows the learning rate value used at each step.

organoid. In the second example, an irregular organoid is seen with an adjacent ambiguous shape (potentially debris or necrotic tissue); while manual annotators excluded this secondary structure, the model incorporated it into the organoid mask. For the third example, the organoid exhibits highly complex features, including tentacle-like extensions and internal cavities; the model remains relatively accurate but shows a slight loss of precision regarding these intricate details and small recesses.

Level 3 - High Complexity and Edge Cases: These challenging scenarios provide insight into the model's behavior under demanding imaging conditions. In images exhibiting low contrast, partial blur, or highly atypical

morphologies, the segmentation task becomes inherently more complex. While minor boundary deviations may occur in such cases, the model often maintains a coherent and biologically relevant delineation of the organoid structure (Figure 5.3). In the first example, an organoid is located at the extreme edge of the well against a very dark background and contains an internal bubble; the model identifies the bubble as a distinct entity (with 72% confidence) while simultaneously detecting the entire organoid (99% confidence), leading to a conflict in instance separation. In the final example, the model demonstrates impressive sensitivity by successfully detecting and segmenting a very small organoid within a low-contrast environment cluttered with cellular debris, even when the target is barely visible to the human eye (98% confidence).

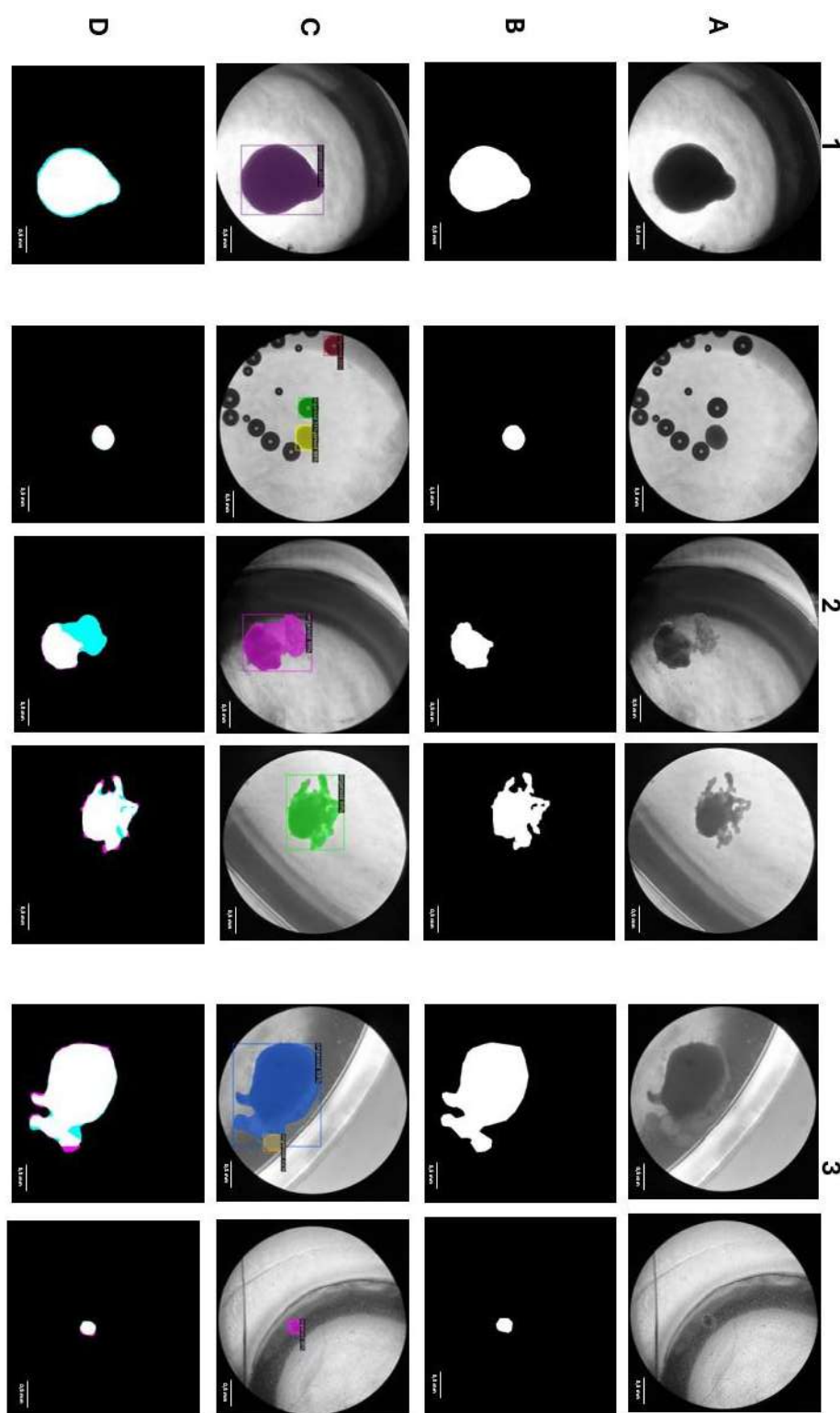


Figure 5: Qualitative evaluation of organoid segmentation performance across varying imaging conditions. The figure displays representative samples organized into three complexity levels: (1) Ideal scenarios featuring high contrast and distinct boundaries; (2) Intermediate complexity with irregular shapes or edge-proximity; and (3) Challenging cases exhibiting low contrast, blurring, or visual artifacts. For each column, four panels illustrate the segmentation process: (A) Original brightfield microscopy image. (B) Manual ground truth binary mask. (C) Automated prediction generated by the Mask R-CNN model, overlaying the segmentation mask, bounding box, and class confidence score. (D) Comparative overlay of the manual (B) and automated (C) masks; white regions indicate agreement (Intersection over Union), while cyan and magenta highlight discrepancies (False Negatives and False Positives). Scale bars: 0.5 mm.

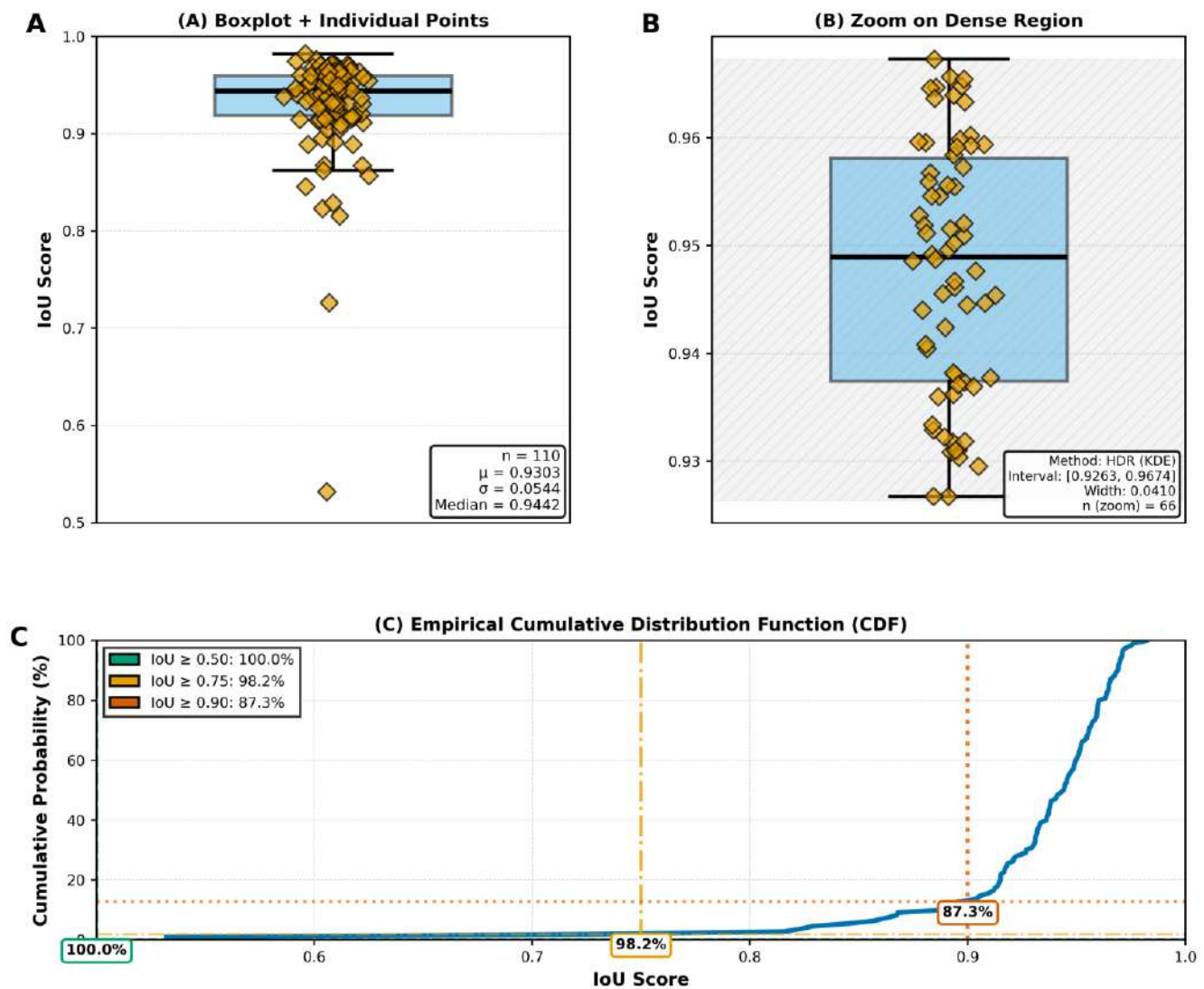


Figure 6: Quantitative assessment of segmentation accuracy via Intersection-over-Union (IoU) analysis. The figure illustrates the statistical distribution of IoU scores derived from the comparison between Detectron2-generated masks and manual ground truth annotations. (A) Global vertical boxplot displaying the overall IoU distribution for the test dataset ($n = 110$), with overlaid jittered orange diamond points representing individual samples to visualize data spread. The central horizontal black line indicates the median IoU (0.944). The lower and upper edges of the sky blue box represent the first ($Q1 = 0.919$) and third ($Q3 = 0.959$) quartiles, respectively. Whiskers extend to show the data range within $1.5 \times IQR$. Statistical summary shows: mean $\mu = 0.930$, standard deviation $\sigma = 0.054$. (B) Magnified view of the distribution's dense zone (shaded region with diagonal hatching), highlighting the concentration of high-performance predictions within the HDR 50% interval [0.926, 0.967]. This zoomed representation uses identical visual encoding (sky blue boxplot, orange diamond points) to emphasize that it constitutes a focused subset of panel A. The interval width of 0.041 captures 60% of the observations. (C) Empirical Cumulative Distribution Function (ECDF) of the IoU scores. The blue curve plots the IoU value (x-axis) against the cumulative probability in percentage (y-axis). Three critical performance thresholds are marked: IoU ≥ 0.50 (green, 100% of images), IoU ≥ 0.75 (yellow, 98.2% of images), and IoU ≥ 0.90 (orange, 87.3% of images), indicated by vertical lines with distinct patterns (dashed, dash-dot, and dotted respectively).

QUANTITATIVE EVALUATION OF INSTANCE SEGMENTATION

The segmentation accuracy generated by Mask R-CNN was quantitatively assessed by comparing the predicted masks against the manual ground truth across the entire test dataset ($n=110$). The statistical distribution of IoU scores is presented in Figure 6.

The global analysis (Figure 6A) reveals high model performance, characterized by a median IoU score of 0.944. As illustrated by the boxplot, the majority of predictions are concentrated within a narrow range. The IQR is 0.040, with 50% of the samples falling between the first quartile ($Q1 = 0.919$) and the third quartile ($Q3 = 0.959$). This 'dense zone' (Figure 6B), defined by the HDR 50% interval [0.926, 0.967], indicates high consistency in segmentation quality for the

majority of images, capturing 50% of observations within a width of 0.041. The whisker analysis (defined by the $1.5 \times$ IQR criterion) identified 7 outliers falling below the threshold of 0.858.

The ECDF depicted in Figure 6C, confirms the model's robustness across various tolerance thresholds. Performance threshold analysis reveals that 100.0%, 98.2%, and 87.3% of the images achieved IoU scores exceeding 0.50, 0.75, and 0.90, respectively.

DISCUSSION

The performance and reliability of the proposed organoid segmentation pipeline were evaluated through a combination of training dynamics analysis, qualitative visual inspection, and quantitative performance metrics (Figures 4, 5, 6). Together, these results provide a comprehensive overview of the Detectron2-based Mask R-CNN model's behavior and its suitability for automated organoid analysis in brightfield microscopy.

The training analysis presented in Figure 4 demonstrates that the Detectron2 model can segment organoids with high precision. The rapid decrease in total loss (from 1.62 to 0.16, Figure 4A, black dashed line), along with the stabilization of the classification (blue curve) and mask loss curves (green curve), indicates that the network successfully learns to identify organoid morphology. In addition, the localization losses via the BBox regression loss (yellow curve, stabilizing around 0.06) and the Region Proposal Network losses (orange and purple curves, remaining stable at approximately 0.002 and 0.003 respectively) remain close to zero from the beginning of training, suggesting that the model can readily identify the spatial positions of organoids within the images.

A critical adjustment phase is observed between iterations 20 and 500. During this interval, the false positive rate increases transiently to 0.94 around iteration 20 (Figure 4C, orange curve). This behavior suggests that, early in training, the model confuses optical artifacts such as cellular debris or air bubbles with actual organoids. Importantly, this limitation is transient and is progressively corrected as training proceeds. After approximately iteration 500, the false positive rate drops to 0.05 (Figure 4C), indicating that the model acquires increased specificity and generally learns to distinguish biological structures from background noise.

This reliability is further supported by the stability of the false negative rate (Figure 4C, blue curve), which remains consistently low throughout training, oscillating between 0.015 and 0.025. Then, mask accuracy reaches 0.97 (Figure 4B, light orange curve), highlighting the robustness of the learned segmentation. Finally, the linear learning rate warm-up strategy (Figure 4D) reaching a maximum of 2.50×10^{-4} at iteration 1,000 contributes to training stability during early iterations.

While early training shows some initial difficulty in discriminating artifacts, the rapid convergence toward stability justifies moving directly to an evaluation of the model's performance on the test set. Qualitative analyses illustrate how this optimized learning translates into concrete segmentation results under diverse imaging conditions. The results confirm that an instance segmentation approach based on the Mask R-CNN architecture is well suited for the automated analysis of organoids (Figure 5). The high degree of spatial agreement observed between automated predictions and manual annotations under standard conditions suggests that the model effectively learns fundamental morphological features (Figure 5.1). Given the model's good performance on typical morphology cases, the qualitative evaluation was deliberately designed to emphasize challenging scenarios: only one example is presented in Level 1 (typical morphology), while five examples are distributed across Levels 2 and 3 to thoroughly assess the model's behavior under intermediate and high complexity conditions.

A key strength of the model is its ability to handle morphological variability, including low-luminosity instances and irregular shapes that challenge traditional thresholding methods (Figure 5.2). Notably, discrepancies observed in intermediate complexity cases reveal an interesting nuance in boundary definition. For non-circular organoids with decreasing density at the periphery, the model tended to include ambiguous structures (potentially debris or necrotic tissue) within the segmentation mask, whereas manual annotators excluded them. However, a critical consideration regarding these discrepancies lies in the nature of the ground truth itself. The reference masks were established based on visual interpretation, and as the annotators are not the primary experimentalists, the lack of expert-level validation for ambiguous cases introduces a degree of subjectivity. Furthermore, the absence of validation from the CellTechs team represents a key limitation regarding the model's readiness for routine deployment within the laboratory workflow.

This also underscores the potential for inter-observer variability in manual annotations, particularly for edge cases where even human judgment faces uncertainty due to imaging artifacts or biological heterogeneity. This suggests that, thanks to generalization, the model may be more sensitive to subtle pixel intensity gradients than the human eye, potentially offering a more consistent definition of 'organoid area' than subjective manual tracing. In cases where organoids are surrounded by multiple bubbles of similar size, the model demonstrates robust detection capabilities, identifying the target organoid with high confidence while also detecting bubbles with lower confidence scores. It is worth noting that the confidence threshold of 0.70, defined in the inference parameters, was deliberately set at a relatively permissive level. This choice intentionally exposes the model's uncertainty boundaries, making detections of ambiguous structures such as air bubbles visible at lower

confidence scores, as illustrated in Figure 5.2. A stricter threshold would have suppressed many of these artifact detections, but at the cost of concealing the model's discriminative behavior. Importantly, the final mask correctly isolates only the organoid, demonstrating the effectiveness of the confidence-based selection process. Furthermore, the model successfully detected minute organoids in low-contrast environments cluttered with cellular debris, demonstrating robustness in scenarios where simple intensity-based detection often fails.

However, the analysis of high-complexity cases (Figure 5.3) provides important insights into the current technical boundaries of the approach. Cases involving organoids positioned at extreme edges of wells against very dark backgrounds, particularly when containing internal bubbles, can lead to instance separation conflicts where both the bubble and the entire organoid are detected as distinct entities (but with 27% lower confidence for the bubble). These edge cases still indicate that performance remains sensitive to the signal-to-noise ratio and image quality. It is also important to note that such high-complexity imaging conditions (e.g., organoids surrounded by numerous bubbles) are considerably rarer in the training dataset, which explains the model's reduced though still good performance in these scenarios. The post-processing strategy of retaining only the single organoid with the highest confidence score clearly favors overall performance metrics. If the task required detecting multiple organoids per image, the complexity and difficulty would increase substantially.

To further substantiate these visual observations, segmentation performance was quantitatively assessed using IoU metrics. The predominance of high IoU values reflects a strong correspondence between automated predictions and manual annotations (Figure 6). The global analysis reveals a high median IoU of 0.944, confirming the precision of the pipeline.

The limited dispersion of IoU values, evidenced by a narrow interquartile range of 0.040, indicates that performance is consistent across the majority of the dataset rather than being driven by a small number of accurate predictions. This 'dense zone' of performance ([0.926, 0.967], capturing 60% of observations) suggests that the learned representations remain robust despite variability in organoid morphology (Figure 6A and 6B). Although 7 outliers fell below the whisker threshold (<0.858), likely corresponding to the complex edge cases discussed previously, the cumulative distribution analysis (Figure 6C) demonstrates that 98.2% of predictions exceed an IoU of 0.75 and 87.3% exceed an IoU of 0.9. This high success rate supports the applicability of the pipeline for routine quantitative analyses and longitudinal monitoring of organoids.

Furthermore, these results underscore a major advantage over manual contouring in Fiji. In addition to improving

reproducibility, the automated pipeline substantially reduces analysis time. Manual correction required approximately 2–4 minutes per image, representing a considerable time investment for large datasets. In contrast, once trained, the model is capable of automatically executing a 110-image test suite within a timeframe of 30 seconds to 4 minutes (depending on internet connection because of the use of Google apps). This time efficiency makes the approach particularly suitable for longitudinal studies and high-throughput experimental designs.

However, while the current pipeline is highly effective for brain organoid analysis, several technical options could further improve its performance. First, using more diverse data augmentation such as vertical flips, scaling, and brightness adjustments would make the model more resilient to variations in light and size. Second, adjusting the Mask R-CNN anchor boxes to better fit small or elongated organoids could reduce localization errors. Additionally, a longer learning rate warm-up or a step-decay strategy would stabilize early training and reduce curve noise. Finally, refining the Non-Maximum Suppression (NMS) and confidence thresholds would help transition the model toward multi-instance detection in crowded wells. These optimizations will enhance precision and help the model handle even more complex imaging environments.

CONCLUSION

This study demonstrates that advanced instance segmentation of brain organoids can be realistically implemented within standard laboratory settings using an open-source deep learning framework. The proposed Detectron2-based Mask R-CNN pipeline directly addresses the practical challenge outlined in the introduction by showing that consistent and reproducible organoid analysis is achievable under realistic laboratory constraints, without requiring dedicated local computational infrastructure. By leveraging transfer learning and cloud-based GPU resources, the pipeline achieves stable convergence, strong generalization across heterogeneous datasets, and consistently high delineation accuracy, as confirmed by training dynamics, qualitative inspection, and quantitative IoU metrics. The predominance of high IoU values with limited performance dispersion further indicates robust mask quality rather than isolated successes. Beyond segmentation accuracy, the pipeline enhances workflow scalability by reducing reliance on manual annotation, thereby limiting operator bias and enabling high-throughput quantitative analyses. The public release of the inference code ensures methodological transparency and reproducibility, facilitating broader adoption within the research community. While certain extreme imaging conditions reveal opportunities for further refinement, future developments will focus on advanced data augmentation, hyperparameter optimization, and architecture fine-tuning to further strengthen the pipeline's resilience. Overall, this work contributes to the integration of accessible AI tools into

organoid research, supporting more standardized, scalable, and reproducible morphometric assessment.

GLOSSARY

Bounding Box (BBox): A bounding box is a rectangular box used to localize an object within an image. It is typically defined by its coordinates, width, and height.

Brightfield microscopy: Optical microscopy technique based on light transmission through the sample, producing greyscale images without fluorescent labeling.

Empirical cumulative distribution function (ECDF): Function that represents the proportion of observations in a dataset with values less than or equal to a given threshold.

Empirical distribution: Distribution of observed values derived directly from measured data, without assuming any underlying theoretical probability model.

Feature Pyramid Network (FPN): A convolutional neural network architecture that enhances object detection at multiple scales by combining feature maps from different layers.

It improves detection of both small and large objects within an image.

Ground truth / Reference mask: Manually annotated segmentation mask considered as the correct representation of an object and used as a benchmark for training and evaluating automated models.

Instance segmentation: Computer vision task that involves detecting and delineating each individual object in an image, assigning a distinct segmentation mask to every object instance.

Interquartile range (IQR): Statistical measure defined as the difference between the third and first quartiles of a distribution, representing the spread of the central 50% of the data.

Intersection-over-Union (IoU): Metric used to quantify the overlap between a predicted segmentation mask and a reference mask, defined as the ratio of the area of their intersection to the area of their union.

Iterations: The total number of complete training cycles the model performs on the dataset.

Loss: A mathematical measure of the difference between the model's prediction and the actual ground truth.

Non-Maximum Suppression (NMS): Algorithm used to eliminate redundant detections by retaining only the prediction with the highest confidence among overlapping objects.

Region of Interest (ROI) / Candidate Regions: A Region of Interest is a selected area in an image that is likely to contain an object. Candidate regions are generated automatically by detection models for further classification and localization.

Region Proposal Network (RPN): A sub-network that identifies potential regions of interest within an image before detailed analysis.

Transfer learning: Machine learning strategy in which a model pretrained on a large dataset is adapted to a new task using a smaller, task-specific dataset.

Warm-up Strategy: An initial training phase where the learning rate increases linearly to stabilize the model's weight adjustments.

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CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

REFERENCES

1. Organoids: Principle, application and perspective. <https://www.the-innovation.org/article/doi/10.59717/j.xinn-life.2024.100088>.
2. Les organoïdes : des organes miniatures aujourd'hui incontournables pour la biologie et la médecine | INRAE. <https://www.inrae.fr/actualites/organoides-organes-miniatures-aujourd'hui-incontournables-biologie-medecine>.
3. in-carta-image-analysis-software.pdf.
4. Russell, S. J. & Norvig, P. *Artificial Intelligence: A Modern Approach*. (Pearson, Boston, 2022).
5. Jordan, M. I. & Mitchell, T. M. Machine learning: Trends, perspectives, and prospects. *Science* **349**, 255–260 (2015).
6. LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* **521**, 436–444 (2015).

7. Lecun, Y., Bottou, L., Bengio, Y. & Haffner, P. Gradient-based learning applied to document recognition. *Proc. IEEE* **86**, 2278–2324 (1998).
8. Deep Residual Learning for Image Recognition. *ResearchGate*
https://www.researchgate.net/publication/286512696_Deep_Residual_Learning_for_Image_Recognition.
9. Rawat, W. & Wang, Z. Deep Convolutional Neural Networks for Image Classification: A Comprehensive Review. *Neural Comput.* **29**, 2352–2449 (2017).
10. facebookresearch. GitHub - facebookresearch/detectron2: Detectron2 is a platform for object detection, segmentation and other visual recognition tasks. *GitHub*
<https://github.com/facebookresearch/detectron2>.
11. Google Colab.
https://colab.research.google.com/drive/16jcaJoc6bCFAQ96jDe2HwtXj7BMD_-m5.
12. COCO - Common Objects in Context.
<https://cocodataset.org/#home>.
13. colab.google. *colab.google* <http://0.0.0.0:8080/>.
14. Rani, A., Ortiz-Arroyo, D. & Durdevic, P. Defect Detection in Synthetic Fibre Ropes using *Detectron2* Framework. *Appl. Ocean Res.* **150**, 104109 (2024).

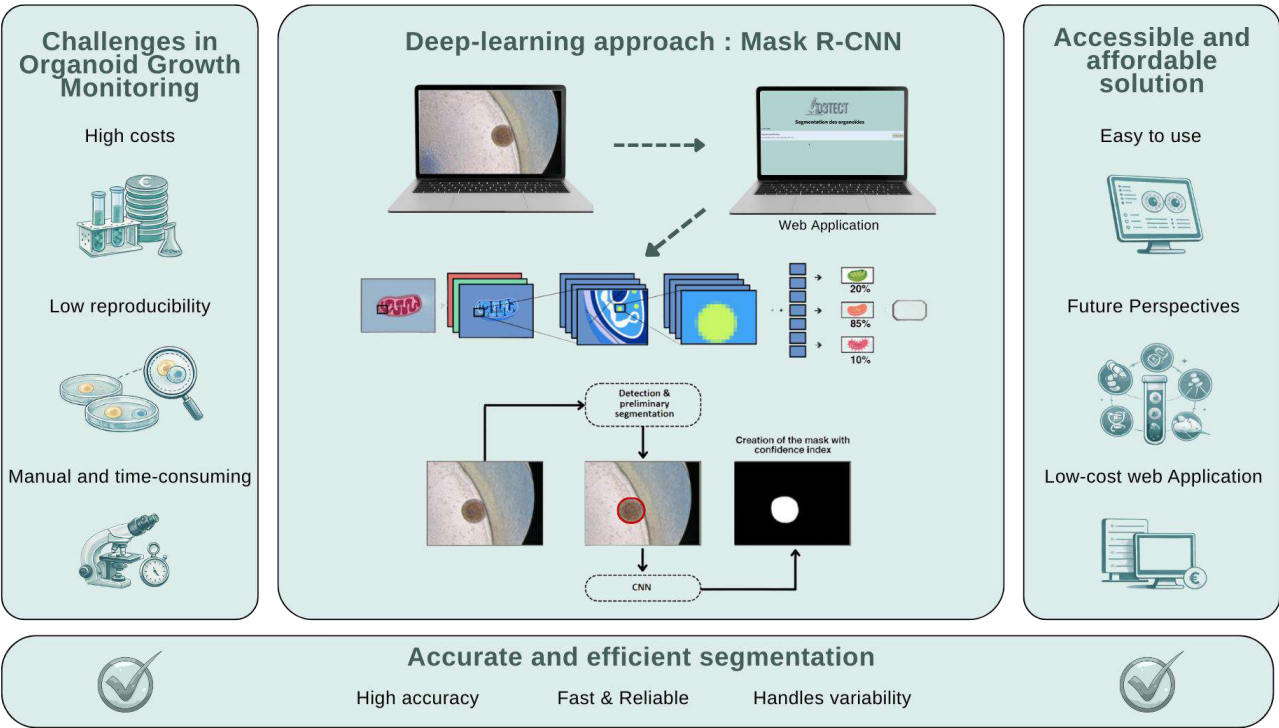
Deep Learning-Based Algorithm for Segmentation of Organoids from Microscopy Images

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VISUAL ABSTRACT



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Abstract – Three-dimensional (3D) cell culture and organoids models have revolutionized biomedical research by providing more physiologically relevant systems than traditional two-dimensional (2D) cultures. However, monitoring organoids growth and morphology remains challenging due to the need for expensive imaging systems or labor-intensive manual segmentation which is time-consuming and prone to observer-bias. In this study, we propose a deep learning-based approach using a Mask R-CNN architecture to automatically detect and segment organoids from microscopy images. The developed algorithm achieves high accuracy and reproducibility while significantly reducing analysis time and costs. This solution enables accessible and efficient monitoring of organoid growth without disrupting their culture environment. Our results demonstrate the potential of this method to democratize computational tools in cell biology and to improve the precision of organoid-based research. Although currently optimized for early-stage organoids, future work will focus on expanding the model to various organoid types and imaging conditions.

Index Terms – Organoids, Deep Learning, Mask R-CNN, Image Segmentation, Microscopy, Computational Biology.

INTRODUCTION

Traditionally, two-dimensional (2D) cell culture systems have been the standard model for studying cellular behaviours. In these systems, cells grow as monolayers on flat, rigid surfaces, which enables easy observation and manipulation. Cell culture remains essential to biological research, as it enables the reproduction and observation of cellular development *in vitro*. This process, characterized by the increase in the number and size of cells, plays a fundamental role in understanding tissue development, regeneration, and repair. Culturing cells outside their organism in a nutrient medium allows scientists to study their structure, behaviors, and interactions (Guertin & Sabatini, 2008).

However, cells grown in simple 2D environments exhibit a simplified architecture that does not reflect the complexity of whole organs. 2D cultures fail to accurately reproduce the three-dimensional (3D) microenvironment of tissues *in vivo*, including spatial organization, cell–cell and cell–matrix interactions, as well as biochemical and mechanical gradients that are essential for maintaining physiological functions. As a result, cellular responses observed in 2D systems often poorly translate to *in vivo* conditions (Suarez-Martinez et al., 2022).

To overcome these limitations, various 3D cell culture techniques have been developed, ranging from scaffolds and hydrogels to more complex structures such as organoids (Suarez-Martinez et al., 2022). An organoid is a 3D multicellular structure composed of different cell types specific to a given organ and capable of recapitulating key aspects of its architecture and function (Habanjar et al., 2021). These miniature structures can be derived from stem cells or induced pluripotent stem cells (iPSCs). iPSCs are adult cells that have been genetically reprogrammed to return to an embryonic-like state, allowing them to differentiate into almost any cell type and thereby reproduce essential organizational and functional properties of human organs (Yi et al., 2021).

Organoids therefore open up considerable perspectives in regenerative medicine, particularly for personalized therapeutic approaches. Because they can be generated from a patient's own cells, organoids are genetically identical to the recipient, which strongly limits immune rejection. This autologous strategy reduces the risk of graft rejection and the need for long-term immunosuppressive treatments, representing a major advantage over conventional transplantation methods (Lancaster & Knoblich, 2014). Consequently, organoids hold strong potential for personalized regenerative therapies and tissue replacement (Clevers, 2016).

Organoids also represent valuable models for biomedical research, enabling the investigation of complex diseases such as cancer or neurodegenerative disorders, including Alzheimer's disease (Dutta et al., 2017). Compared with conventional 2D cultures, organoids better reproduce tissue heterogeneity, 3D architecture, and physiologically relevant cell interactions. In cancer research, patient-derived organoids preserve genetic alterations and drug-response profiles, making them powerful tools for disease modeling and therapeutic screening. Similarly, brain organoids provide a unique platform to study early pathogenic mechanisms of neurodegenerative diseases, including neuronal differentiation defects and protein aggregation processes (Kim et al., 2020).

Given the increasing complexity of organoid-based models, their quantitative analysis requires advanced computational tools capable of handling large imaging datasets and complex biological structures. In recent years, deep learning has been successfully implemented to study a wide range of biological objects, particularly through medical imaging and microscopy. For example, convolutional neural networks have achieved expert-level performance in radiological image analysis, such as cancer detection and classification in mammography, computed tomography, and histopathological images, enabling more objective and reproducible diagnosis (Litjens et al., 2017). In parallel, deep learning has been widely adopted in cell biology for automated segmentation and analysis of cells and tissues from microscopy images, significantly outperforming traditional image-processing approaches in terms of accuracy and scalability (Ronneberger et al., 2015) (Moen et al., 2019). These successful applications highlight the relevance of deep learning for analyzing complex biological systems and strongly support its implementation for automated detection and segmentation of organoids.

A critical step in organoid analysis is segmentation, which consists of precisely outlining each organoid within microscopy images. Accurate segmentation is essential because it enables the extraction of quantitative morphological parameters such as surface area, perimeter, diameter, and shape descriptors. These metrics are fundamental for monitoring organoid growth, assessing developmental dynamics, and comparing experimental conditions. However, segmentation remains challenging due to variability in organoid morphology, imaging conditions, and the presence of visual artifacts.

Despite their potential, monitoring organoid growth still relies either on expensive imaging systems and proprietary software or on time-consuming manual segmentation, which is prone to observer bias and limits scalability. These constraints restrict the accessibility of organoid-based analyses, particularly in laboratories with limited resources. The challenge is therefore to develop an automated, low-cost, and

reliable solution capable of tracking organoid growth accurately without disturbing their culture environment.

The guiding question for this article is the following: How can organoid segmentation be automated, at low cost, in order to optimize the speed, reliability, and precision of analyses in cell biology?

In this study, we propose a deep learning-based approach using a Mask R-CNN architecture to automatically detect and segment organoids from bright-field microscopy images. Our method enables precise, reproducible, and rapid segmentation across diverse imaging conditions while significantly reducing analysis of time and cost. We demonstrate that the proposed approach achieves high segmentation accuracy and robust generalization when trained on heterogeneous datasets from multiple laboratories. By integrating this model into an accessible analysis pipeline, our work provides an efficient and affordable solution for quantitative organoid monitoring, contributing to the democratization of computational tools in cell biology.

MATERIALS AND METHODS

DATABASE

The database used in this study consists of microscopy images of human brain organoids obtained from three distinct sources: "Laboratory A" (CellTechs) (« CellTechs », s. d.), a partner research platform that provided experimental imaging data from its own work; "Laboratory B" (University Hospital Heidelberg) (Schröter et al., 2024) and "Laboratory C" (Dokuz Eylül University) (Kahveci et al., 2024), two additional laboratories whose datasets were collected from open-access repositories indexed on PubMed. Due to confidentiality reasons, Laboratory A's data will not be shown. All characteristics such as the origin of their organoids are described in the two articles.

The dataset includes images acquired under different experimental conditions, magnifications, and lighting settings, in order to capture the morphological variability of early-stage organoids. The images were primarily obtained using bright-field microscopy, representing organoids at various stages of growth and differentiation.

Flexible selection criteria were applied: all images in which one or more organoids were visible were retained, even when they contained artifacts like air bubbles, reflections, or variations in lighting. This choice allowed our deep learning model to learn to distinguish organoids from visual disturbances commonly present in real experimental conditions. However, duplicate images and those showing excessive noise or severe deformation were excluded to maintain the consistency and representativeness of the dataset.

The combination of images from Laboratories A, B, and C allowed us to test and compare different dataset random compositions to determine which mix produced the best-performing model. In total, the database includes 276 microscopy images, divided into 230 for training and 46 for validation to ensure robustness and reproducibility of the model's performance.

THE DEEP-LEARNING MODEL: MASK R-CNN

As mentioned earlier, in our study we used a Mask R-CNN model for automatic detection and segmentation of organoids in particular. It is useful to briefly recall the principle of convolutional neural networks (CNNs), on which it is based, as well as the concept of deep learning. Deep learning refers to a branch of artificial intelligence inspired by the functioning of the human brain. The approach of Mask R-CNN relies on artificial neural networks organized into multiple layers, capable of automatically learning from large amounts of data (Robert, 2020).

In the case of CNNs, these networks are specifically designed to analyze images: the first layers detect simple features such as lines, edges, and textures, while the deeper layers combine this information to recognize more complex visual structures. This hierarchical process allows the model to understand the visual content of an image, although a standard CNN alone cannot precisely locate objects or define their exact shape.

Mask R-CNN is an extension of these basic architectures. It not only identifies the presence of an object but also determines its position and outlines by producing a segmentation mask for each detected instance. In practice, the model analyzes the image to extract meaningful visual features and then proposes regions where objects are likely to be found. For each region, it performs two simultaneous operations: it classifies the object and defines a bounding box around it, while also generating a detailed pixel-by-pixel mask that outlines its shape.

This approach allows the Mask R-CNN model to detect and segment multiple objects simultaneously within the same image. It is particularly well suited to our dataset, which contains several organoids of varying sizes and shapes, sometimes close together or partially overlapping. The model can therefore distinguish each organoid individually and produce an accurate and reliable segmentation, which is essential for calculating morphological parameters such as area, perimeter, diameter, and circularity (*matterport/Mask_RCNN*, 2017/2025).

Finally, a transfer learning strategy was used to optimize the model. A Mask R-CNN model pre-trained on a large image dataset was retrained on our own annotated images, which reduced both training time and data requirements while maintaining a high level of precision for organoid segmentation.

PREPROCESSING AND ANNOTATION METHODS

Before training, the images were preprocessed and annotated to create a dataset suitable for supervised learning a method in which the model learns from labeled examples to automatically recognize and segment new, similar images.

The manual outlining of organoids was performed using Fiji software (Fiji, s. d.) an open-source platform widely used for biological image analysis. The microscopy images were first imported into Fiji, each visible organoid was carefully outlined using the “*freehand selections*” tool. After verifying that the cutout line created is as close as possible to the outline of the organoid, the selection was refined by applying “*Edit > Selection > Fit spline*”. The resulting outline was then converted into a binary mask and saved in PNG format. These masks were used as reference annotations for training and evaluating the model.

To increase the amount of data available and improve the model's generalization ability, a simple rotation-based data augmentation strategy was applied. This transformation introduces slight variations in organoid orientation without altering their morphology, helping the model adapt to diverse imaging conditions.

The dataset was then divided into training and validation subsets to enable model optimization and performance monitoring during learning. Regarding split ratios, we chose 80% of images and masks for training and 20% of images and masks for validation. The test images were selected separately: some were similar to those used for training and validation, while others were new images that the model had never seen before. This approach made it possible to assess the model's ability to generalize its predictions to unseen data.

ADAPTATION OF THE PRE-TRAINED MASK R-CNN MODEL TO OUR DATASET

A custom pipeline was developed to automatically detect and segment organoids from microscopy images using Python ver. 3.7.12 (*Welcome to Python.Org*, 2025). The development environment was based on Anaconda (Contributors, s. d.) for package management and Visual Studio Code (*Visual Studio Code - Code Editing. Redefined*, s. d.) for code implementation and execution. The Mask R-CNN model, built with TensorFlow library version 1.15.0 (*TensorFlow*, s. d.), was employed for organoid segmentation, as it enables both object detection and pixel-level instance segmentation, allowing each organoid to be identified and outlined individually within an image.

Prior to model training, a preprocessing step ensured that all input images shared the same format and dimensions. A dedicated Python function was implemented to read entire folders of microscopy images and automatically convert them into a standardized format compatible with the model. In addition, a verification step was included before model

initialization to confirm the presence and integrity of the corresponding binary masks for each image.

The pipeline integrated into several complementary libraries:

- Scikit-image for image manipulation and transformation (Van Der Walt et al., 2014),
- NumPy for array operations and mathematical computations (NumPy, s. d.),
- Matplotlib for visualization of intermediate and final results (Matplotlib — Visualization with Python, s. d.),
- H5py for managing structured data and model storage (HDF5 for Python — h5py 3.15.1 documentation, s. d.).

This integrated workflow enables the extraction of quantitative measurements for each segmented organoid, including surface area, perimeter, and diameter, providing a robust, reproducible, and efficient framework for quantitative organoid analysis.

Model training was performed in Python using the TensorFlow library within the Anaconda environment. The base implementation of the Mask R-CNN architecture was adapted from the open-source repository published by Matterport on GitHub (*matterport/Mask_RCNN*, 2017/2025). The model was trained for 10 epochs, one epoch being a complete iteration over the whole dataset, using transfer learning from pre-trained weights on the COCO dataset (Lin et al., 2014), a large-scale database commonly used for computer vision tasks. This approach allowed the network to reuse generalized visual representations before adapting to organoid segmentation. Only the final layers specific to the dataset were retrained.

Each model configuration, corresponding to the different dataset compositions, was trained once to ensure comparability across results. Training was executed on a workstation equipped with an Intel Core i9-13900HX (2.20 GHz) processor, 32 GB of RAM, and an NVIDIA GeForce RTX 4090 Laptop GPU (16 GB VRAM), providing sufficient computational resources for stable and efficient optimization. All scripts and trained model weights were stored and versioned on GitHub, ensuring full traceability and reproducibility of the work. The trained model, source code, the associated web application and a user's guide are publicly available on GitHub at the following link: <https://github.com/Shehanieudu/Detect.git>.

DATA ANALYSIS

The intersection over union (IoU) is obtained by dividing the area of overlap between masks by the area of union of the masks. The IoU was calculated on a small sample size after each model training in order to select the best model.

To further optimize performance, the model used a multi-task loss function that combines errors from five distinct metrics, including positioning and classification. By comparing its prediction to the manual annotations, the model computes a global loss score at each epoch. This metric is recorded as training loss and validation loss to guide learning and allowed us to choose the best model. It should be noted that due to the small sample size of models, statistics are not applicable.

RESULTS

QUALITATIVE ASSESSMENT OF ORGANOID SEGMENTATION

The performance of the Mask R-CNN model was first evaluated qualitatively by comparing automatic segmentation outputs with manual annotations across a variety of image conditions (Fig.1). The dataset included images with single organoids, multiple organoids, heterogeneous morphologies, and visual artifacts such as air bubbles or uneven illumination.

In well-cropped images containing a single organoid (Fig.1A), the model generated segmentation masks that closely matched the manually annotated references, with clear and continuous delineation of organoid boundaries. This high level of agreement is reflected by an IoU value of 90.0%, indicating that the automated segmentation is almost entirely overlapped with the manually defined organoid area. Comparable results were obtained for images presenting more complex organoid shapes or internal texture variations (Fig.1B), with an IoU of 90.1%. These results indicate that the model successfully learned the visual features associated with early-stage organoids despite variability in morphology and imaging conditions.

In contrast, images containing artifacts such as air bubbles (Fig.1C) showed lower segmentation performance (IoU = 36.9%), although the model was generally able to distinguish organoids from non-biological structures. However, occasional misdetections occurred when these artifacts closely appeared similar to organoid contours, particularly when their size approached that of actual organoids, leading to visual ambiguity for the model. Images displaying strongly bilobed or partially fused organoids (Fig. 1D) showed a reduced segmentation agreement compared to simpler cases, with an IoU of 83.9%. A notable limitation was identified in these cases, where automatic segmentation failed to correctly separate adjacent structures, highlighting the model's sensitivity to extreme morphological complexity. Finally, in images containing multiple organoids within the same field of view (Fig.1E), the model successfully detected and segmented several instances simultaneously, achieving a high IoU of 91.6%. This result highlights the model's ability to handle multi-instance segmentation while maintaining accurate boundary definition.

Overall, these qualitative results indicate that the proposed deep learning approach performs robustly for isolated and

multiple organoids under standard imaging conditions, while more complex visual scenarios, such as the presence of artifacts or fused morphologies, lead to reduced segmentation agreement. The range of IoU values observed across Figure 1 reflects the sensitivity of segmentation performance to image complexity and morphological ambiguity. IoU was computed by comparing the binary segmentation mask generated by the model with the corresponding manually annotated mask on a pixel-by-pixel basis and defined as the ratio between the number of overlapping pixels (intersection) and the total number of pixels belonging to either mask (union).

QUANTITATIVE EVALUATION OF MODEL PERFORMANCE

To quantitatively assess model performance and generalization, several training configurations were tested using datasets originating from Laboratory A, B, and C, either individually or in combination. Training and validation performance and loss values are summarized in Figure 2 and Table 1.

When models were trained on individual laboratory datasets, training performance ranged from 77.02% (Laboratory B) to 80.24% (Laboratory A), with corresponding validation performances between 70.73% and 82.73% (Table 1A). These results indicate a moderate level of generalization but also reveal variability linked to dataset-specific characteristics.

Combining datasets from multiple laboratories led to a noticeable improvement in validation performance. Models trained on combined datasets (A+B, A+C, B+C) showed higher robustness, with validation performances reaching up to 79.16%. The best overall performance was achieved when training on the full combined dataset (A+B+C), which resulted in a validation performance of 85.53%, exceeding the corresponding training performance (83.44%). This suggests

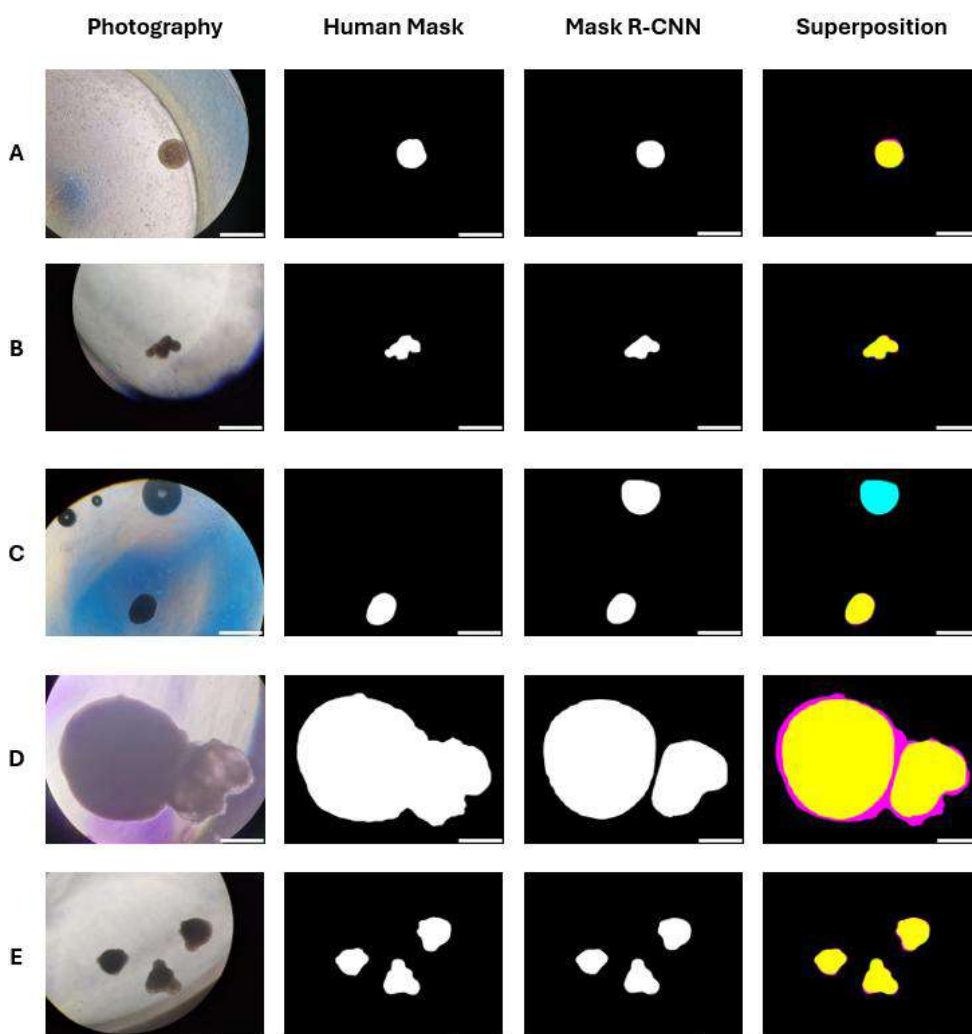


Figure 1: Comparison between human-created masks and predicted masks across different types of organoid images. (A) Well segmentation of a round organoid. (B) Well segmentation of an irregular-shaped organoid. (C) Image containing an organoid and a bubble. (D) 'Bilobed' organoid. (E) Three organoids in a single picture. In the comparison column, human-created masks are highlighted in magenta; predicted masks are highlighted in blue, and the superposition is highlighted in yellow. IoU from superpositions A to E are as follows: 90.0%; 90.1%; 36.9%; 83.9%; 91.6%. Scale bar set at 500 pixels.

improved generalization and reduced overfitting when data heterogeneity is increased.

However, notably, the A+B combination was an exception to this trend, exhibiting an unexpectedly low validation performance of 60.14% with a substantial performance drop of 39.86% compared to its training performance. This anomaly could be attributed to several factors, including possible incompatibilities between the experimental protocols of laboratories A and B, significant differences in image characteristics (contrast, brightness, resolution), or potential overrepresentation of certain organoid phenotypes in the training set that are not present in the validation set.

Loss analysis further supports the observations of Fig.2B and 2D with Table 1B. Models trained on single datasets exhibited

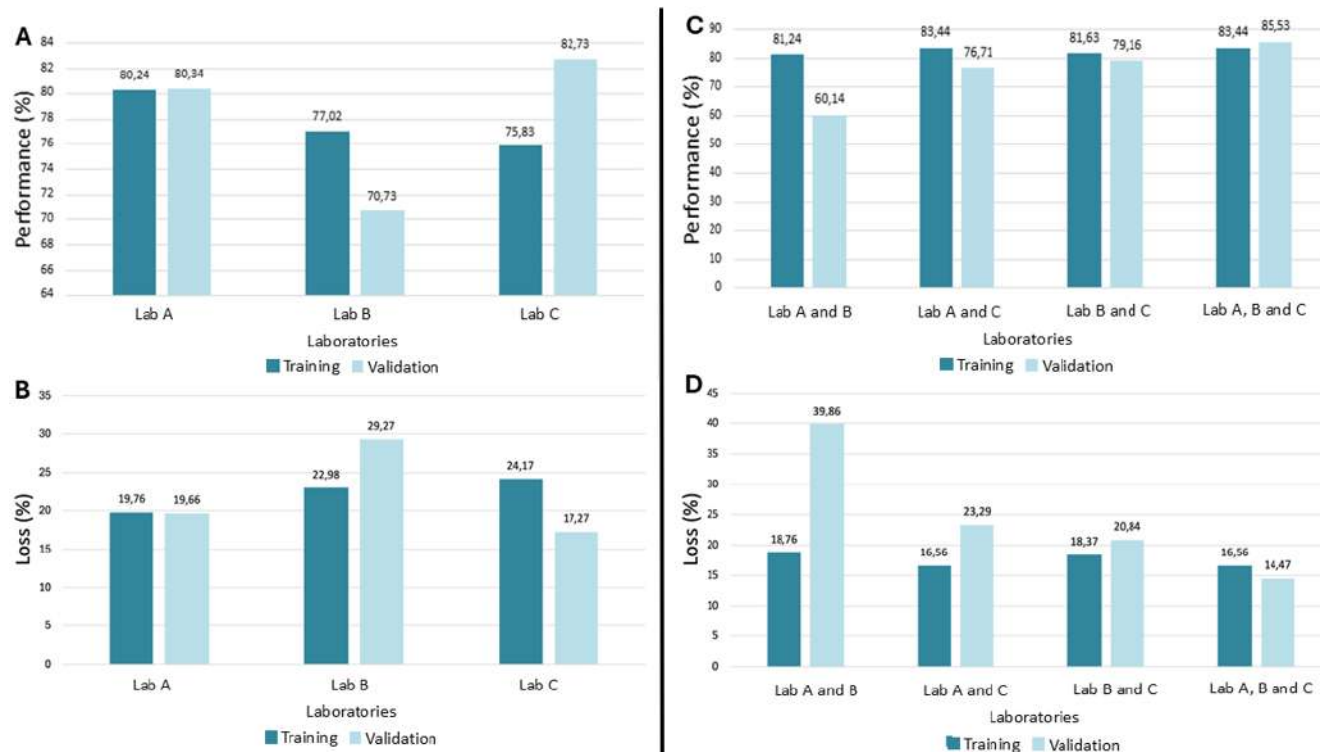


Figure 2: Training and validation performance and loss diagrams for different deep learning model configurations. The figure compares model accuracy and loss across different dataset configurations to assess the influence of data source combinations on training stability and generalization performance. (A) Training performance of models trained separately on datasets from laboratories A, B, and C. (B) Training losses of the corresponding models. (C) Validation performance of models trained on combined datasets (A+B, B+C, A+C, A+B+C). (D) Validation losses corresponding to the combined training datasets.

higher validation losses, particularly for Laboratory B. In contrast, combined datasets consistently showed lower validation losses, with the lowest value observed for the A+B+C configuration (14.47%). The reduced difference between training and validation loss in this configuration indicates improved stability and generalization of the model.

EXTRACTION OF MORPHOLOGICAL MEASUREMENTS

Once segmentation was completed, the generated masks were used to extract quantitative morphological parameters for each detected organoid, including surface area, perimeter, diameter, and circularity.

These measurements were computed automatically for all segmented instances, providing standardized and reproducible metrics that can be used to monitor organoid growth over time. The extracted data were structured and made available for download in both CSV and Excel formats, facilitating further analysis and integration into existing experimental workflows.

Compared to manual segmentation, the automated approach significantly reduced processing time while maintaining consistent measurements across images. The ability to extract quantitative features directly from segmentation outputs, combined with easy access to the results in common data

formats, demonstrates the practical relevance of the proposed method for high-throughput organoid analysis and longitudinal studies.

DISCUSSION

INFLUENCE OF IMAGE QUALITY ON SEGMENTATION PERFORMANCE

The qualitative results clearly show that image quality plays a significant role in segmentation accuracy. In fact, we can see that under optimal imaging conditions (good focus, homogeneous illumination, absence of reflections, always keep the same methodology), the model produced segmentation masks that closely matched manual annotations. These conditions contribute to reproducibility and enable the model to reliably learn the visual characteristics specific to organoids.

However, images affected by dark reflections or strong illumination occasionally led to incorrect detections. In this type of case, the model sometimes interpreted dark regions or shadows as organoids, probably because the model is based on what it knows and visually associates these shadows to a real organoid. This limitation highlights the sensitivity of deep learning models to image artifacts and underlines the

importance of standardized image acquisition protocols to improve robustness.

CHALLENGES REGARDING BUBBLES AND BILOBED OR FUSED ORGANOIDS

One recurring source of segmentation error concerns the presence of air bubbles in hand-taken microscopy images. This limitation does not originate from inadequate training or annotation errors but rather reflects inherent constraints of the model. Indeed, Mask R-CNN primarily relies on visual features such as shape, texture, and brightness, without any biological understanding of the objects being segmented. Since bubbles and organoids share several morphological characteristics (roundness, defined borders, intensity patterns), the model lacks sufficient discriminative information to reliably distinguish between them. One potential approach to mitigate this issue would be to increase the number of training images containing such artifacts, allowing the model to better learn discriminative patterns between bubbles and true biological structures.

A second limitation was observed in the segmentation of bilobed or partially fused organoids. In such cases, the model may incorrectly interpret the morphological separation between lobes, leading to two organoids being detected as one. Potential improvements could involve applying image preprocessing strategies, such as morphological filtering, smoothing operations, or brightness normalization, to enhance the visual continuity of bilobed organoids during the training phase. However, this issue would require a larger dataset containing diverse examples of bilobed and fused organoids. Together, these observations illustrate inherent

limitations of instance-based segmentation models when confronted with complex or ambiguous biological morphologies.

IMPACT OF DATASET COMPOSITION ON MODEL GENERALIZATION

The training performance improved with the addition of images from multiples laboratories, suggesting that model prediction accuracy is intrinsically linked to the diversity and heterogeneity of the dataset. This trend suggests that diversity acts as a regularization mechanism against overfitting. When trained on a single laboratory, the model risks “over-learning” laboratory-specific signatures, such as luminosity or camera characteristics, which reduces the model’s generalizability outside the trained dataset. In the triplet configuration, the model has more resources to ignore those non-biological variables, making the model more robust to varied imaging conditions, provided the image quality remains sufficient. Moreover, significant gaps between training losses and validation losses (or training performance and validation performance), may imply the need to regularize the dataset or to ensure that the training and validation datasets are statistically equivalent.

Furthermore, the gap between model predictions and manual masks may be explained in part due to inter-observer variability, inherent in our multi-user annotation process across six members. Even if the same standardized protocol was followed, subjective interpretations inevitably arise, particularly with irregularly shaped organoids. In such cases, the precise delineation of organoids or overlapping edges can vary between individuals, consequently, the model learns an ‘average’ of these annotations. However, the large volume of

the dataset helps mitigate the issue; by training on a diverse range of masks, the model reduces the weight of individual inconsistencies, allowing the global learning process to converge toward a more stable prediction of organoid morphology.

INTEGRATION INTO A WEB-BASED APPLICATION

To facilitate accessibility and usability, the trained model was integrated into a web application that allows users to upload single microscopy images, multiple images, or entire folders for batch processing (Fig.3). Each picture has to be vigorously taken with the same objective on the microscope, the same distance between the organoid and the camera and

Table 1. Training and validation performances and losses of the Deep-learning models according to the dataset composition. (A) Training and validation performances (%) of models trained on datasets from laboratories A, B, and C, individually or in combination (A+B, A+C, B+C, A+B+C). (B) Corresponding training and validation losses (%) for the same dataset configurations.

A	Laboratory	Training performance (%)	Validation performance (%)	Training - Validation (%)
	Lab A	80,24	80,34	-0,1
	Lab B	77,02	70,73	6,29
	Lab C	75,83	82,73	-6,9
	Lab A and B	81,24	60,14	21,1
	Lab A and C	83,44	76,71	6,73
	Lab B and C	81,63	79,16	2,47
	Lab A, B and C	83,44	85,53	-2,09

B	Laboratory	Training loss (%)	Validation loss (%)	Training - Validation (%)
	Lab A	19,76	19,66	0,1
	Lab B	22,98	29,27	-6,29
	Lab C	24,17	17,27	6,9
	Lab A and B	18,76	39,86	-21,1
	Lab A and C	16,56	23,29	-6,73
	Lab B and C	18,37	20,84	-2,47
	Lab A, B and C	16,56	14,47	2,09



Figure 3: Screenshot of the built webapp. Complete demonstration of the web application in french at <https://youtu.be/KWkiU70KAJI>.

finally the same zoom on the camera. For each input, the application generates segmented outputs along with quantitative morphological measurements and provides the associated confidence score for each detected organoid. The interface displays visual overlays of the segmentation masks on the original images, enabling rapid verification and interpretation of the results. This implementation demonstrates the feasibility of deploying the proposed solution as an accessible, low-cost tool for routine organoid monitoring without requiring advanced computational expertise.

CONCLUSION

In conclusion, this study presents a robust and accessible deep learning-based pipeline for the automatic segmentation of early-stage brain organoids from bright-field microscopy images. By using a Mask R-CNN architecture trained on diverse datasets originating from multiple laboratories, our method provides accurate and reproducible segmentation while enabling the automated extraction of key morphological parameters and significantly reducing analysis time and manual bias.

Although the current model was optimized and validated on brain organoids, preliminary observations indicate that similar performance can be achieved on other early-stage 3D structures, such as gastruloids (data not shown), suggesting that the proposed framework is broadly applicable to a wide range of early-stage organoid models. Nevertheless, due to intrinsic differences in morphology, texture, and growth dynamics across organoid types, we will have to create

dedicated training and validation datasets will be required for each specific model to ensure optimal performance. Future work will therefore focus on extending this approach to additional organoid systems and imaging conditions, reinforcing its potential as a generalizable and low-cost tool for quantitative analysis in biology.

CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

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REFERENCES

- CellTechs. (s. d.). *La recherche à SupBiotech*. Consulté 24 octobre 2025, à l'adresse <https://www.supbiotech.fr/recherche-innovation-biotechnologie/laboratoire-celltechs/>
- Clevers, H. (2016). Modeling Development and Disease with Organoids. *Cell*, 165(7), 1586-1597. <https://doi.org/10.1016/j.cell.2016.05.082>
- Contributors, S. W. (s. d.). *Spyder | The Python IDE that scientists and data analysts deserve*. Spyder IDE. Consulté 30 octobre 2024, à l'adresse <https://www.spyder-ide.org>
- Dutta, D., Heo, I., & Clevers, H. (2017). Disease Modeling in Stem Cell-Derived 3D Organoid Systems. *Trends in Molecular Medicine*, 23(5), 393-410. <https://doi.org/10.1016/j.molmed.2017.02.007>
- Fiji. (s. d.). ImageJ Wiki. Consulté 24 octobre 2025, à l'adresse <https://imagej.github.io/software/fiji/index>
- Guertin, D. A., & Sabatini, D. M. (2008). Chapter 12—Cell Growth. In J. Mendelsohn, P. M. Howley, M. A. Israel, J. W. Gray, & C. B. Thompson (Éds.), *The Molecular Basis of Cancer (Third Edition)* (p. 169-175). W.B. Saunders. <https://doi.org/10.1016/B978-141603703-3.10012-3>
- Habanjar, O., Diab-Assaf, M., Caldefie-Chez, F., & Delort, L. (2021). 3D Cell Culture Systems : Tumor Application, Advantages, and Disadvantages. *International Journal of Molecular Sciences*, 22(22), 12200. <https://doi.org/10.3390/ijms222212200>
- HDF5 for Python—H5py 3.15.1 documentation. (s. d.). Consulté 24 octobre 2025, à l'adresse <https://docs.h5py.org/en/stable/>
- Kahveci, B., Polatli, E., Bastanlar, Y., & Guven, S. (2024). OrganoLabeler : A Quick and Accurate Annotation Tool for Organoid Images. *ACS Omega*, 9(46), 46117-46128. <https://doi.org/10.1021/acsomega.4c06450>
- Kim, J., Koo, B.-K., & Knoblich, J. A. (2020). Human organoids: Model systems for human biology and medicine. *Nature Reviews Molecular Cell Biology*, 21(10), 571-584. <https://doi.org/10.1038/s41580-020-0259-3>
- Lancaster, M. A., & Knoblich, J. A. (2014). Organogenesis in a dish: Modeling development and disease using organoid technologies. *Science*, 345(6194), 1247125. <https://doi.org/10.1126/science.1247125>
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO : Common Objects in Context. In D. Fleet, T. Pajdla, B. Schiele, & T. Tuytelaars (Éds.), *Computer Vision – ECCV 2014* (p. 740-755). Springer International Publishing. https://doi.org/10.1007/978-3-319-10602-1_48
- Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60-88. <https://doi.org/10.1016/j.media.2017.07.005>
- Matplotlib—Visualization with Python. (s. d.). Consulté 24 octobre 2025, à l'adresse <https://matplotlib.org/>
- Matterport/Mask_RCNN. (2025). [Python]. Matterport, Inc. https://github.com/matterport/Mask_RCNN (Édition originale 2017)
- Moen, E., Bannon, D., Kudo, T., Graf, W., Covert, M., & Van Valen, D. (2019). Deep learning for cellular image analysis. *Nature Methods*, 16(12), 1233-1246. <https://doi.org/10.1038/s41592-019-0403-1>
- NumPy. (s. d.). Consulté 24 octobre 2025, à l'adresse <https://numpy.org/>
- Robert, J. (2020, septembre 28). Deep Learning ou Apprentissage Profond : Qu'est-ce que c'est ? *DataScientest*. <https://datascientest.com/deep-learning-definition>
- Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net : Convolutional Networks for Biomedical Image Segmentation* (arXiv:1505.04597). arXiv. <https://doi.org/10.48550/arXiv.1505.04597>
- Schröter, J., Deininger, L., Lupse, B., Richter, P., Syrbe, S., Mikut, R., & Jung-Klawitter, S. (2024). A large and diverse brain organoid dataset of 1,400 cross-laboratory images of 64 trackable brain organoids. *Scientific Data*, 11(1), 514. <https://doi.org/10.1038/s41597-024-03330-z>
- Suarez-Martinez, E., Suazo-Sanchez, I., Celis-Romero, M., & Carnero, A. (2022). 3D and organoid culture in research : Physiology, hereditary genetic diseases and cancer. *Cell & Bioscience*, 12(1), 39. <https://doi.org/10.1186/s13578-022-00775-w>

- TensorFlow*. (s. d.). Consulté 15 février 2025, à l'adresse <https://www.tensorflow.org/?hl=fr>
- Van Der Walt, S., Schönberger, J. L., Nunez-Iglesias, J., Boulogne, F., Warner, J. D., Yager, N., Gouillart, E., & Yu, T. (2014). scikit-image: Image processing in Python. *PeerJ*, 2, e453. <https://doi.org/10.7717/peerj.453>
- Visual Studio Code—Code Editing. Redefined*. (s. d.). Consulté 24 octobre 2025, à l'adresse <https://code.visualstudio.com/>
- Welcome to Python.org*. (2025, février 11). Python.Org. <https://www.python.org/>
- Yi, S. A., Zhang, Y., Rathnam, C., Pongkulapa, T., & Lee, K.-B. (2021). Bioengineering Approaches for the Advanced Organoid Research. *Advanced Materials*, 33(45), 2007949. <https://doi.org/10.1002/adma.202007949>

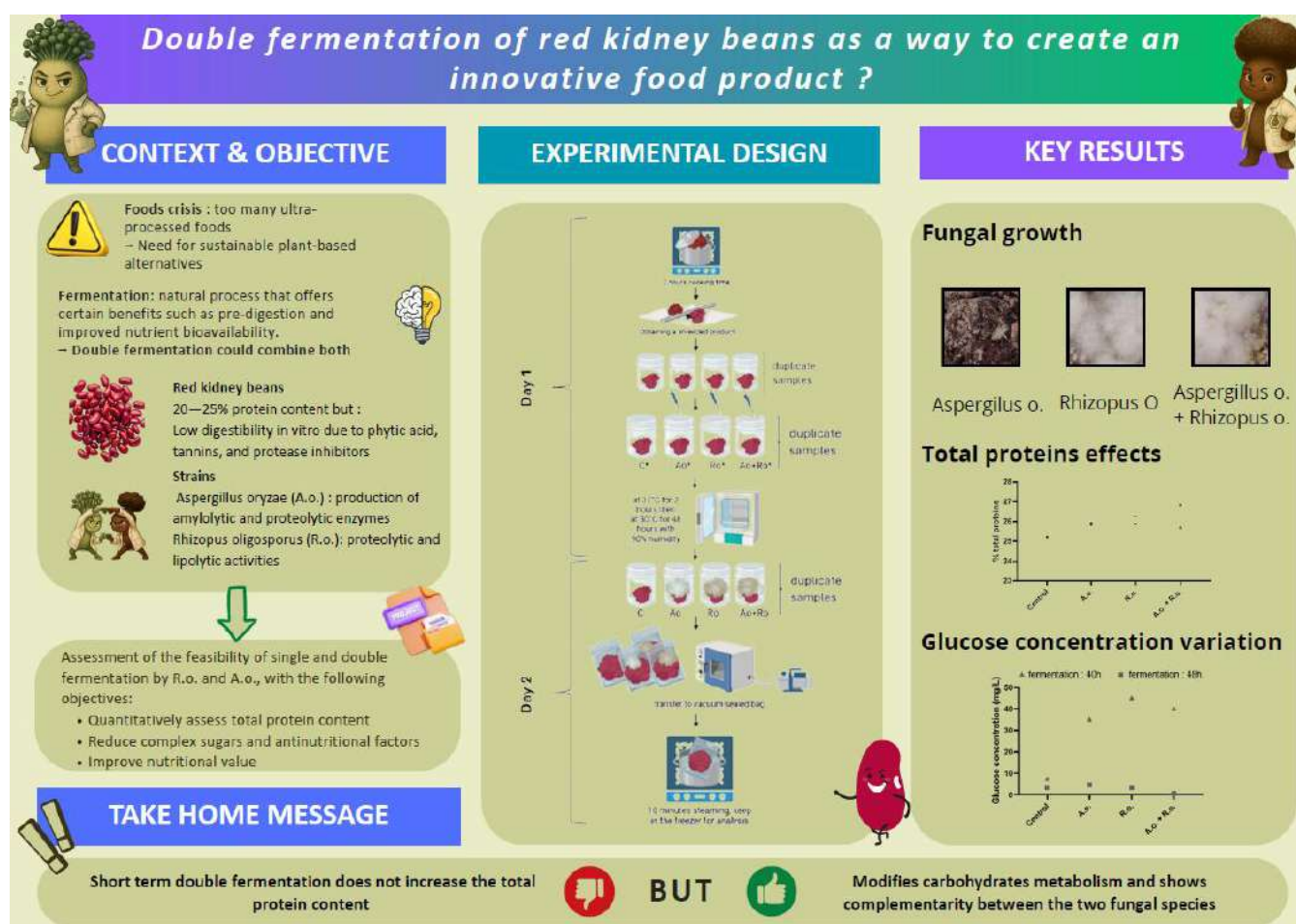
Double Fermentation of Red Kidney Beans as a Way to Create an Innovative Food Product?

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VISUAL ABSTRACT



Double Fermentation of Red Kidney Beans as a Way to Create an Innovative Food Product?

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Abstract – The increasing consumption of ultra-processed foods raises major public health concerns, highlighting the need for sustainable and nutritionally improved alternatives. In this study, a simultaneous double fermentation of red kidney beans using *Aspergillus oryzae* and *Rhizopus oligosporus* was investigated as a strategy to enhance the nutritional value of plant-based proteins. Single and double fermentations were performed under solid-state conditions for 48h to analyse fungal growth, total protein content, and glucose dynamics. Distinct species-specific growth patterns were observed, with homogeneous colonization by *Rhizopus oligosporus* and more heterogeneous development by *Aspergillus oryzae*. Despite visible differences in fungal growth, total protein content remained stable across conditions, as assessed using the Kjeldahl method. In contrast, glucose monitoring revealed notable differences in carbohydrate metabolism, with *Rhizopus oligosporus* showing the highest capacity for complex sugar degradation after 48h. These results suggest that short-term fungal fermentation primarily induces metabolic reorganization of carbohydrates rather than quantitative changes in protein content. Overall, this work emphasizes the potential of double fungal fermentation as a promising approach for the valorisation of legume-based substrates and the development of innovative, sustainable fermented plant protein ingredients.

Index Terms – Double fermentation, *Rhizopus oligosporus*, *Aspergillus oryzae*, red kidney beans, Kjeldahl, plant-based proteins, glucose, metabolism, valorisation.

INTRODUCTION

Over the past few decades, people's eating habits have changed significantly due to the industrialization of the food industry and the accelerated pace of modern life. According to INSEE ¹, between 1960 and 2010, the time spent preparing meals was reduced by 25% and, as a result, this study shows that the composition of grocery lists changed during this same

period. There has been a 5% decrease in purchases of raw products such as meat, fruit, and vegetables and, conversely, a 4% increase in purchases of sugary products and of quick, convenient, and cheaper options, known as prepared meals. This clearly shows that home-cooked meals and fresh foods have gradually been replaced. Foods produced through complex industrial processes and typically containing multiple additives, artificial flavors, texturizing agents, fats, sugars, and salt aimed at reducing production costs while enhancing shelf life and taste, are referred to as ultra-processed foods ². Easy to find and tasty, they now occupy an important place in the modern diet ³.

However, this change in the way we eat has serious effects on our health ⁴. While problems such as obesity, type 2 diabetes, and heart diseases are already linked to ultra-processed foods, their effects on the digestive system are now receiving more attention ³. Healthy digestion relies on a balance among the organs of the digestive system, digestive enzymes, and the gut microbiota ⁵. Excessive consumption of fatty, sweet, or salty foods can disrupt this balance leading to problems such as slow digestion, gut microbiota imbalance, acid reflux, bloating, and disorders such as irritable bowel syndrome (IBS) ^{6,7}. Moreover, the additives and emulsifiers found in these foods may damage the intestinal barrier, and make the digestive tract more permeable, potentially leading to chronic inflammation ⁸.

FERMENTATION AND PLANT-BASED PROTEINS VALORISATION

Given the negative effects of ultra-processed foods on health, there is a growing need for alternatives that improve nutritional quality while limiting heavy processing.

In this context, fermentation appears to be a relevant solution to enhance the value and digestibility of plant proteins.

Fermentation is a traditional biotechnological process that harnesses the metabolic activity of microorganisms such as bacteria, yeasts, and filamentous fungi, which can transform

organic substrates thanks to their specific enzymes. During this process, microbial enzymes (proteases, amylases, phytases) break down macromolecules into smaller, more accessible compounds. This mechanism reduces antinutritional factors such as phytic acid, tannins, and trypsin inhibitors, while improving protein digestibility and amino acid bioavailability^{9,10}. According to Senanayake et al.¹⁰, the in vitro protein digestibility of legumes such as lentils or chickpeas can increase by 15 to 35% after fermentation, depending on the strains and environmental conditions. Phytate degradation can sometimes reach up to 80%, thereby enhancing mineral availability. Moreover, microbial proteolysis results in the formation of low-molecular-weight peptides, including bioactive sequences with antioxidant and antihypertensive potential, whose activity may increase by up to 50% compared to non-fermented controls⁹.

Beyond the nutritional benefits, fermentation also improves sensory properties. Senanayake et al.¹⁰ mention an approximate 40% reduction in compounds responsible for the undesirable flavours associated with legumes, along with a measurable improvement in texture and palatability, linked to the microbial production of pleasant volatile compounds. Together, these effects present fermentation as a sustainable and effective approach to enhance the nutritional and functional value of plant proteins, while limiting their level of processing. These results support the growing interest in optimizing fermentation processes, especially by combining microorganisms with complementary enzymatic functions.

Among the different strategies explored to improve the nutritional quality of plant proteins, double fermentation, has gained considerable attention due to its synergistic effects. This approach involves combining several microorganisms with complementary enzymatic profiles to enhance the degradation of macromolecules and the release of nutrients.

DOUBLE FERMENTATION

In this context, *R.o.* and *A.o.* occupy a central place among filamentous fungi used in traditional food fermentations. Their association is commonly applied in processes such as tempeh or miso production, where it enables a more complete hydrolysis of proteins and carbohydrates¹⁰. On the one hand, *R.o.* is known for its proteolytic and lipolytic activities, which facilitate the breakdown of proteins and lipids while maintaining substrate cohesion. *A.o.*, on the other hand, produces a wide range of amylolytic and proteolytic enzymes that accelerate the conversion of starch and proteins into simple sugars and free amino acids.

A study by Starzyńska-Janiszewska et al.¹¹ showed that double fermentation of grass pea (*Lathyrus sativus*) by *R.o.* and *A.o.* led to an 18% increase in the final product's protein content, a 42% decrease in simple sugars, and a 1.4-fold increase in total free amino acid content, along with a notable improvement in antioxidant activity. These results confirm that double fermentation can more effectively enhance the

nutritional profile of legumes than monoculture fermentation, due to the enzymatic complementarity of the two fungi.

The success of double fermentation depends on both the microorganisms used and the chosen substrate. Therefore, selecting a legume with high potential but known limitations, such as red kidney beans, is essential to properly evaluate this approach.

CHOICE OF SUBSTRATE AND STUDY OBJECTIVES

Legumes are a major source of plant proteins, but their use remains limited due to low digestibility and the presence of antinutritional compounds. Red kidney beans (*Phaseolus vulgaris*) have a high protein content, ranging from 20 to 25% of dry matter, but an in vitro digestibility often below 70%, due to the presence of phytic acid, tannins, and protease inhibitors^{9,10}. These compounds reduce the bioavailability of amino acids and limit the nutritional value of beans, despite their strong potential as a sustainable protein ingredient.

Thus, double fermentation appears as a promising strategy to overcome these limitations, relying on the enzymatic synergy between different microbial strains^{10,11}. The combination of *R.o.* and *A.o.*, already recognized for their complementary effects in double fermentation, could improve the digestibility and functional profile of red kidney beans by increasing the content of assimilable proteins, reducing simple sugars, and generating bioactive peptides.

This study evaluates the feasibility and nutritional impact of a simultaneous double fermentation of red kidney beans by *R.o.* and *A.o.*, with the objective of:

- quantitatively assessing total proteins content,
- reducing the content of simple sugars and antinutritional factors,
- and improving the overall nutritional value of the fermented substrate.

Ultimately, this work contributes to the development of high-value-added fermented plant protein ingredients, tailored to meet population nutritional needs, particularly in the context of the transition toward more sustainable diets.

MATERIALS AND METHODS

STRAINS

Spores of *Aspergillus oryzae* (*Koji-Kin*), a culinary-grade strain from the brand Kensho, were purchased from Amazon (reference: B07HF4J4CG).

Spores of *Rhizopus oligosporus*, a culinary-grade strain from the brand Tempehtation, were provided by the Sup'Biotech culinary laboratory.

RED KIDNEY BEANS

All red kidney beans used in the experiments were purchased in a dry state from the commercial retailer Grand Frais (France) (reference: 132585).

*FERMENTATION***Preparation of the substrate**

Red kidney beans are soaked for 24h, then drained and vacuum-sealed in bags. The beans are cooked for 2h in boiling water and then mashed.

Preparation of the inoculum

The inoculums are prepared in sterile physiological saline solution (0.9% NaCl) under sterile conditions (Table 1).

Table 1: Inoculum preparation parameters for A.o. and R.o. cultures.

Conditions	A.o.	R.o.	A.o. + R.o.
Inoculation concentrations (per gram of substrate)	10 ⁴ spores	10 ⁴ spores	10 ⁴ + 2.10 ⁴ spores
Substrate weight (g) per jar	200	200	200
Number of jars	2	2	2

FERMENTATION PARAMETERS

The substrate and the inoculum are placed in clean and sterile glass jars (size: diameter 9.5cm × height 10cm), in layers. The jars were incubated (incubator Memmert) at 37°C for 2h, then again at 30°C for 48h, protected from light and at 90% humidity¹⁰.

PROTEIN MEASUREMENT

The measurement of protein content was performed using the Kjeldahl method, according to the NF EN ISO 5983-2¹² standard, at the INRAE UE EASM laboratory. The samples were first freeze-dried for 72h and then ground using a Retsch ZM 200 mill fitted with a 0.5mm sieve. The water content of each sample was evaluated, followed by the measurement of total protein content using the conversion factor 6.25 (N × 6.25). Finally, the residual dry matter content was determined according to the NF ISO 6496¹³ standard, after drying for 4h at 103°C.

GLUCOSE DOSAGE

The 3,5-dinitrosalicylic acid (DNS) method is a simple and cost-effective spectrophotometric technique for quantifying reducing sugars such as glucose. It works by measuring the red brown coloured product formed when DNS reagent reacts with reducing sugars under alkaline and heated conditions, with absorbance measured at 530nm – 540nm.

PREPARATION OF DNS REAGENT

DNS (128848-5G, Sigma-Aldrich) reagent was prepared by adding successively 1g of DNS, 20mL of 2 M sodium

hydroxide solution (NaOH, SX0607N-6), and 30g of sodium potassium tartrate tetrahydrate (SR255.250G) into a 100mL beaker. The volume was adjusted to 100mL with distilled water (100mL, 10977-035), and the mixture was homogenized using a magnetic stirrer until a clear solution was obtained.

STANDARDS ANALYSIS

D-glucose stock solution (0.005 mol/L) was used to prepare standards with volumes of 0mL, 0.2mL, 0.4mL, 0.6mL, 0.9mL, and 1mL, diluted to 1.5mL with distilled water. DNS reagent (1mL) was added, tubes were sealed and heated at 100°C for 5 minutes, then cooled in ice for 2 minutes. Samples were diluted to 10mL with distilled water, mixed, and left 15 minutes at room temperature. Absorbances were measured at 530nm using a spectrophotometer with a DNS/water blank as reference. A standard curve was plotted from absorbance versus glucose concentration.

SAMPLES ANALYSIS

The fermented and non-fermented samples (1mL each) were analysed using the same protocol as the standard analysis. A volume of 0.5mL of each sample was taken using graduated pipettes with a pipette bulb and introduced into separate PYREX tubes, then adjusted to 1.5mL with 1mL of distilled water. Then, 1mL of the DNS reagent was added to each tube, which was sealed with cotton wool and aluminium foil. The tubes underwent the same thermal treatment as the standard curve: heating in a water bath at 100 °C for 5 minutes, cooling in ice, adjusting the final volume to 10mL with distilled water, and resting for 15 minutes at room temperature. The absorbances were measured at 530nm using the UV-Visible spectrophotometer. The glucose concentrations in the samples were determined by interpolation on the previously obtained standard curve.

GRAPHS AND STATISTICAL ANALYSIS

Graphs and statistical analyses were performed using the GraphPad Prism software (version 8.0.2). Statistical analyses were carried out using non-parametric ANOVA tests (Kruskal–Wallis test). The significance threshold was set at p = 0.05.

RESULTS

This section presents the experimental outcomes of the single and double fungal fermentations of red kidney beans. The results focus on three main aspects: macroscopic fungal growth under different inoculation conditions, total protein content as an indicator of nutritional composition, and glucose concentration as a marker of carbohydrate metabolism. Together, these analyses aim to assess the impact of fermentation time and fungi selection on substrate transformation, and to provide a comprehensive evaluation of the biochemical changes induced by fungal fermentation.

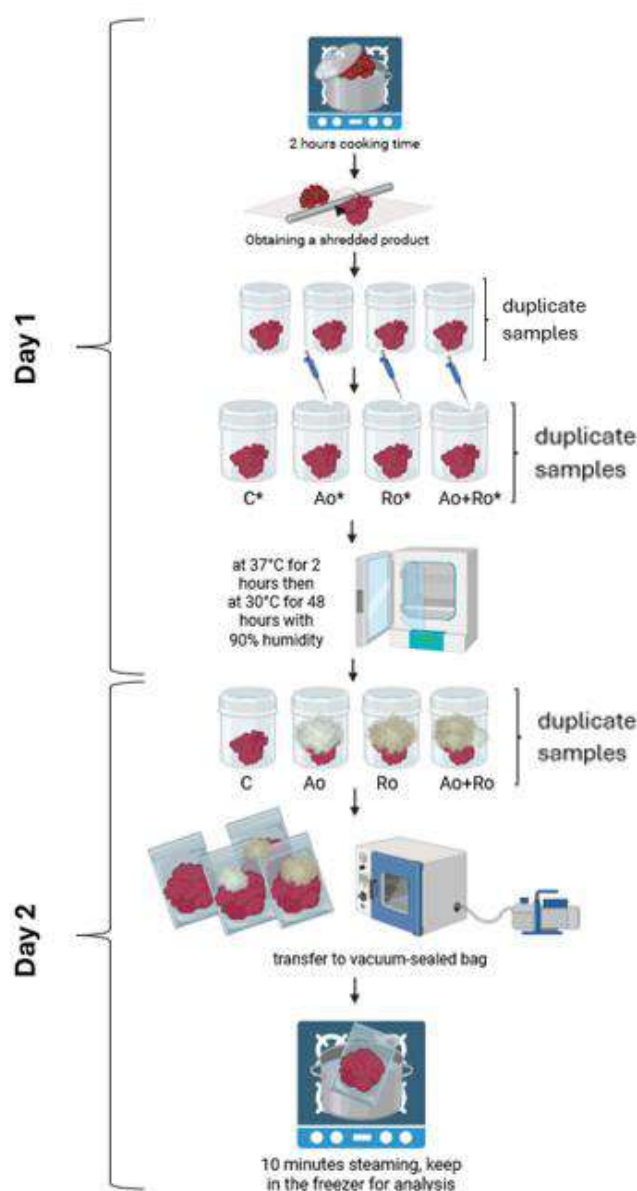


Figure 1: Experimental design of the fermentation process (Source: Diagram created using the Biorender website). C*: red beans only (substrate). A.o.*: 10^4 spores of *Aspergillus oryzae* per gram of substrate. R.o.*: 2.10^4 spores of *Rhizopus oligosporus* per gram of substrate. A.o.+R.o.*: 10^4 spores of *A. oryzae* and 2.10^4 spores of *R. oligosporus* per gram of substrate.

MACROSCOPIC ANALYSIS OF THE FERMENTATION CONDITIONS

As shown in Figure 1, the different experimental conditions of beans were incubated for 48h at 30°C. At the end of the incubation period, fungal growth was observed in all inoculated conditions. Figure 2 presents the visual observations obtained after fermentation. In the non-inoculated condition (Figure 2: Condition A), fungal growth of white coloration was observed in one of the two replicates (Figure 2 : A1), whereas the second replicate (Figure 2 : A2) showed no visible signs of contamination. For the condition

inoculated with *A.o.* (Figure 2 : Condition B), yellow fungal growth was observed in both replicates. However, heterogeneity in growth intensity was noted: replicate B1 exhibited a well-developed yellow fungal layer on the surface, while replicate B2 showed a more limited growth. In the condition inoculated with *R.o.* (Figure 2: Condition C), homogeneous fungal development was observed in both replicates, characterized by the formation of a continuous white mycelial layer.

Finally, in the dual fermentation condition (Figure 2: Condition D), the concomitant presence of white and yellow mycelia was observed in both replicates.

PROTEIN CONTENT ASSESSMENT USING THE KJELDAHL METHOD

To further investigate whether these differences in fungal development were associated with changes in nutritional composition, total protein content was quantified for the samples presented in Figure 2 using the Kjeldahl method. To avoid any bias, we excluded the value of the infected control (Figure 2: A1). Total protein analysis (Figure 3) revealed no statistically significant differences between the experimental conditions, as indicated by the Kruskal–Wallis test ($p = 0.6286$). In the control group, protein content of 24.49% (Figure 2: A2) was measured. Samples inoculated with *A.o.* showed protein contents of 25.85% (Figure 2: B1) and 25.91% (Figure 2: B2), while samples fermented with *R.o.* exhibited values of 25.88% (Figure 2: C1) and 26.28% (Figure 2: C2). Finally, the combined fermentation condition (Figure 2: *A.o.* + *R.o.*) resulted in protein contents of 25.70% (Figure 2: D1) and 26.83% (Figure 2: D2).

Overall, despite observable differences in fungal growth patterns across conditions, protein content remained relatively stable, suggesting that the fermentation processes applied did not significantly impact total protein levels under the tested conditions.

DNS GLUCOSE DOSAGE

The glucose concentrations analysis (Figure 4) shows variations in glucose concentration measured after 40h and 48h of fermentation for the different studied conditions. After 40h of fermentation, glucose concentrations remained low overall for all experimental conditions. The control medium has a concentration close to 0mg/L. The media inoculated by the microorganisms show lower values, between 0 and 5mg/L. Among these, *R.o.* cultures and the *A.o.* + *R.o.* co-culture have the lowest concentrations, at this stage, the differences between treatments remain moderate. On the other hand, after 48h of fermentation, a notable change in the glucose concentration profile is observed under experimental conditions. The control medium did not show a significant increase in glucose concentration, reaching only about 8mg/L. Individual cultures of *A.o.* and *R.o.* show high glucose concentrations, close to 35mg/L and 45mg/L respectively. Similarly, the co-culture of *A.o.* + *R.o.* is clearly

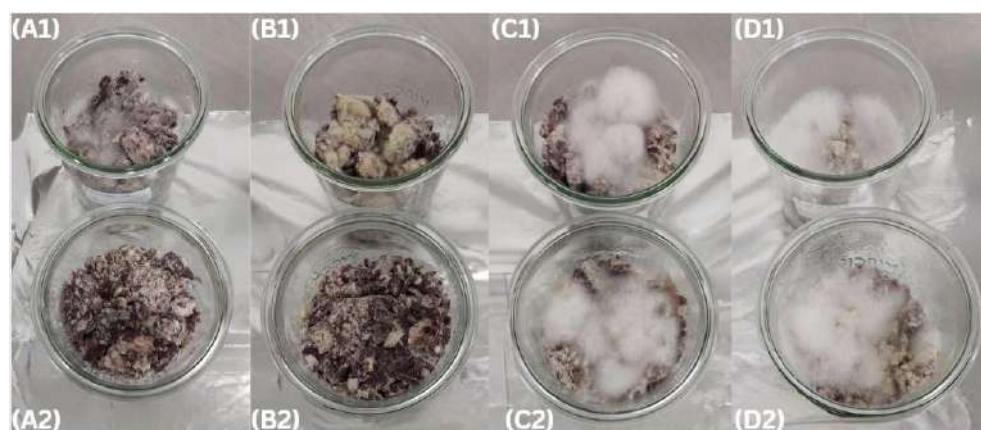


Figure 2: Macroscopic photographs of red kidney beans fermentation under different inoculation conditions. (A1)&(A2) Non-inoculated control jars. (B1)&(B2) Jars inoculated with *Aspergillus oryzae*. (C1)&(C2) Jars inoculated with *Rhizopus oligosporus*. (D1)&(D2) Jars co-inoculated with *Aspergillus oryzae* and *Rhizopus oligosporus*.

distinguishable from the control condition. After 48h of fermentation, the glucose concentration reaches about 40mg/L.

DISCUSSION

This study aimed to evaluate the impact of single and double fungal fermentations using *A.o.* and *R.o.* on a red kidney bean-based substrate, by combining macroscopic observations of fungal development with quantitative analysis of total protein content and glucose content.

The visual differences observed in Figure 2 between fermentation conditions highlight species-specific growth behaviors under solid-state fermentation. The condition *A.o.* displayed a characteristic yellow pigmentation¹¹, consistent with its known production of secondary metabolites and surface-oriented mycelial expansion. However, the notable variability between replicates suggests that its growth is highly dependent on microenvironmental parameters such as moisture gradients, oxygen availability, and inoculum homogeneity. This sensitivity is well documented for filamentous fungi¹⁴ and represents a key challenge for reproducibility at an industrial scale.

In contrast, *R.o.* exhibited homogeneous and robust colonization across replicates, forming a continuous white mycelial network. This growth pattern is typical of *R.o.*¹⁵ and underlies its industrial relevance, particularly in traditional fermentations such as tempeh, where uniform substrate binding is required. The consistency observed suggests that the applied fermentation conditions were well optimized for this species, resulting in reliable substrate colonization.

The dual fermentation condition revealed the simultaneous presence of both fungal morphologies, indicating that *A.o.* and *R.o.* were able to coexist without strong competitive exclusion. This observation suggests potential metabolic complementarity, as both species are known to secrete distinct

enzymatic repertoires involved in carbohydrate and protein hydrolysis. Such interactions are of particular interest in biotechnological applications, where mixed cultures can enhance substrate valorization through synergistic enzyme production.

Despite these clear differences in fungal development, total protein content (Figure 3) did not vary significantly between experimental conditions. This result indicates that, within the 48h fermentation period, fungal growth did not lead to measurable protein enrichment.

One possible explanation is that the synthesis of fungal biomass proteins is counterbalanced by proteolytic degradation of substrate proteins, resulting in stable net protein levels¹⁶. Additionally, part of the nitrogen assimilated by fungi may be diverted toward non-protein nitrogenous compounds, such as free amino acids, peptides, or secondary metabolites.

It is also important to consider the methodological limitations of the Kjeldahl assay, which quantifies total nitrogen rather than true protein content. As a result, potential qualitative improvements, such as enhanced protein digestibility, changes in amino acid composition, or reduction of antinutritional factors, are not captured by this analysis. Therefore, while total protein levels remain unchanged, fermentation may still improve the nutritional and functional quality of the substrate¹⁷.

Overall, these results suggest that short-term fungal fermentation primarily influences microbial growth dynamics and substrate colonization rather than total protein concentration. From a bioprocess engineering perspective, extending fermentation time, optimizing inoculum ratios, or coupling protein quantification with functional assays (e.g., amino acid profiling, in vitro digestibility, or enzyme activity measurements) would provide a more comprehensive evaluation of the biotechnological potential of single and double fungal fermentations.

The monitoring of glucose concentrations (Figure 4) provides complementary insight into the metabolic activity of the fungi during fermentation. At 40h, glucose levels remained low for all conditions, suggesting rapid assimilation of released sugars during the early growth phase, particularly for *R.o.* and the co-culture, which is consistent with their high metabolic activity. After 48h, inoculated conditions showed a notable increase in glucose concentration compared to the control, indicating intensified enzymatic hydrolysis of complex

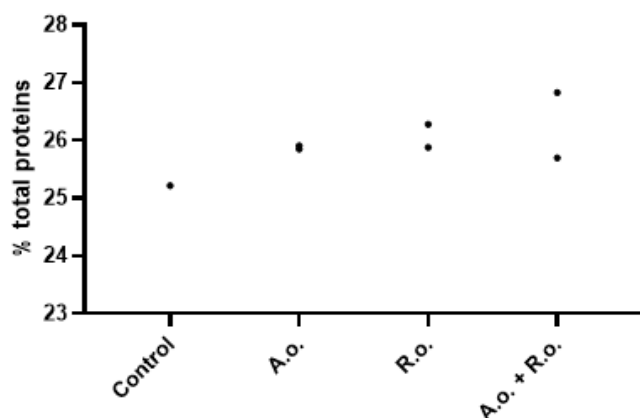


Figure 3: Effect of fermentation on total proteins by Kjeldahl method. Each dot represents one jar (refer to figure 2). Test Kruskal-Wallis, $p = 0,6286$, total proteins percentage in red beans after 48h of fermentation.

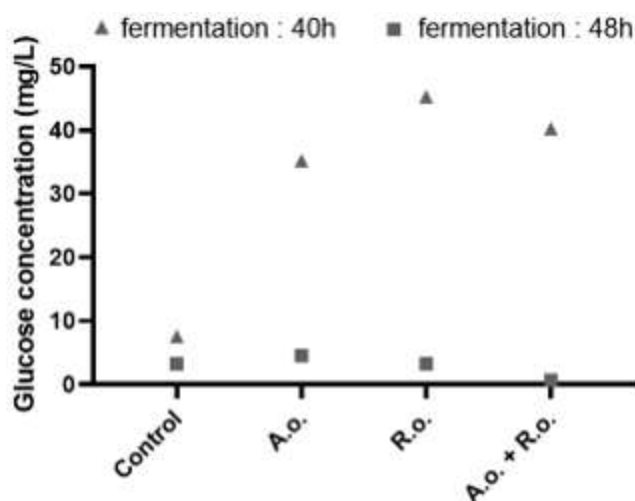


Figure 4: Effect of fermentation on glucose concentration. Each dot represents one jar from two different experimentations. DNS dosage after 40h and 48h of fermentations.

carbohydrates. This accumulation suggests that, at this stage, glucose release exceeded microbial consumption. Among the tested systems, *R.o.* exhibited the highest glucose concentration, confirming its strong capacity for complex sugar degradation¹⁸.

Overall, these results reinforce the role of fermentation time and microbial selection in shaping carbohydrate metabolism and support the suitability of *R.o.* for efficient polysaccharide degradation in legume-based fermentations¹⁹.

CONCLUSION

This study demonstrated that fungal fermentation of beans using *A.o.* and *R.o.*, whether applied individually or in combination, led to distinct fungal growth patterns without significantly affecting total protein content after 48h of incubation. While *R.o.* showed homogeneous and reproducible substrate colonization, *A.o.* exhibited more

variable growth, highlighting species-specific sensitivities to solid-state fermentation conditions.

The absence of significant changes in protein content suggests that short-term fermentation primarily influences microbial development rather than overall protein concentration. However, this does not exclude qualitative improvements in protein functionality or nutritional value, which were not assessed by the Kjeldahl method.

From a biotechnological perspective, future work should focus on extending fermentation duration, optimizing mixed-culture strategies, and integrating complementary analytical approaches such as amino acid profiling, protein digestibility assays, and enzymatic activity measurements. In parallel, particular attention should be paid to the potential synthesis of undesirable or toxic compounds (e.g., mycotoxins or other secondary metabolites) that may arise during prolonged fermentation. Careful monitoring of fermentation conditions and safety-related parameters will therefore be essential to ensure that nutritional improvements are achieved without compromising product safety. These investigations would provide deeper insight into the potential of fungal fermentations to enhance the nutritional and functional properties of legume-based substrates, with promising applications in the development of fermented plant-based foods and sustainable protein sources.

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CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

REFERENCES

1. Cinquante ans de consommation alimentaire : une croissance modérée, mais de profonds changements - Insee Première - 1568. <https://www.insee.fr/fr/statistiques/1379769#graphique-figure2>.
2. Monteiro, C. A. *et al.* The UN Decade of Nutrition, the NOVA food classification and the trouble with ultra-processing. *Public Health Nutr.* **21**, 5–17 (2018).
3. Aliments dits ultratransformés : mieux comprendre leurs effets potentiels sur la santé | Anses - Agence nationale de sécurité sanitaire de l'alimentation, de l'environnement et du travail. <https://www.anses.fr/fr/content/aliments-dits-ultratransformes-mieux-comprendre-leurs-effets-potentiels-sur-la-sante>.
4. Pas si super – C'est quoi un aliment ultra-transformé ? · Inserm, La science pour la santé. *Inserm* <https://www.inserm.fr/c-est-quoi/pas-si-super-cest-quoi-un-aliment-ultra-transforme/>.
5. Gill, P. A., Inniss, S., Kumagai, T., Rahman, F. Z. & Smith, A. M. The Role of Diet and Gut Microbiota in Regulating Gastrointestinal and Inflammatory Disease. *Front. Immunol.* **13**, 866059 (2022).
6. Bilal, M., Ashraf, S. & Zhao, X. Dietary Component-Induced Inflammation and Its Amelioration by Prebiotics, Probiotics, and Synbiotics. *Front. Nutr.* **9**, (2022).
7. Whelan, K., Bancil, A. S., Lindsay, J. O. & Chassaing, B. Ultra-processed foods and food additives in gut health and disease. *Nat. Rev. Gastroenterol. Hepatol.* **21**, 406–427 (2024).
8. Cara, G. La consommation de certains additifs alimentaires émulsifiants serait associée à un risque accru de diabète de type 2. *Salle de presse de l'Inserm* <https://presse.inserm.fr/la-consommation-de-certains-additifs-alimentaires-emulsifiants-serait-associee-a-un-risque-accru-de-diabete-de-type-2/68327/> (2024).
9. Marco, M. L. *et al.* Health benefits of fermented foods: microbiota and beyond. *Curr. Opin. Biotechnol.* **44**, 94–102 (2017).
10. Senanayake, D., Torley, P. J., Chandrapala, J. & Terefe, N. S. Microbial Fermentation for Improving the Sensory, Nutritional and Functional Attributes of Legumes. *Fermentation* **9**, 635 (2023).
11. Starzyńska-Janiszewska, A., Stodolak, B., Duliński, R. & Mickowska, B. The influence of inoculum composition on selected bioactive and nutritional parameters of grass pea tempeh obtained by mixed-culture fermentation with *Rhizopus oligosporus* and *Aspergillus oryzae* strains. *Food Sci. Technol. Int.* **18**, 113–122 (2012).
12. ISO 5983-2:2009. *ISO* <https://www.iso.org/fr/standard/52199.html> (2009).
13. ISO 6496:1999. *ISO* <https://www.iso.org/fr/standard/12871.html> (1999).
14. Pandey, A. Solid-state fermentation. *Biochem. Eng. J.* **13**, 81–84 (2003).
15. Jennessen, J., Schnürer, J., Olsson, J., Samson, R. A. & Dijksterhuis, J. Morphological characteristics of sporangiospores of the tempe fungus *Rhizopus oligosporus* differentiate it from other taxa of the *R. microsporus* group. *Mycol. Res.* **112**, 547–563 (2008).
16. Frias, J., *et al.* Impact of Fermentation on the Nutritional Quality, Bioactive Compounds and Potential Health Properties of Legumes. (2009).
17. Mariotti, F., Tomé, D. & Mirand, P. P. Converting nitrogen into protein--beyond 6.25 and Jones' factors. *Crit. Rev. Food Sci. Nutr.* **48**, 177–184 (2008).
18. Agista, A. Z. *et al.* Investigation of *Rhizopus oligosporus* Metabolites in Fermented Wheat Bran and Its Bio Function in Alleviating Colitis in Mice Model. *Metabolites* **14**, 359 (2024).
19. Zhang, Y. *et al.* Solid-state fermentation with *Rhizopus oligosporus* RT-3 enhanced the nutritional properties of soybeans. *Front. Nutr.* **9**, 972860 (2022).

Valorization of Orange Peels for Hesperidin-Enriched Butter with Hypocholesterolemic Potential

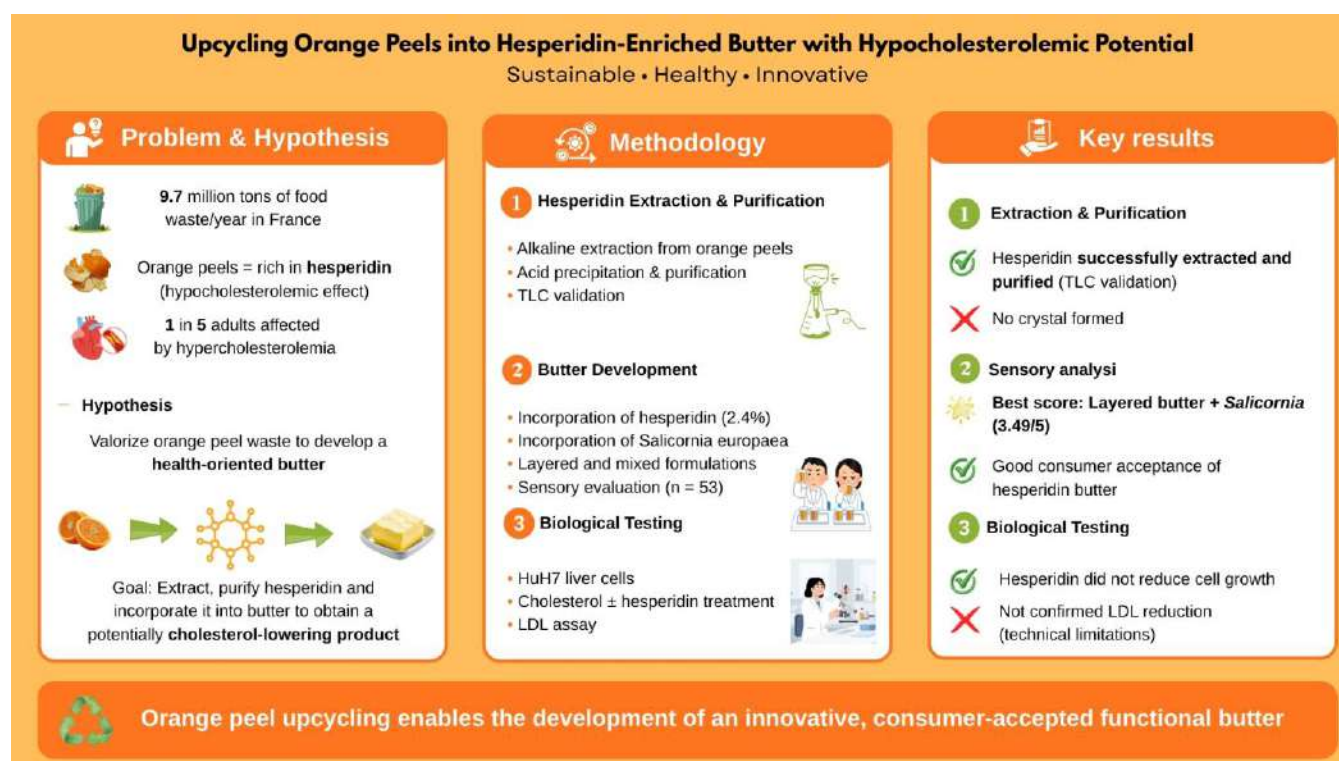
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VISUAL ABSTRACT



Valorization of Orange Peels for Hesperidin-Enriched Butter with Hypcholesterolemic Potential

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Abstract – Due to a growing environmental concern, the valorization of agri-food co-products represents a major opportunity in terms of nutrition and economy. This study focuses on the extraction of hesperidin, a bioflavonoid found in orange peels, with hypocholesterolemia properties, to enrich a traditional butter. Hesperidin was extracted using a solvent-safe alkaline method and purified. Hesperidin presence and purification was confirmed by Thin-Layer Chromatography (TLC). Several butter formulations were developed, including a layered version and a mixture with a wild plant named *Salicornia europaea*. Butters were submitted to a sensory analysis involving fifty-three participants. The results indicated a preference for the *Salicornia*-enriched butter and good acceptance of the hesperidin-layered butter. To validate the health benefits, the effect of hesperidin was tested on HuH7 liver cells. Results shown an increase in cell viability, the effect of cholesterol decrease could not be confirmed due to technical limitations associated with the cell line and assay kit. Despite these biological limitations, the project successfully demonstrated the potential of upcycling orange peels into an innovative and accepted food product.

Index Terms – hesperidin, butter, orange-peels, cholesterol, extraction, HuH7 cells.

INTRODUCTION

Today, food waste management has become a major issue in protecting our planet. In 2023, it is estimated that the agri-food industry was responsible for the loss of 9.7 million tonnes of food in France across the entire food chain, of which 5.9 million tonnes are classified as inedible (Déchets et gaspillage alimentaires en France en 2023, s. d.). This includes fruit by-products, corresponding mainly to useful waste from fruit, which represent an economic loss for

businesses, but also a reduction in potential nutritional value for consumers (Kumar et al., 2020). Certain types of waste, such as orange peels, are particularly rich in bioactive molecules that are beneficial to health, for instance bioflavonoids (Bioflavonoïdes, s. d.). From a sustainable ethic perspective, the recovery of these molecules from plant waste represents a promising way, combining nutrition, health, and sustainability.

Bioflavonoids are bioactive metabolites and the most abundant polyphenolic compounds found in plants. The phenolic functions of which they are composed provide antioxidant or oxidizing properties (Stoclet & Schini-Kerth, 2011). Among them, we are interested in hesperidin, a glycoside formed by the combination of two structural entities linked together by an O-glycosidic bond: an osidic part (rutinose) and an aglycone part (hesperetin) (Hespéridine, 2024). This molecule abundant in citrus fruits, has antioxidant, anti-inflammatory and anti-carcinogenic properties (Ogunro, 2025). The characteristic that draws our attention is its cholesterol-lowering effect.

In fact, according to “Santé Publique France” 12 million of French people, corresponding to 1 in 5 adults, suffer from hypercholesterolemia (Hypercholestérolémie, s. d.). This metabolic disorder has various causes and results in abnormally high blood cholesterol levels, i.e. total cholesterol above 2g/L (Menir, 2025). This cholesterol is divided into two main types: LDL cholesterol (known as “bad cholesterol”) and HDL cholesterol (“good cholesterol”) (Janapala & Reddivari, 2025). HDL-C (High-Density Lipoprotein Cholesterol) captures excess cholesterol in the blood and transports it to the liver to be eliminated with bile. Its action can also contribute to beneficial cardiovascular effects (Kosmas et al., 2022). On the other hand, LDL-C (Low-Density Lipoprotein Cholesterol) corresponds to a low-density lipoprotein that transports cholesterol to peripheral tissues. However, when the latter is present in excess, it

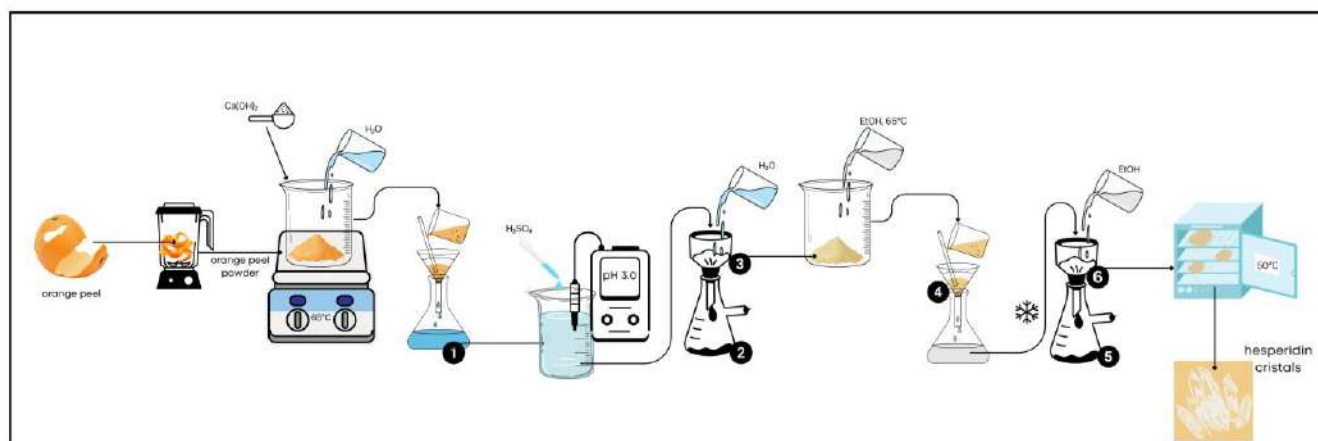


Figure 1. Extraction and purification of hesperidin from orange peel. Schematic representation of the extraction and purification process of hesperidin from orange peel. The peels are dried, ground, and mixed with water and calcium hydroxide at 65 °C. After filtration (1) and pH adjustment to 3 using sulfuric acid, the precipitate is vacuum filtered (2) and washed (3), dissolved in hot ethanol and filtered (4), cooled, vacuum filtered (5) and washed (6), before being recrystallized to obtain pure hesperidin crystals, which are oven dried.

accumulates on the walls of the arteries as fatty deposits, contributing to health problems, particularly atherosclerosis which increases the risk of cardiovascular disease (Cholest rol LDL : sympt mes et traitements, s. d.).

Our molecule of interest, hesperidin, therefore has an important role to play in the fight against hypercholesterolemia, based on previously conducted *in-vitro* and *in-vivo* studies in rats. It is capable of reducing total cholesterol, LDL-C, and triglyceride levels (Gence, 2019). After ingestion, it is converted by intestinal bacteria via a process of deglycosylation into its active form, hesperitin. Once absorbed, hesperitin is released into the bloodstream in the form of glucuronide and sulphate conjugates, where it exerts its cholesterol-lowering effects by modulating lipid metabolism (Khorasani et al., 2023).

In order to apply its cholesterol-lowering properties, we considered incorporating hesperidin into a food product: butter. This approach seems promising because this product is often associated with high cholesterol and saturated fat content. Thus, people concerned with maintaining low cholesterol levels and cardiovascular health could find this food as an interesting alternative to consuming traditional butter. Furthermore, due to its high consumption in France and its place in dietary habits (Le beurre, production et consommation, s. d.), this new product represents an innovative opportunity with a strong potential for acceptance by the public. Although there are already many margarines (vegetable fats) on the market aimed at people who want to reduce their cholesterol, they are often rejected because they lack the original taste of butter, which is an important factor in the French diet. Moreover, they contain trans fats or partially hydrogenated oils which are harmful components for human health. (Rivers, 2017). That's why the product we wish to offer would meet the expectations of our future customers, who are eager to consume "healthier" butter.

We aim to investigate how to valorize hesperidin contained in orange peels into a food-product with a hypocholesterolemic effect.

Our objective is to combine the taste satisfaction of traditional milk-based butter with proven benefits on health, thanks to the incorporation of hesperidin which adds nutritional and environmental value to the final product. Three distinct experiments are being conducted to achieve this objective. Firstly, hesperidin is extracted and purified from orange peels to add value to this agri-food by-product. These two actions are based on several methods implemented in order to recover the purest possible hesperidin for use in subsequent applications. In a second experiment, the incorporation of hesperidin into a manufactured food product, in this case butter, is studied. Several formulation trials are carried out, followed by a sensory analysis to evaluate the organoleptic characteristics of the product and determine whether it is accepted and meets consumer expectations. Finally, the cholesterol-lowering properties of hesperidin are analysed on human liver cells to confirm the biological efficacy of the molecule.

MATERIAL AND METHODS

HESPERIDIN EXTRACTION AND PURIFICATION

The aim of this first experiment was to evaluate a simplified and solvent-safe extraction and crystallization protocol for hesperidin, avoiding the use of toxic organic solvents such as petroleum ether or methanol. By avoiding those toxic organic solvents, we ensure the final product is safe for human consumption and compatible with food industry standards. The goal was to obtain pure hesperidin crystals from orange peel using a water-based alkaline extraction and acid precipitation process (Figure 1). The orange peel were sourced from local school canteens and bakeries. However, as

they were collected after the fruit had been cleaned or consumed, their organic status could not be determined.

Preparation of Orange Peel Powder

Orange peels were dried in a hot-air oven at 50–60°C for 24 hours, ground into a fine powder (~1 mm particle size), and sieved if necessary to ensure homogeneity (Manikyam *et al.*, 2022).

Alkaline Extraction

One hundred grams of orange peel powder were mixed with 500 mL of distilled water in a 1L beaker. Five grams of calcium hydroxide [Ca(OH)₂] (Merck, 239232-100G) were added (Hendrickson *et al.*, 1954), and the mixture was heated at 60–70°C with continuous stirring for 2 hours. Under these alkaline conditions, Ca(OH)₂ deprotonates the phenolic groups of hesperidin, forming a soluble calcium salt that facilitates its extraction from the orange peel. The hot mixture was then filtered to obtain the filtrate containing the calcium salt of hesperidin.

Acidic Precipitation

The filtrate was acidified with diluted sulfuric acid (Merck, 258105-100ML) to reach approximately pH 3.0 in order to reprotonate the phenolate groups and precipitate hesperidin. The solution was allowed to stand at room temperature for 2 hours to promote the precipitation of hesperidin. The precipitated crystals were collected by vacuum filtration using a Büchner funnel and washed with cold distilled water (Hendrickson *et al.*, 1954; Hendrickson *et al.*, 1955).

Purification and Crystallization

The crude hesperidin crystals were dissolved in a minimal volume of pure (> 96.3%) hot ethanol (50–60°C) (Grosseron, 9006802) (Hendrickson *et al.*, 1955), as hesperidin is soluble in ethanol, allowing its separation from insoluble impurities. The hot solution was filtered to remove insoluble impurities, allowed to cool slowly to room temperature, and then refrigerated at 4°C for 12 hours to induce crystallization. The resulting crystals were filtered under vacuum, rinsed with cold ethanol (Grosseron, 9006802), and dried under vacuum at 50°C to yield purified crystalline hesperidin (Manikyam *et al.*, 2022).

Thin-Layer Chromatography (TLC) Analysis

Samples of the orange peel extract, crude crystals, and purified hesperidin standard were dissolved in 70% ethanol (Grosseron, 9006802) (1:1 w/v). The mobile phase consisted of butanol (Sordalab, SI019.1L): acetic acid 99%(Sigma, A6283-1L): distilled water (60 : 20 : 20 v/v). Silica gel plate with UV indicator (Macherey-Nalgel, ALUGRAM SIL G UV25) were used, with a baseline drawn 1 cm from the edge, each sample was spotted at regular intervals (Hendrickson *et al.*, 1957). Pure Hesperidin (Bulk Supplements, UGS:HESP100) was used as reference. The plate was placed in the chromatography chamber and developed for 20–30 minutes. After removal, the solvent front was marked, and the

plate was examined under UV light at 254 nm for band visualization (Hendrickson *et al.*, 1959).

BUTTER PREPARATION

Food Materials

The cream used for butter preparation was a commercial full-fat liquid cream (30% fat; Intermarché, France). It was stored at room temperature in a dry pantry until use. The *Salicornia europaea* (Christine Le Tennier, France) used for enrichment was obtained from and supplied preserved in brine containing water, salt, and citric acid. It was stored at ambient temperature in its original sealed container. The hesperidin powder (pure extract from *Citrus sinensis*) was purchased from Bulk Supplements (USA Bulk Supplements, UGS:HESP100) and stored in a cool, dry environment until incorporation. In fact, the hesperidin used for our experiment is commercial.

Preparation of Control Butter

Butter was produced by mechanical churning of liquid cream at ambient temperature (21 ± 1°C). The cream was poured into the stainless-steel bowl of a planetary mixer (Kenwood, UK) equipped with a wire whisk. The mixture was initially agitated at low speed (~200 rpm) to incorporate air and promote homogenization. The speed was then progressively increased to induce phase inversion, resulting in the separation of the fat phase (butter) and the aqueous phase (buttermilk).

Once visible agglomeration of fat globules and yellowing of the mixture occurred, agitation speed was reduced (~100 rpm) to minimize splashing and to allow complete coalescence of the fat mass. The churning was considered complete when the fat had coalesced around the whisk, and buttermilk was clearly separated.

The butter and buttermilk mixture was then poured through a sterile muslin cloth placed over a collection vessel. The butter was manually pressed through the cloth to remove residual buttermilk and subsequently rinsed under cold running water to eliminate remaining whey.

Incorporation of Hesperidin

Hesperidin was incorporated into the previously prepared butter base. The butter was slightly heated in a water bath to facilitate the incorporation of the molecule of interest. So, it was softened in a stainless-steel bowl using a spatula until a malleable pomade consistency was obtained. Then, we added 2.4% of hesperidin powder manually and we mixed to ensure homogeneous distribution while avoiding mechanical shear that could degrade the compound. This quantity was defined by previous experiment (not shown) that lead us to this optimal quantity to match the highest quantity of hesperidin with the acceptance of consumers. The mixture was spread into a uniform layer of approximately 5 mm-thick between sheets of food-grade paper and stored under refrigeration (4°C) until use.

Table 1. Analysis of participants characteristics.

Participant characteristics			
Category	Variable	Effective (n = 53)	Percentage
Age	19 - 24 years	33	62,30%
	25 - 30 years	9	17%
	31 - 50 years	9	17%
	≥ 51 years	2	3,8%
Gender	Female	40	75,5%
	Male	11	20,8%
	Other / Non-binary / Would rather not say	2	3,8%
Expertise level	Amateur (No tasting)	41	77,4%
	Regular (1 to 5 tastings)	10	18,9%
	Expert (> 5 tastings)	2	3,8%
Influencing factors	Perfume wearing	23	43,4%
	Smoker (1h before)	1	1,9%
	Coffee (1h before)	4	7,5%

Incorporation of *Salicornia europaea*

The *Salicornia* preparation was incorporated into the hesperidin-enriched butter. The goal is to satisfy consumers that appreciate salted butter without adding salt which, in high quantity, can damage health (Brand, 2022). In this purpose, we chose *Salicornia*, a wild plant halophilic that grows mainly by the sea and on soils particularly rich in salt (Limongelli, 2022). The preserved *Salicornia* was rinsed under cold running water to remove brine, blotted dry with absorbent paper, and finely chopped into pieces of approximately 2 mm. The pieces were gently incorporated into the butter pomade to maintain their structural integrity. The mixture was then shaped into a 5 mm-thick layer between sheets of food-grade paper and refrigerated (4°C) until use.

Preparation of Layered Butter Samples

Layered butter samples were prepared by assembling a central layer of hesperidin-enriched butter (with or without *Salicornia*) between two outer layers of plain butter to achieve a natural visual appearance. Each layer was pre-formed to a uniform thickness of 5 mm and stacked carefully on a flat surface between food-grade papers. The layered structure was gently pressed to ensure adhesion between layers without excessive deformation. The final product was wrapped and refrigerated at 4°C until sensory evaluation.

ORGANOLEPTIC TEST

Samples

Four butter samples were prepared for sensory evaluation.

- Control sample: butter without additives.
- Mixed hesperidin sample: consisted of butter containing powdered hesperidin homogeneously mixed.

- Layered hesperidin sample: three-layer structure designed to preserve a natural external appearance: a central layer enriched in hesperidin, enclosed between two layers of plain butter.
- Layered hesperidin sample with *Salicornia europaea*: same sample as layered hesperidin sample with addition of *Salicornia*. This configuration was chosen to minimize the visual impact of the hesperidin's natural ochre-brown coloration, which could otherwise alter consumer perception.

All samples were molded into identical shapes and coded with random three-digit numbers to ensure a blind evaluation. (Codex Alimentarius, 2001)

Participants

Fifty-three individuals participated in the sensory test. Participation was voluntary, and all subjects provided informed consent for the anonymous use of their data for research purposes, in accordance with ethical standards for sensory analysis. Most of the participants were non expert in tasting. And 75.5% are female. Conditions were not optimum for the tasting because 43.4% of the participants were wearing perfume (Table 1).

Experimental Conditions

The sensory evaluation took place in a classroom setting at SupBiotech on April 16 and 17, 2025. Sessions were organized in 30-minute intervals throughout both days. Participants were seated sufficiently apart to prevent visual or verbal interaction. A standardized introductory briefing was delivered before each session, explaining the purpose of the study, the blind tasting procedure, and the importance of refraining from communication during the evaluation.

All butter samples were removed from refrigeration 10 minutes before tasting to reach a comparable serving temperature. The four butters were presented in a fixed order across all sessions, with their number identification. Each participant received a glass of water and standardized portions of traditional baguette bread obtained from the same bakery. Although the bread originated from different batches on the two test days, its type and source remained constant.

Sensory Evaluation Procedure

The evaluation was conducted using an online questionnaire (Microsoft Forms) with anonymized data collection. The form included preliminary questions on participants' general health and butter consumption habits, followed by the sensory assessment of the four samples. Each sample was evaluated successively based on predefined sensory attributes, with three to seven descriptors proposed for each question.

The attributes and corresponding options were as follows:

- **Color:** very pale (white), pale (off-white with a slight yellow tint), light yellow, golden yellow, light green, dark green, brownish, orange;
- **Appearance:** natural, homogeneous, grainy, glossy, matte;
- **Odor:** pleasant, delicate, neutral, cloying, aggressive;
- **Texture:** smooth, grainy, rubbery/plastery, fibrous, creamy;
- **Flavor intensity:** weak, medium, strong;
- **Perceived aromas:** dairy, citrus, vegetal, marine, salty, none, other;
- **Specific tastes:** sweet, salty, bitter, acidic, umami, none;
- **Balance of flavors:** well balanced, slightly unbalanced, overly marked in one direction, no particular flavor or no opinion.

Finally, participants can give an overall liking by choosing a score ranging from 1 to 5 hearts (1 = “dislike”, 5 = “like very much”) and express their opinion on the samples tasted or give advice on the progress of the organoleptic test.

Data Analysis

Responses were exported from Microsoft Forms and analyzed using descriptive statistics. For each butter sample, the proportion of responses associated with each descriptor was calculated to determine the most frequently cited sensory terms. The overall liking score was summarized using the median, deemed more representative of the central tendency due to the presence of divergent opinions and the relatively small panel size ($n = 53$).

EFFICIENCY TEST

Cell Culture

The HuH7 (human hepatoma cells) cell culture was obtained from the INSERM institute (thanks to Nassim Arouche, U1197) and used at passage twenty-two (P22). They were first thawed in a water bath then placed in a twelve-well plate with 2mL of Dulbecco's Modified Minimum Essential Medium, supplemented with 20% Fetal Bovine Serum (SVF) and 1% Penicillin/Streptomycin (warmed at 37°C). Culture medium was a product of Sigma-Aldrich (C6164), and additives came from Sigma for Penicillin/Streptomycin (P4333) and from BIOSER for SVF (FB-1280). Cultures were incubated at 37°C in 95% air/ 5% CO₂ humidified atmosphere. After 24 hours, the medium was changed for 10% of SVF. The cells were followed until they reached 80-90% confluence in order to start their induction with different compounds.

Treatment with Hesperidin and Cholesterol

Hesperidin (Bulk Supplements, UGS:HESP100) and cholesterol (Sigma-Aldrich, C3045) were dissolved in Ethanol 95% to obtain the final concentration in cell culture medium of 40µM for cholesterol (Guo et al., 2022) and 50µM for hesperidin (Morin et al., 2008). Matched control conditions used an equivalent amount of ethanol, with a final concentration of 0,9%/well.

We added treatment in cell culture medium after incubation, as follow (Table 2):

Table 2. Conditions performed in the cellular LDL-cholesterol-lowering effect test.

Cells	Mix	Mix	Mix
Hesperidin	Cholesterol	Cholesterol	Cholesterol
Prophylactic	Prophylactic	Therapeutic	Therapeutic

For the well “Cells”, we let the HuH7 for 48 hours with ethanol, a final concentration of 0,9%. More control wells were “Hesperidin” and “Cholesterol” wells, which were incubated for 48 hours with hesperidin or cholesterol (respectively). Finally, we had two different conditions of interaction between cholesterol and hesperidin. In the prophylactic wells, we added hesperidin and then changed the medium after 24 hours and put cholesterol for an additional 24 hours. The “Therapeutic” wells were the opposite; we treated with cholesterol for the first 24 hours and then changed the medium for hesperidin.

Microscopic Observation

We performed microscopic observations (Bio-Rad, Zoe Fluorescent Cell Imager 1450031), during our experiments to follow the growth rate. We especially performed it before and after the treatment of hesperidin and cholesterol.

Sonication and LDL Assay

For the sonication (Bioblock, Vibracell 75041), we performed waves of 30% amplitude. Each well was treated with three

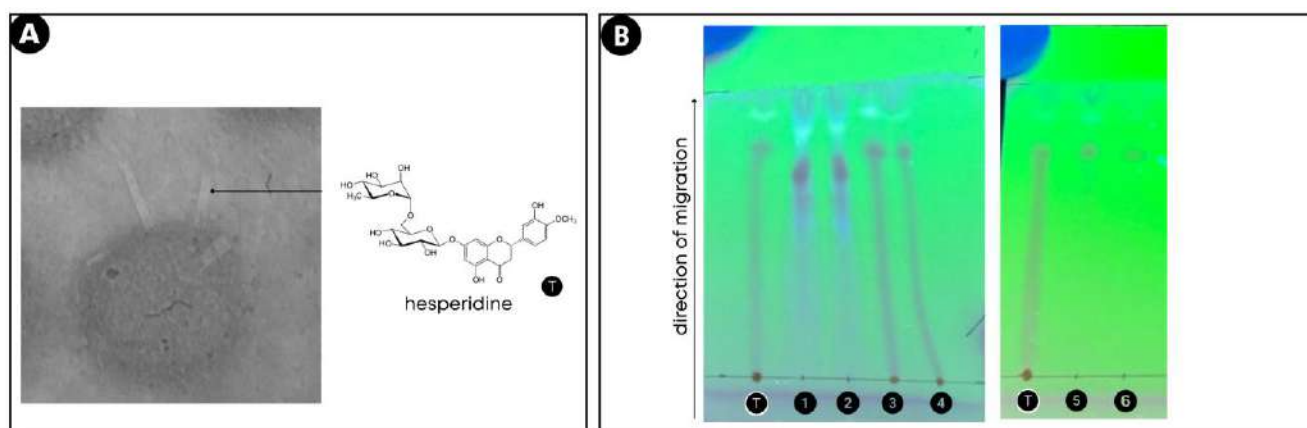


Figure 2: Identification of hesperidin from orange peel. (A) Photograph of hesperidin crystals obtained at step 3 and its corresponding chemical structure. (B) Thin-layer chromatography (TLC) analysis of the fractions obtained during the purification process. The reference spot (T) corresponds to standard hesperidin, and lanes (1–6) represent extracts from steps 1 to 6 of the process (cf. Figure 1)

rounds of 20s on and off, and we kept the samples in ice all the time. We used the study (Ilias et al., 2021) as a reference to elaborate this protocol and check the good destruction of HuH7 with microscopic observations.

For the LDL assay, we followed the Low-density Lipoprotein cholesterol colorimetric assay kit from Elabscience (Ref : E-BC-K205-96). We measured the absorption at 546nm as recommended by the kit's protocol, with a plate reader (Thermoscientific, Multiscan Go).

RESULTS

HESPERIDIN EXTRACTION AND PURIFICATION

The extraction of hesperidin from orange peel and its qualitative analysis were performed according to the protocol illustrated in Figure 1. The results regarding crystal morphology and chromatographic purity are presented in Figure 2.

Figure 2-A shows the hesperidin obtained after the acidic precipitation step. After pH adjustment and vacuum filtration, a solid material was observed on the Büchner funnel membrane. Macroscopic observation revealed the presence of irregular crystal-like structures mixed with a fine powder, unevenly distributed on the membrane.

Figure 2-B shows the thin-layer chromatography (TLC) analysis of the fractions obtained during the different steps of the purification process (Figure 1). Lane T corresponds to the hesperidin standard (Bulk Supplements, UGS:HESP100). Lanes 1 to 6 represent the fractions collected after each purification step (Figure 1). Several lanes show smear lines, indicating the presence of multiple compounds. A spot with a similar migration distance to the standard (T) is observed in some fractions, suggesting the presence of hesperidin.

The retention factors for our spot (R_f), calculated for each lane (Figure 2-B) are as follows:

- Standard (T): $R_f = 0.76$
- Fractions 1 and 2: $R_f = 0.70$
- Fractions 3, 4, and 5: $R_f = 0.78$
- Fraction 6: $R_f = 0.74$

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This experiment shows similarity to the control with all the samples for the odor, flavor intensity, and specific taste profiles. There is no major deviation in these sensory dimensions once tasting begins (Fig. 3c, Fig. 3e, Fig. 3g).

We can determine the sensory profile of the control butter, to which we refer for comparison for the three other samples. First, about the general aspect, the color is mainly seen as yellow toned with no atypical shade. The appearance is mostly described as natural, homogeneous, and glossy; and the odor is perceived as neutral, pleasant and delicate. Then during the degustation, the texture is described as smooth and creamy. The perceived aromas and specific tastes are dairy or none; no dominant taste emerges. This sample is considered well balanced by the participants.

This experiment can be used to determine the sensory profile of the layered butter with addition of *Salicornia*, which is the butter with the best overall score (3,49/5). The aspect is described as different from the control with a more greenish color, more grainy appearance, and less homogeneous. The grainy aspect is similar to the three samples with hesperidin. Then, the odor and aroma are much more vegetal, but the citrus aroma does not emerge. Furthermore, salt and marine notes are detected by the participants. This product is the least balanced flavor profile among the four despite the best overall grade.

Finally, it allows us to compare the layered butter and mixed butter with the addition of hesperidin in both. First, the mixed



Figure 3. Sensory profile of the four butter samples based on panelists' responses. Each panel (a–h) represents the percentage of responses for the selected descriptors within each sensory attribute: (a) color, (b) appearance, (c) odor, (d) texture, (e) flavor intensity, (f) perceived aromas, (g) specific tastes, and (h) balance of flavors. Data represents the distribution of responses among 53 participants. Asterisks indicate significant differences from the control sample according to a Z-test: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Values without asterisks are not significantly different ($p > 0.05$).

butter shows a strong brown perceived coloration where the layered butter remains visually closer to the control. Once tasting begins, both samples have very similar profiles (texture, flavor intensity, perceived aromas, specific tastes, and balanced flavors). If we compare the overall liking, these two samples obtained extremely close scores: 3,40/5 for the layered butter and 3,43/5 for the mixed hesperidin butter.

EFFICIENCY TEST

Hesperidin and Cholesterol Promote Cell Viability

As a reminder, cells were first incubated for 24 hours before the beginning of the experiment (Fig. 4A). The viability of HuH7 cells pretreated with cholesterol and/or hesperidin was monitored 24 hours and 48 hours after the beginning of the experiment (Fig. 4B). First, all the cells including the controls have grown between 24 hours and 48 hours. But, we have seen an increase in the number compared with the cells alone for the cholesterol, hesperidin and mix group. In fact, it is known that cholesterol is a viability promoter and we have been able to confirm it in this experiment. Qualitatively for hesperidin, we observed the same increase in cell number.

Reduction of LDL-Cholesterol Level by Hesperidin

As a second analysis, we performed a quantitative colorimetric assay of LDL levels in media of sonicated cells (Fig. 4C). The goal was to observe if there was an intra or extracellular decrease in LDL levels in the presence of hesperidin. However, due to technical problems that will be developed in the discussion, there was no difference between groups. The values were at zero or very low for each group, which did not allow us to reach a conclusion regarding the ability of hesperidin to reduce LDL under prophylactic, therapeutic or simultaneous conditions.

DISCUSSIONS

HESPERIDIN EXTRACTION AND PURIFICATION

The extraction and purification of hesperidin from orange peel were followed at each step by macroscopic observations and by thin-layer chromatography (TLC), in order to evaluate the efficiency of the protocol. The protocol aimed to obtain solid hesperidin crystals at the end of the process, which would allow the calculation of the extraction yield. However, this objective was not fully achieved.

During the acidic precipitation step, the formation of a precipitate was observed on the Büchner funnel membrane after the first filtration. Macroscopic observation showed the presence of irregular crystalline structures together with a fine powder. This suggests a partial precipitation of the compound of interest, possibly mixed with impurities. The uneven distribution of the solid on the membrane indicates that the filtration or precipitation conditions were not optimal, which may have caused material loss or incomplete recovery of the product.

The TLC analysis provided a crucial method for monitoring the evolution of purity across the different process steps. The hesperidin standard served as the primary reference, displaying a single, well-defined spot with an R_f of 0.76.

In contrast, the initial fractions (Lanes 1 and 2) presented a distinct migration behavior ($R_f = 0.70$) accompanied by significant smear lines. This indicates the presence of a complex mixture of compounds and confirms that the crude extract obtained from the initial non-selective extraction was heavily laden with impurities. While Lane 3 began to show migration characteristics closer to the standard, the presence of smearing persisted in these early stages.

However, as the purification process progressed specifically following the dissolution in hot ethanol, cooling, and filtration steps (Lanes 4 to 6), the chromatographic profiles showed a marked improvement. The smear lines and secondary spots diminished, while a major spot appeared with retention factors ($R_f = 0.74 - 0.78$) comparable to that of the hesperidin standard ($R_f = 0.76$). This shift indicates a successful progressive purification of the target compound and the effective removal of interfering matrix components.

Therefore, although the macroscopic observation suggested that solid crystal recovery was limited, the TLC results confirm that hesperidin was successfully extracted and significantly purified during the process.

However, because solid crystals could not be recovered in sufficient amount at the end of the protocol, it was not possible to calculate the extraction yield. This limitation represents a weakness of the study and may be explained by losses during filtration steps or by incomplete crystallization. Although crystalline structures were observed, they could not be isolated and dried as expected. In this protocol, non-food-garded ethanol was selected as the safest solvent currently available for effective purification. While it is widely used in the food industry as a food-grade extraction solvent, the purified hesperidin was mainly obtained in liquid form (dissolved in ethanol). To ensure the final product is suitable for direct human consumption, a total removal of residual ethanol (e.g., via vacuum evaporation) would be required. This highlights that while the process is 'human-safe' in its nature, further optimization is needed to reach a ready-to-use solid food ingredient.

In conclusion, although solid crystals were not obtained in sufficient quantity, the analyses performed, particularly TLC, show that hesperidin was successfully extracted and purified from orange peel. The main limitations, related to crystallization and solid recovery, highlight the need to optimize the experimental conditions. Additional analyses, such as microscopic or structural characterization of the crystals, as well as modifications of the protocol to improve product recovery, would be useful from perspectives to strengthen the validation of the method.

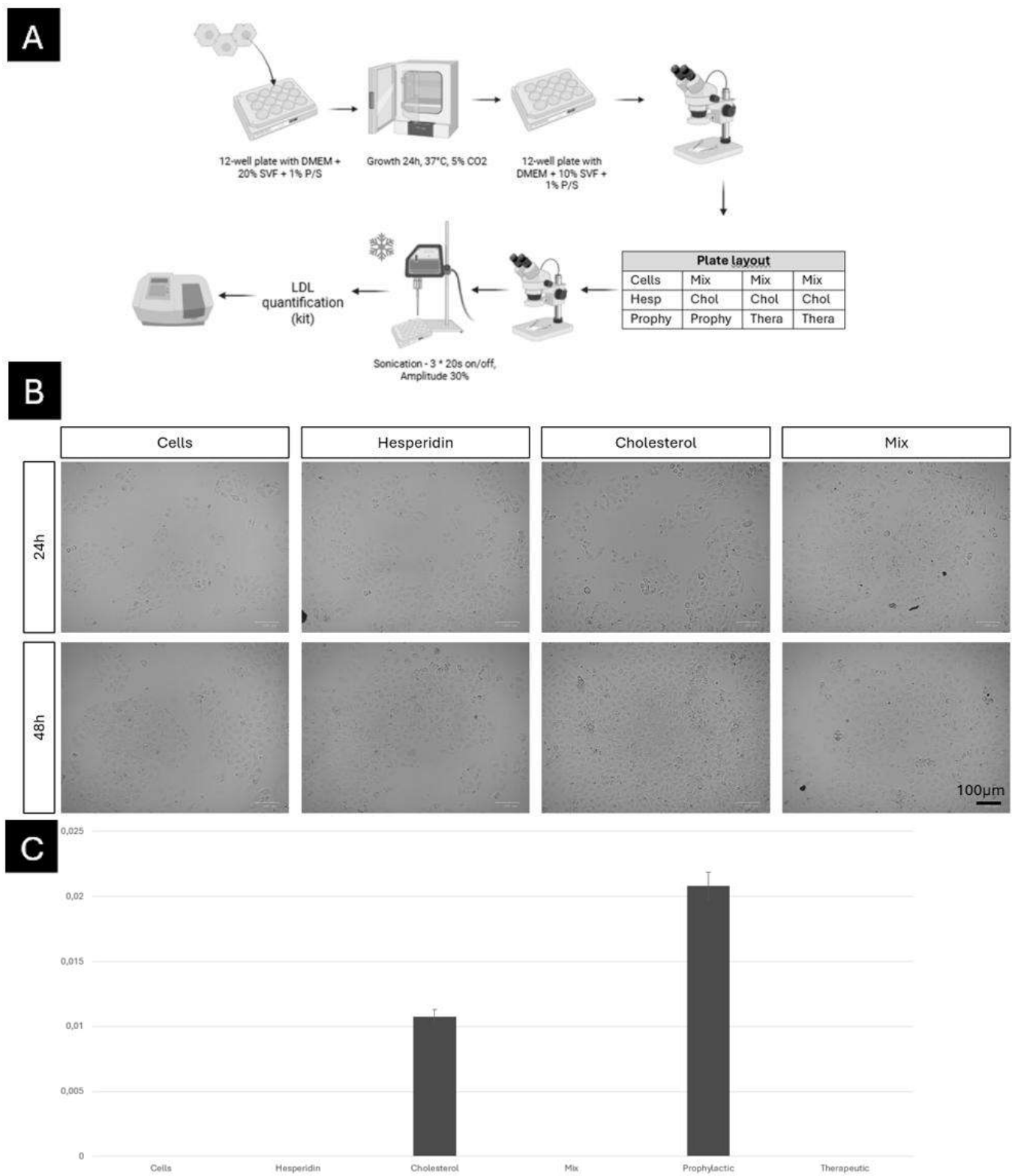


Figure 4. (A) Schematic representation of the LDL-cholesterol-lowering effect assay of hesperidin on HuH7 cells (B) Observation of HUH7 cells growth with or without treatment with cholesterol (40µM) and hesperidin (50µM), after 24h or 48h of culture, under optical microscope, scale bar shown in the figure (C) Dosage of LDL in HuH7 cells with or without treatment with cholesterol (40µM) and hesperidin (50µM) (Mean±SD).

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The control butter shows characteristics of traditional butter (yellow color, natural and homogeneous appearance, neutral and delicate odor, smooth and creamy texture) and the flavor is perceived as well-balanced. It receives a high overall liking score (4.04/5), which indicates that it meets customer expectations.

The butter with *Salicornia* is the most appreciated sample. This result is coherent with the strong preference for salty flavors expressed by the panel, with 94% of participants indicating that they like salty tastes in the preliminary questionnaire. This preference can be paralleled with broader consumer trends in France, where a growing portion of the market is dedicated to salted butter. (Made in FR, 2024)

The hesperidin-enriched butters differ primarily by the visual aspect. The layered formulation looks more like the control, whereas the mixed formulation is darker and browner and less uniform. Then, during tasting, both hesperidin samples have relatively similar sensory profiles despite these visual differences: texture, flavor intensity, perceived aromas, specific tastes, and flavor balance remain largely comparable. Their overall liking scores are very close (3.43 for the mixed sample and 3.40 for the layered one), which suggests that the main factor influencing preference variations between the two butters is appearance. These results demonstrate how important visual aspects are, and we need to keep this data in mind for consumer buying intent. Even when different formulations provide comparable taste experiences, a butter that looks like the control is more likely to obtain favorable reactions.

Finally, the absence of citrus notes and the appearance of a vegetal flavor in the hesperidin samples is intriguing and can highlight issues regarding hesperidin flavor in a fat matrix. Hesperidin comes from citrus fruits; however, when it is added to butter, its aroma expression can change. This observation suggests that the sensory contribution of hesperidin may differ depending on the food matrix and require further investigation.

Another explanation can come from the fact that there is bias in our organoleptic test. Indeed, several factors may have influenced the results: 7.5% of the participants had consumed coffee one hour before the tasting, and 43% were wearing perfume. Furthermore, due to organizational constraints, the sessions were conducted during the whole day, including after meal, which could have impacted the test outcomes.

EFFICIENCY TEST

In our experiments, we cultured the cells in an enriched medium of cholesterol to mimic a pathological hypercholesterolemic environment. However, during the growing protocol we observed differences between cells treated by cholesterol, versus treated with cholesterol and

hesperidin, versus untreated conditions. As previously said, the cells cultured with cholesterol and/or hesperidin looked like they had a higher growth rate. This observation is coherent with the important and well-established role of cholesterol as an essential structural component of cellular membranes, and his role in proliferation (De Castro Fonseca et al., 2018).

Moreover, we found the same effect as cholesterol for hesperidin. It also promotes cell proliferation by influencing antioxydants factors and anti-apoptotic properties, as described in some articles (Li et al., 2025). This is interesting because although we could not observe the cholesterol-lowering effect of hesperidin, it does not prevent cell growth.

Finally, when cholesterol and hesperidin are together, we observe a synergy between the two molecules. As we discovered this effect during the experiments by microscopic observation, it was not possible to study further or quantify the differences. Therefore, the two components together improved the cell viability for the HuH7.

As seen in the results, the LDL-cholesterol-lowering effect on cells did not show any effect. In fact, several elements can explain this result. First, we used HuH7 cells to perform our analysis. This was not the cell type found in the literature for this test. Indeed, the HepG2 cells are preferred to highlight differences in cholesterol levels (Guo, 2022 & Ilias, 2021) because it is a standard hepatocyte for the study of liver functions. However, due to an availability problem, we conducted our assay on HuH7 and HepaRG (results not shown) cells. These two types correspond respectively to hepatocellular carcinoma cells (representing a pathologic condition) and hepatocytes progenitors (which do not express the same factors as hepatocytes). This could be the first reason for not achieving expected results.

Next, the LDL quantification kit was not the most appropriate one for cells. In fact, it was a kit intended for serum, plasma and animal tissue samples. But, it was not possible to get a cell kit for the experiment. This could also explain the negative results obtained.

Moreover, another possible explanation for the absence of detectable differences in LDL levels in this experiment could be related to the cell lysis protocol. As explained previously, we chose to sonicate the cells before using the kit of enzymatic detection. However, the sonication parameters were adapted from different published protocols, as we did not find any equivalent on HUH7 cells. We notably saw experiments on HepG2 cells (Xie et al., 2009). They first washed the cells with PBS before resuspending in isopropanol and then sonic for 20 minutes at 4°C. They did not indicate the power ranges (amplitude) (Xie et al., 2009). But in our conditions, performed sonication for 20 minutes by conditions was too much to ensure reproducibility since it was a manual operation. We found another article, where authors

homogenized HepG2 cells by 2 minutes sonication at 20% amplitude on ice bath (Ilias et al., 2021). This protocol was shorter, so we decided to perform it at different amplitudes on our cells. We determined the 30% amplitude as the better one to destructure our cells without destroying every component. Indeed, an insufficient sonication could have resulted in incomplete cell lysis and therefore an inadequate release of LDL. But excessive sonication could destroy lipid structures and compromise the effectiveness of the enzymatic test. To conclude, one hypothesis of the low LDL differences observed could be due to an excessive sonication that deteriorates the LDL structure and hinders the enzymatic mechanism of detection.

CONCLUSION

In conclusion, the hesperidin-enriched butter that we produced represents a real opportunity in the agri-food sector. Through this project, we succeeded in implementing various processes to aim our main objective: to valorize hesperidin contained in orange peels into a food product.

First, hesperidin was successfully extracted from orange peels. Macroscopic observations revealed the presence of hesperidin crystals. In a second step, purity could be demonstrated through analytical evidence. The Rf analysis, following thin-layer chromatography (TLC), between the purified sample and the standard confirms the presence of the compound of interest. Furthermore, several tests incorporating commercial hesperidin into butter were carried out. Sensory analysis and data collected from the general public revealed that the butter with a layer of hesperidin and *Salicornia* is highly appreciated compared to a control butter. The organoleptic characteristics are fulfilling: the taste profile of the laboratory-produced butter is of good quality. Finally, the last aspect of the project is to combine the production of a food product with a cholesterol-lowering effect. This property was analyzed on human liver cells. Unfortunately, the biological efficacy of hesperidin was not confirmed by this experiment. It remains to be demonstrated, based on the proven effects in rats: (Gence, 2019) these may suggest that the action is similar in humans. Future research on these results could be conducted.

PERSPECTIVES

Based on our results and the limits we found, we see several ways to improve the project. First, there is a key technical issue with our extraction step because of the use of ethanol. This solvent is not guaranteed to be food-grade, which we need to add the extract directly to a consumer product. Also, it could be interesting to further explore our crystallization technique, reviewing the process to induce better crystallization. Our results could be improved until obtain pure crystals, so ensure the additive is safe for food use. Secondly, for the finished product, customer feedback shows a chance to expand the range. Indeed, the perspective is to make a product with no added salt, to keep health benefits but with a

different flavoring (not *Salicornia*) that copies the taste of semi-salted butter. Further trials could be judged by a panel of testers in order to strengthen the reproducibility of the results obtained. Third, at the biology level, our in-vitro method needs a full review. We must repeat the tests on a more suitable cell type, which will better fit the real-life effect of hypercholesterolemia in the human body. We also need to update the cell's disruption (lysis) steps so the LDL assay could be performed efficiently. Finally, it could be interesting to study what happens to the molecule in the body. We should check how hesperidin breaks down in the digestive system. This will help us understand how the body absorbs it and how it really works in living organisms.

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CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

REFERENCES

- Bioflavonoïdes. (s. d.). In *Nutrimea — Ingrédients*. Retrieved January 10, 2026, from <https://www.nutrimea.com/fr/ingredients/38-bioflavonoides>
- Brand, A., Visser, M. E., Schoonees, A., & Naude, C. E. (2022). Replacing salt with low-sodium salt substitutes (LSSS) for cardiovascular health in adults, children and pregnant women. *Cochrane Database of Systematic Reviews*, 8(8), CD015207. <https://doi.org/10.1002/14651858.CD015207>
- Chef Philippe. (2017). *Faire son beurre maison*. Meilleur du Chef. Retrieved January 21, 2026 from <https://www.meilleurduchef.com/fr/recette/faire-son-beurre-maison.html>
- Cholestérol LDL : Symptômes et traitements. (s. d.). Elsan. Retrieved October 11, 2025, from

<https://www.elsan.care/fr/pathologie-et-traitement/maladies-endocriniennes/cholesterol-ldl-causes-traitements>

- Codex Alimentarius. (2001). *Directives Codex pour l'évaluation organoleptique en laboratoire du poisson et des mollusques et crustacés (CAC/GL 31-1999)*. FAO/OMS.
- De Castro Fonseca, M., França, A., Florentino, R. M., Fonseca, R. C., Filho, A. C. M. L., Vidigal, P. T. V., Oliveira, A. G., Dubuquoy, L., Nathanson, M. H., & Leite, M. F. (2018). Cholesterol-enriched membrane microdomains are needed for insulin signaling and proliferation in hepatic cells. *American Journal of Physiology-Gastrointestinal and Liver Physiology*, 315(1), G80–G94. <https://doi.org/10.1152/ajpgi.00008.2018>
- Déchets et gaspillage alimentaires en France en 2023. (s. d.). Données et études statistiques pour le changement climatique, l'énergie, l'environnement, le logement et les transports. Retrieved October 11, 2025, from <https://www.statistiques.developpement-durable.gouv.fr/dechets-et-gaspillage-alimentaires-en-france-en-2023>
- Gence, L. (2019). *Étude des propriétés nutrition-santé d'un concentré d'agrumes enrichi en hespéridine et β-cryptoxanthine : bioaccessibilité des caroténoïdes et effets santé impliqués dans la prévention du diabète de type 2* (Thèse de doctorat). Université de La Réunion. <https://theses.fr/2019LARE0009>
- Guo, J., Li, Y., Yuan, Y., Li, X., Li, X., Jiang, X., Bai, W., & Jiao, R. (2022). Cholesterol-lowering activity of 10-gingerol in HepG2 cells is associated with enhancing LDL cholesterol uptake, cholesterol efflux and bile acid excretion. *Journal of Functional Foods*, 95, 105174. <https://doi.org/10.1016/j.jff.2022.105174>
- Hendrickson, R., & Kesterson, J. W. (1954). Recovery of citrus glucosides. *Proceedings of the Florida State Horticultural Society*, 67, 199-202. <https://journals.flvc.org/fshs/article/download/101807/97742>
- Hendrickson, R., & Kesterson, J. W. (1955). Purification of crude hesperidin. *Proceedings of the Florida State Horticultural Society*, 68, 121-123. <https://journals.flvc.org/fshs/article/download/101651/97595>
- Hendrickson, R., & Kesterson, J. W. (1957). Chemical analysis of citrus bioflavonoids. *Proceedings of the Florida State Horticultural Society*, 70, 196-203. <https://journals.flvc.org/fshs/article/download/101458/97402>
- Hendrickson, R., Kesterson, J. W., & Edwards, G. J. (1959). Hesperidin in Orange Juice and Peel—Extracts Determined by UV Absorption. *Proceedings of the Florida State Horticultural Society*, 72, 258-262. <https://journals.flvc.org/fshs/article/download/101230/97174>
- Hespéridine. (2024). In *Wikipédia*. <https://fr.wikipedia.org/w/index.php?title=Hesp%C3%A9ridine&oldid=213688353>
- Hypercholestérolémie. (s. d.). Retrieved October 11, 2025, from <https://www.santepubliquefrance.fr/maladies-et-traumatismes/maladies-cardiovasculaires-et-accident-vasculaire-cerebral/hypercholesterolemie>
- Ilias, A. N., Ismail, I. S., Hamzah, H., Mohidin, T. B. M., Idris, M. F., & Ajat, M. (2021). Rebaudioside A enhances LDL cholesterol uptake in HEPG2 cells via suppression of HMGCR expression. *Reports of Biochemistry and Molecular Biology*, 10(3), 477–487. <https://doi.org/10.52547/rbmb.10.3.477>
- Ilias, A. N., Ismail, I. S., Hamzah, H., Mohidin, T. B. M., Idris, M. F., Malaysia, P. B. D. P. U. P. S. I. 3. T. M. P., & Ajat, M. (2021). Rebaudioside A enhances LDL cholesterol uptake in HEPG2 cells via suppression of HMGCR expression. *Reports of Biochemistry and Molecular Biology*, 10(3), 477–487. <https://doi.org/10.52547/rbmb.10.3.477>
- Janapala, U. S., & Reddivari, A. K. R. (2025). Low Cholesterol Diet. In *StatPearls*. StatPearls Publishing. <http://www.ncbi.nlm.nih.gov/books/NBK551722/>
- Kesterson, J. W., & Hendrickson, R. (1952). The glucosides of citrus. *Proceedings of the Florida State Horticultural Society*, 65, 223-225. <https://journals.flvc.org/fshs/article/download/102008/97942>
- Khorasanian, A. S., Fateh, S. T., Gholami, F., Rasaei, N., Gerami, H., Khayyatzadeh, S. S., Shiraseb, F., & Asbaghi, O. (2023). The effects of hesperidin supplementation on cardiovascular risk factors in adults : A systematic review and dose-response meta-analysis. *Frontiers in Nutrition*, 10, 1177708. <https://doi.org/10.3389/fnut.2023.1177708>
- Kosmas, C. E., Sourlas, A., Guzman, E., & Kostara, C. E. (2022). Environmental Factors Modifying HDL Functionality. *Current Medicinal Chemistry*, 29(10), 16871701. <https://doi.org/10.2174/0929867328666210714155422>
- Kumar, H., Bhardwaj, K., Sharma, R., Nepovimova, E., Kuča, K., Dhanjal, D. S., Verma, R., Bhardwaj, P., Sharma, S., & Kumar, D. (2020). Fruit and Vegetable Peels :

- Utilization of High Value Horticultural Waste in Novel Industrial Applications. *Molecules*, 25(12), 2812. <https://doi.org/10.3390/molecules25122812>
- Le beurre, production et consommation. (s. d.). Ministère de l'Agriculture et de la Souveraineté alimentaire. Retrieved October 11, 2025, from <https://agriculture.gouv.fr/infographie-le-beurre-production-et-consommation>
- Li, A., Nie, T., Wang, X., & Ma, J. (2025). Hesperidin ameliorates hepatocyte cell cycle disruption, apoptosis, and necroptosis in mice with nonalcoholic fatty liver disease via the VEGFA-PI3K/AKT pathway. *International Immunopharmacology*, 163, 115304. <https://doi.org/10.1016/j.intimp.2025.115304>
- Limongelli, F., Crupi, P., Clodoveo, M. L., Corbo, F., & Muraglia, M. (2022). Overview of the Polyphenols in Salicornia: From Recovery to Health-Promoting Effect. *Molecules (Basel, Switzerland)*, 27(22), 7954. <https://doi.org/10.3390/molecules27227954>
- Made in FR. (2024). *Le marché du beurre en France*. <https://madeinfr.fr/etudes-de-marche/agroalimentaire-alimentation/le-marche-du-beurre-en-france/>
- Manikyam, H. K., Tripathi, P., Patil, S. B., Lamichhane, J., Chaitanya, M., & Patil, A. R. (2022). Extraction, purification, and quantification of hesperidin from the immature Citrus grandis/maxima fruit Nepal cultivar. *Asian Journal of Natural Product Biochemistry*, 20(1), Article 1. <https://doi.org/10.13057/biofar/f200105>
- Menir, O. (2025, 24 mai). *Hypercholestérolémie : guide complet 2025 – Symptômes, traitements et innovations*. LeMedecin.fr. <https://lemedecin.fr/medical/pathologies/hypercholestérolémie.html>
- Morin, B., Nichols, L. A., Zalasky, K. M., Davis, J. W., Manthey, J. A., & Holland, L. J. (2008). The citrus flavonoids hesperetin and nobiletin differentially regulate low density lipoprotein receptor gene transcription in HEPG2 liver cells3. *Journal of Nutrition*, 138(7), 1274–1281. <https://doi.org/10.1093/jn/138.7.1274>
- Ogunro, O. B. (2025). An updated and comprehensive review of the health benefits and pharmacological activities of hesperidin. *Biochemical and Biophysical Research Communications*, 772, 151974. <https://doi.org/10.1016/j.bbrc.2025.151974>
- Rivers, C. (2017, juin 30). Le beurre ou la margarine : Quoi choisir? *Service alimentaire Gordon*. <https://gfs.ca/fr-ca/ideas/le-beurre-ou-la-margarine-quoi-choisir/>
- Stoclet, J.-C., & Schini-Kerth, V. (2011). Flavonoïdes alimentaires et santé humaine. *Annales Pharmaceutiques Françaises*, 69(2), 7890. <https://doi.org/10.1016/j.pharma.2010.11.004>
- Xie, Z., Liang, G., Zhang, L., Wang, Q., Qu, Y., Gao, Y., Lin, L., Ye, S., Zhang, J., Wang, H., Zhao, G., & Zhang, Q. (2009). Molecular mechanisms underlying the cholesterol-lowering effect of Ginkgo biloba extract in hepatocytes: a comparative study with lovastatin. *Acta Pharmacologica Sinica*, 30(9), 1262–1275. <https://doi.org/10.1038/aps.2009.126>

Towards Green Biotechnology: Mycoremediation of Azo Dyes by *Phanerochaete chrysosporium*

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VISUAL ABSTRACT



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Abstract – The textile industry is a major source of environmental pollution due to the discharge of synthetic dyes into wastewater, particularly azo dyes, which are toxic, persistent, and resistant to conventional treatments. Mycoremediation, based on fungal enzymatic activity, offers a promising and sustainable alternative. This study evaluated the potential of *Phanerochaete chrysosporium* for the degradation of the azo dye Unicert Blue 05601-J, in comparison with *Pleurotus ostreatus* (positive control) and *Aspergillus oryzae* (negative control). Decolorization efficiency was assessed in liquid (Potato Dextrose Broth, PDB) and solid (Potato Dextrose Agar, PDA) media at different dye concentrations to determine the influence of culture conditions on fungal activity. *P. chrysosporium* induced visible azo dye decolorization, particularly in solid medium where clear halos were observed. In liquid systems, absorbance generally decreased over time; however, results showed variability, including fluctuations and occasional negative values, suggesting methodological limitations, measurement constraints, and non-optimized experimental conditions. Overall, these findings support the potential of *P. chrysosporium* for azo dye decolorization and highlight the importance of culture conditions and experimental optimization for fungal mycoremediation processes.

Index Terms – Mycoremediation, Environment pollution, Textile wastewater, Azo dyes, Ligninolytic enzymes, *Phanerochaete chrysosporium*, *Pleurotus ostreatus*, *Aspergillus oryzae*, Decolorization, Solid and liquid medium.

INTRODUCTION

The textile industry is considered one of the most polluting industries in the world. Every year, large amounts of synthetic dyes are released into the environment. It is estimated that 10-20% of these dyes are lost during the dyeing process and end up in wastewater (Berradi et al., 2019). A large part of them belong to the azo dye family, which is defined by the presence

of one or more azo bonds ($-N=N-$) linking aromatic rings. Because of their stability, low cost and wide range of colors, azo dyes are widely used. However, they raise serious environmental and health concerns. Many of them resist mycoremediation and can produce toxic, mutagenic or even carcinogenic compounds during partial degradation (Lellis et al., 2019).

Today, wastewater containing azo dyes is mainly treated with physico-chemical methods such as coagulation, adsorption, or oxidation. Even if these treatments are regulated in Europe by the Water Framework Directive (Directive 2000/60/EC) and the REACH regulation (Regulation (EC) No 1907/2006), they do not always ensure complete azo dye removal.

For this reason, more sustainable biological solutions are being explored. Mycoremediation is one of these approaches. It uses fungi and their extracellular enzymes to break down complex pollutants such as azo dyes. Among ligninolytic fungi, *Phanerochaete chrysosporium* is often studied because it produces strong oxidative enzymes like lignin peroxidase (LiP) and manganese peroxidase (MnP). These enzymes are able to attack the stable aromatic structures found in azo dyes (Kersten et al., 2007; Ram Lakhan Singh et al., 2015; Indumal S. Herath et al., 2024). Thanks to its rapid growth and its well-known enzymatic system, this fungus is commonly used as a model organism in biodegradation studies. However, its efficiency can vary depending on culture conditions, and comparisons between solid and liquid systems are still limited. In addition, the performance of fungal systems may depend on several parameters such as dye concentration, oxygen availability, and the physical structure of the culture medium. Solid systems can promote closer contact between fungal biomass and the pollutant, while liquid systems may allow better diffusion but also introduce variability in measurements. Understanding these differences is important when considering future industrial or environmental applications.

In this study, we evaluated the decolorization capacity of *Phanerochaete chrysosporium* toward the azo dye Unicert Blue 05601-J under different culture conditions. This approach makes it possible to observe not only the fungal activity but also the influence of the culture environment. To support the experimental design, *Pleurotus ostreatus*, a white-rot fungus known for its ability to degrade azo dyes, was used as a positive control based on previous studies (Ram Lakhan Singh et al., 2015). In contrast, *Aspergillus oryzae*, which does not show significant ligninolytic activity related to azo dye degradation, was used as a negative control (data not shown). By comparing these strains under different culture conditions, this study aims to better understand the potential of fungal-mediated azo dye decolorization for textile wastewater treatment.

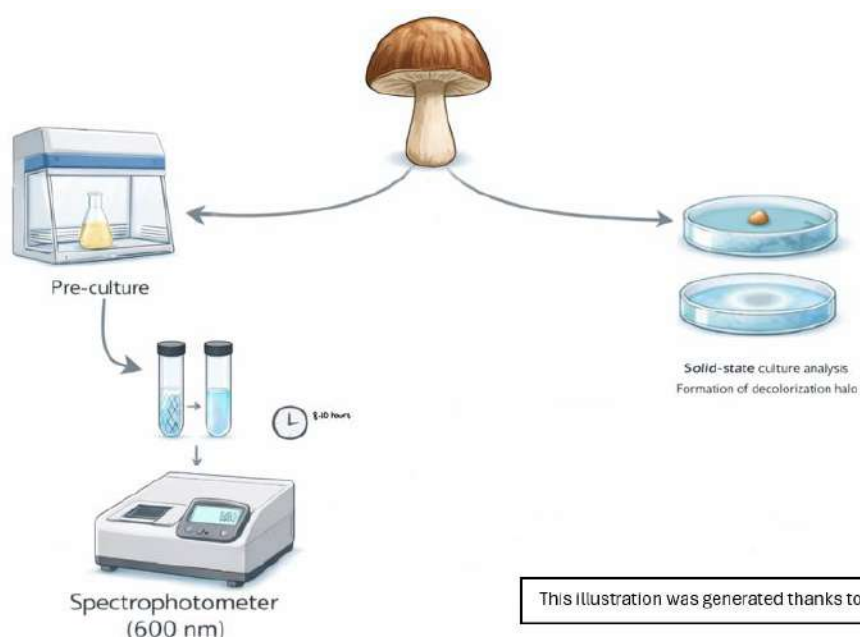


Figure 1. Schematic representation of the experimental protocol for fungal culture and mycoremediation analysis. The workflow illustrates the initial pre-culture of fungal strains (*P. chrysosporium*, *P. ostreatus*, and *A. oryzae*) in a sterile environment. The process involves two parallel evaluation tracks: liquid culture analysis using spectrophotometry at 600 nm to monitor turbidity and solid-state culture on PDA medium to observe the formation of radial decolorization halos over 5 days incubation period.

MATERIEL AND METHOD

This section describes the materials, strains, culture conditions, experimental setup, and analytical methods used to evaluate fungal degradation of the azo dye. The experimental workflow is summarized in a schematic diagram presented as Figure 1.

MICROBIAL STRAINS

The microbial strains used in this study included *Phanerochaete chrysosporium*, a white rot fungus that was selected as the main strain for the degradation of azo dyes. This strain was supplied by Cirm Fungi (INRAE, National Research Institute for Agriculture, Food and Environment) under the reference BRFM 387. The second microorganism studied was *Pleurotus ostreatus*, a basidiomycete fungus recognized as a positive control for its azo dye degradation capabilities. The latter was obtained from the SupBiotech laboratory LRPIA (Laboratoire de Recherche en Ingénierie Agroalimentaire). Finally, *Aspergillus oryzae*, a filamentous fungus, was used as a negative control (data not shown) due to its inability to degrade azo dyes. It was also supplied by the SupBiotech laboratory LRPIA. Preliminary trials were also conducted using *Saccharomyces cerevisiae*. However, these results were ultimately excluded from the primary analysis as the available strain did not correspond to the ligninolytic profiles described in the reference literature (Antošová Z et al., 2018).

DYE AND SOLUTION PREPARATION

The experiments used a blue azo dye, UNICERT BLUE 05601-J, for which a stock solution at a concentration of 750 mg/L was prepared by dissolving 7.5 mg of dye in 10 mL of distilled water, then sterilized using a 20 µm filter. Working solutions at 10, 30, 50 and 100 mg/L were then prepared by dilution in the corresponding growth media.

CULTURE MEDIA

All strains were grown on two types of media to ensure consistent experimental conditions. For solid culture, PDA (Potato Dextrose Agar) was prepared by dissolving 39 g of powder in 1 L of distilled water. For liquid culture, PDB (Potato Dextrose Broth) was used, prepared by dissolving 24 g of powder in 200 mL of distilled water. Before inoculation, these media were enriched with UNICERT BLUE 05601-J at the concentration required for the tests.

CONSERVATION AND ACTIVATION

The strains stored in cryotubes at -80°C were reactivated according to a specific procedure. First, the cryotube was thawed in a water bath maintained at a temperature of 25 to 30°C. Before opening the tube, the outside was disinfected with 70% ethanol to ensure optimal aseptic conditions. A small aliquot was then inoculated onto PDA culture medium plates, again under aseptic conditions. The plates were incubated at a temperature between 26 and 30°C until colony

growth was visible. This step confirmed the viability of the strains and the morphology of the colonies. Finally, the identity and morphological integrity of the strains were verified by macroscopic and microscopic observations, using lactophenol blue dye for better visualization. This rigorous method ensures that the reactivated strains meet the expected quality criteria.

EXPERIMENTAL SETUP

Microbial cultures were placed on Petri dishes for solid media and in Erlenmeyer flasks for liquid media, containing the azo dye at the intended concentrations. For solid media, the azo dye was incorporated before pouring the medium into the Petri dishes. For liquid media, the azo dye was added to the PDB bottles before inoculation. The concentrations tested for each strain were as follows: for *Phanerochaete chrysosporium* and *Pleurotus ostreatus*, the concentrations were 10, 50, and 100 mg/L, while for *Aspergillus oryzae*, used as a negative control (data not shown), the concentration was 30 mg/L. The latter being defined as “detectable concentration” by the legislation (Regulation (EC) No 1907/2006). Each experimental condition was performed in duplicate to ensure reproducibility. The concentration of 50 mg/L, selected as the main condition of interest, was tested in triplicate or more to support comparative analysis and statistical interpretation.

INCUBATION CONDITIONS

All cultures were incubated at 30°C, a temperature optimal for fungal growth and enzyme activity relevant to azo dye degradation. Humidity was controlled to prevent drying of solid cultures.

MONITORING AND ANALYSIS

Macroscopic observations were performed once a day and microscopic observations once a week experience, focusing on the color and texture of the colonies, the growth rate, and the presence or absence of halos. The evaluation of azo dye degradation was performed in different ways. On agar plates, the diameter of the production zones was measured using a ruler. In liquid cultures, absorbance was determined at 600 nm using a spectrophotometer after centrifugation, in comparison with blank controls (medium plus dye without inoculation). This wavelength was selected as it falls within the absorption spectrum of the blue azo dye Unicert Blue 05601-J, allowing for the tracking of relative concentration changes over time. While a full calibration curve was not established, the decrease in absorbance at this wavelength serves as a direct proxy for dye degradation according to the Beer-Lambert principle, which relates light attenuation to the concentration of the absorbing species. However, due to some artifacts, data was considered with strong reservations. Finally, sampling was performed by taking standardized quantities using a sterile scalpel, specifically for fungal colonies. All

measurements were performed independently for each replicate.

STRAIN PRESERVATION

Cultures were prepared for long-term storage by mixing with 15-20% glycerol, supplied by SIGMA under the reference G9012-1L, and storing in cryotubes at -80°C.

EXPERIMENTAL DESIGN AND CONTROLS

Experiments were performed with defined replicate numbers for each condition. For *Phanerochaete chrysosporium*, 15 independent replicates were conducted at 50 mg/L to ensure statistical robustness, while single replicates were performed at 10 mg/L and 100 mg/L. Negative controls consisted of dye-containing medium without fungi and *Aspergillus oryzae* cultures ($n = 1$ each). Positive controls consisted of *Pleurotus ostreatus* cultures tested at 10 mg/L ($n = 2$), 50 mg/L ($n = 3$) and 100 mg/L ($n = 2$).

STATISTICAL ANALYSIS

Statistical analyses were performed using R Studio. Wilcoxon and Mann-Whitney tests were performed because the number of samples is small, making it difficult to verify or assume normality and thus rendering ANOVA, a parametric test, unreliable. First, a nonparametric statistical analysis was performed to evaluate the effect of concentration on fungal discoloration diameter of *Phanerochaete chrysosporium* and *Pleurotus ostreatus*: values measured at D0 were compared to those at D4 using a Wilcoxon test. Next, to compare discoloration between *Phanerochaete chrysosporium* and *Pleurotus ostreatus*, the distributions of diameters (or D0-D4 variations) were compared using a Mann-Whitney test. Given the limited number of replicates, these statistical analyses were used primarily to identify general trends rather than to draw definitive conclusions.

RESULTS

DECOLORIZATION OF UNICERT BLUE 05601-J IN LIQUID PDB MEDIUM

Time-course evolution of dye absorbance

The ability of the different fungal strains to decolorize Unicert Blue 05601-J in liquid PDB medium was evaluated by monitoring absorbance at 600 nm over five days at dye concentrations of 10, 50, and 100 mg/L. (Figure 2A-B)

For *Phanerochaete chrysosporium*, absorbance values showed a time- and concentration-dependent evolution. At 10 mg/L, absorbance slightly decreased between day and day 1), then increased on day 2, before dropping more markedly on day 3 and slightly increasing again on day 4. At 50 mg/L, a progressive decrease in absorbance was observed throughout the week, and at 100 mg/L, absorbance decreased more sharply, from day 0 to day 4.

For *Pleurotus ostreatus*, absorbance values were lower overall and showed a more gradual decrease over time. At 10 mg/L, absorbance increased from day 0 to day 1, then decreased on day 2 and 3, before slightly increasing on day 4. At 50 mg/L, absorbance steadily decreased from day 0 to day 2, a slight increase was measured on day 3 followed by a lower value on day 4. At 100 mg/L, a decrease was also observed between day 0 to day 2, followed by low and relatively stable values on day 3 and day 4.

Table 1. Time-course evolution of absorbance at 600 nm for *Phanerochaete chrysosporium* in liquid PDB medium. * Negative absorbance values are discussed in section Discussion.

Concentration (mg/L)	Absorbance 600nm				
	Day 0	Day 1	Day 2	Day 3	Day 4
10	0,549	0,478	0,544	0,267	0,340
50	0,817	0,636	0,700	0,298	0,490
100	0,915	0,433	0,339	-0.250*	0,298

Table 2. Time-course evolution of absorbance at 600 nm for *Pleurotus Ostreatus* in liquid PDB medium.

Concentration (mg/L)	Absorbance 600nm				
	Day 0	Day 1	Day 2	Day 3	Day 4
10	0,131	0,194	0,078	0,007	0,210
50	0,320	0,189	0,029	0,094	0,060
100	0,423	0,234	0,034	0,049	0,080

Comparison between early incubation peak and degradation phases

To highlight the period corresponding to the most pronounced decrease in absorbance, values were measured on day 0 and day 3 and compared for each strain and each dye concentration (Table 3, Figure 2). For both *Phanerochaete chrysosporium* and *Pleurotus ostreatus*, a clear decrease in absorbance was observed at all tested concentrations. The relative decrease ranged from 51.4% to 127.3% for *P. chrysosporium* and from 70.6% to 94.7% for *P. ostreatus*.

Table 3. Comparison of absorbance values between early incubation (day 0) and degradation phase (day 3). * Negative absorbance values are discussed in section Discussion.

	Concentration (mg/L)	Absorbance 600nm Day0	Absorbance 600nm Day3	Absolute decrease	Relative decrease (%)
<i>P. chrysosporium</i>	10	0.549	0.267	0.282	51.4%
	50	0.817	0.298	0.519	63.5%
	100	0.915	-0.250	1.165*	127.3%*
<i>P. ostreatus</i>	10	0.131	0.007	0.124	94.7%
	50	0.320	0.094	0.226	70.6%
	100	0.423	0.049	0.374	88.4%

Variability of absorbance measurements

Despite the overall decrease in absorbance observed for both strains, temporal fluctuations were noted, particularly with a re-increase in absorbance values on day 4 for several conditions. Negative absorbance values were observed under certain conditions, particularly at higher dye concentrations and during mid-incubation (day 3) (Figure 2B). Although these values consist of artifacts not expected under ideal spectrophotometric conditions, they were consistently recorded in specific samples.

DECOLORIZATION OF UNICERT BLUE 05601-J ON SOLID PDA MEDIUM

Visual assessment of dye decolorization

The decolorization of Unicert Blue 05601-J by fungal strains was evaluated on solid PDA medium supplemented with the dye (Figure 3A and 3B). Prior to inoculation, the agar displayed a uniform blue coloration. During incubation, a progressive transition from blue to translucent, whitish coloration was observed in the medium surrounding the fungal colonies.

For *Phanerochaete chrysosporium*, the formation of clear, decolorization halos around the fungal colonies was observed, particularly at low and intermediate dye concentrations. These halos increased in size over time. *Pleurotus ostreatus* also induced decolorization zones, although they appeared less extended and less intense. In contrast, no visible decolorization was observed for *Aspergillus oryzae*, used as a negative control negative (data not shown).

Measurement of decolorization halo diameters

To quantify decolorization on solid medium, the diameter of the discoloration halos was measured at Day 0 and at Day 4 for each strain and dye concentration (Figure 3C). Halo diameters increased between Day 0 and Day 4 for both strains. *Phanerochaete chrysosporium* showed a highly significant increase (2.2 → 5.9 cm, Wilcoxon, $p = 4 \times 10^{-4}$), while *Pleurotus ostreatus* increase (0.15 → 2.6 cm) did not reach significance ($p = 0.081$), indicating higher variability.

For *Phanerochaete chrysosporium*, decolorization was detected early at 10 mg/L, with halo diameters ranging from 1.7 to 1.8 cm at Day 0. At Day 4, the diameters increased, reaching values between 7.6 and 8.7 cm. At 50 mg/L, initial diameters at Day 0 ranged from 1.1 to 1.9 cm while final diameters varied between 4.4 and 8.7 cm. Although greater variability was observed at this concentration, a marked increase in decolorization diameter was consistently measured over time. At 100 mg/L, no decolorization was observed at Day 0, and a limited halo of approximately 2.6 cm was measured at Day 4.

For *Pleurotus ostreatus*, decolorization halos were observed at all concentrations but remained smaller than those measured for *Phanerochaete chrysosporium*. At 10 mg/L, the halo diameter increased from 0.8 cm at Day 0 to 4.3 cm at

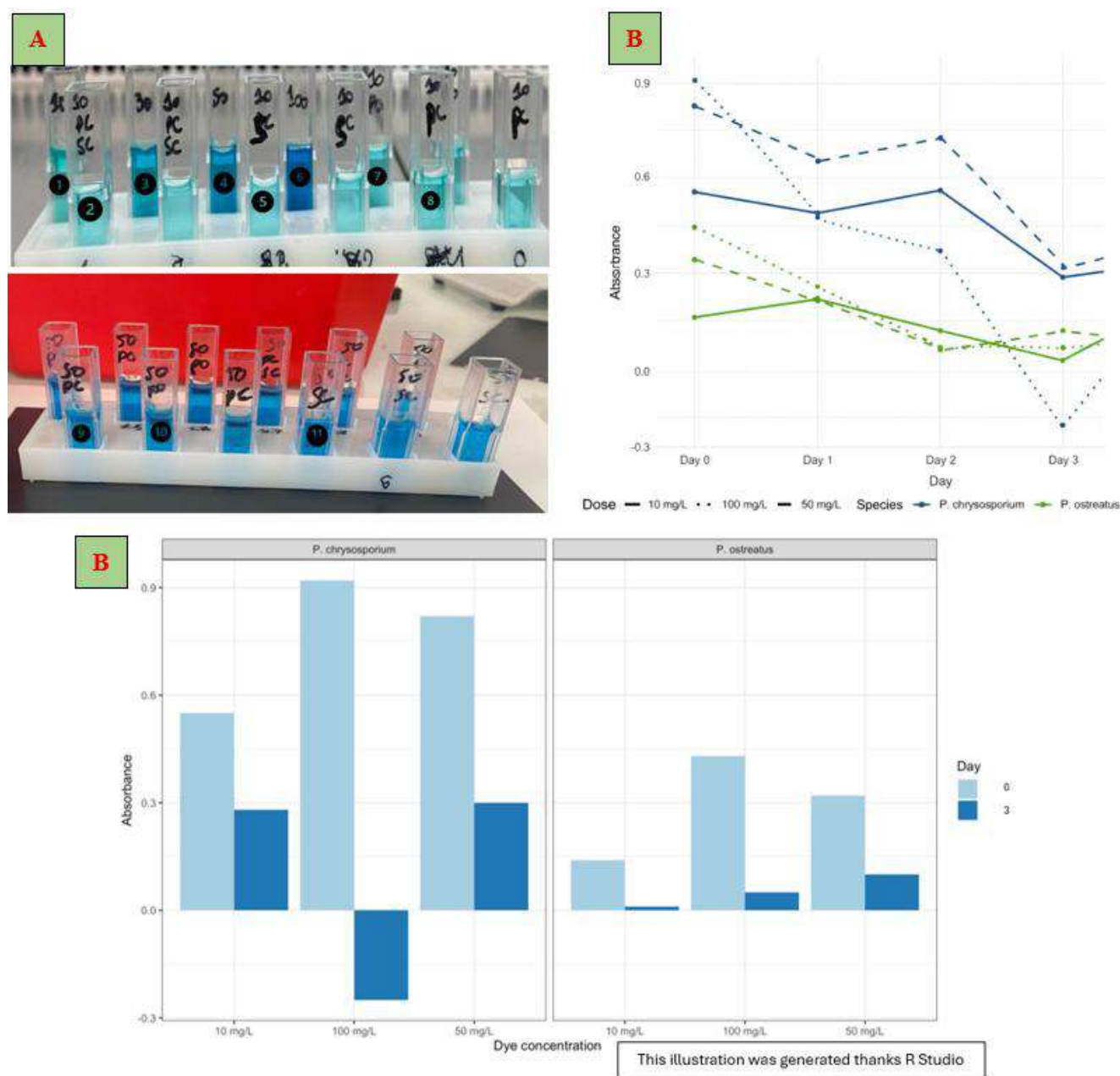


Figure 2. Quantitative analysis of Unicert Blue 05601-J decolorization in liquid PDB medium. (A) Experimental configurations at T0: Visual representation of test tubes at the start of the experiment containing various concentrations of 1293 UNICERT BLUE 05601-J (10, 50, and 100 mg/L). Tubes contain *P. chrysosporium* (PC), *P. ostreatus* (PO), or controls (*A. oryzae* and abiotic medium). Preliminary tests involving *S. cerevisiae* (SC) are also visible. (B) Comparative analysis: The upper panel displays the absorbance at 600 nm for *P. chrysosporium* and *P. ostreatus* over a five-day incubation period at concentrations of 10, 50, and 100 mg/L. Note the outlier recorded for *P. chrysosporium* at 100 mg/L on Day 3 ($A = -0.250$). The bottom panel compares absorbance values specifically between Day 0 and Day 3 across all tested concentrations.

Day 4. At 50 mg/L, diameters ranged from 0.5 to 1.0 cm at Day 0 and increased to 3.3–3.7 cm at Day 4. At 100 mg/L, only minimal decolorization was observed, with diameters increasing from 0.2 cm to 1.0 cm over the incubation period.

Effect of azo dye concentration and strain comparison

For both fungal strains, an increase in dye concentration resulted in a decrease in decolorization efficiency as reflected by smaller halo diameters at higher concentrations. On Day 4,

Phanerochaete chrysosporium produced significantly larger decolorization halos than *Pleurotus ostreatus* (5.9 cm vs 2.6 cm, Mann-Whitney, $p = 0.034$), confirming that *Phanerochaete chrysosporium* exhibits superior decolorization capacity after 4 days of incubation (Figure 3D). At all tested concentrations and incubation times, *Phanerochaete chrysosporium* exhibited larger decolorization halos than *Pleurotus ostreatus*.

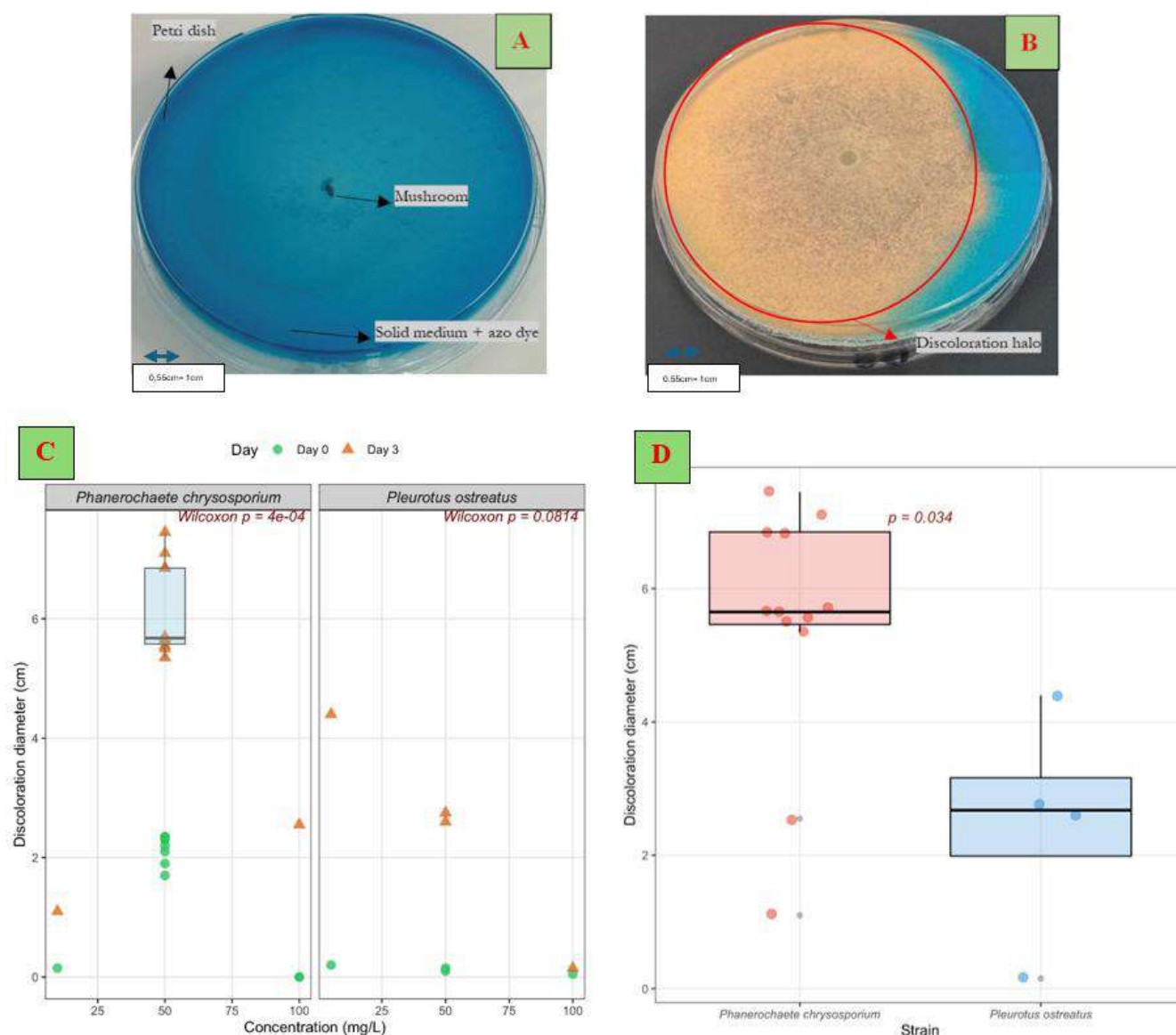


Figure 3. Qualitative and quantitative assessment of azo dye degradation on solid PDA medium. (A) Control plate at Day 1 showing uniform distribution of 1293 UNICERT BLUE 05601-J dye prior to significant fungal growth. (B) Representative image of the final incubation day showing the formation of a decolorization halo around the fungal colony. (C) Comparative analysis of the decolorization diameter (cm) for *P. chrysosporium* and *P. ostreatus* across different concentrations (10, 50, and 100 mg/mL). Time-course evolution from Day 0 to Day 4 was analyzed using the Wilcoxon signed-rank test (*P. chrysosporium*: $p = 4e-04$; *P. ostreatus*: $p = 0.081$). (D) Comparative decolorization efficacy of *P. chrysosporium* and *P. ostreatus* at Day 4 in liquid culture. Statistical analysis using the Mann-Whitney U test reveals a significant difference in decolorization diameter between the two fungal strains ($p = 0.034$).

DISCUSSION

The objective of this study was to evaluate how efficiently *Phanerochaete chrysosporium* can decolorize azo dyes, comparing its performance with *Pleurotus ostreatus* and *Aspergillus oryzae*. Our findings indicate a significant higher decolorization capacity of *P. chrysosporium* on a solid medium compared to *Pleurotus Ostreatus* under the tested conditions (Figure 3C). We also evaluated statistical significance of fungi's decolorization between different azo dye concentration. Our results show better decolorization at 50mg/L and inhibitory effects at 100 mg/L (Figure 3D). Azo

dye concentration could indeed have inhibitory effect according to literature. (Rajhans et al., 2021) Furthermore, the clear formation of degradation zones on PDA medium demonstrates the suitability of solid culture systems for reliably assessing fungal-mediated dye degradation, providing a more stable environment for observation than liquid systems.

However, the experiments conducted in liquid PDB medium revealed significant methodological challenges, most notably the recording of negative absorbance values reaching -0.250 for *P. chrysosporium* at 100 mg/L. Theoretically, absorbance

should remain positive; these anomalies, alongside temporal fluctuations in the data, are interpreted as methodological artifacts. Several factors, like optical interference, likely contributed to these results. Indeed, the absence of systematic filtration meant that fungal micro-aggregates or suspended mycelia could scatter light unevenly, distorting spectrophotometric readings at 600 nm. It could also be linked to background correction. In that regard, the use of a single blank (medium and dye) failed to account for the background turbidity or the potential release of fungal metabolites and pigments at the end of the growth cycle.

These liquid-phase variations (Figure 2B) contrast sharply with the visible efficiency on solid PDA medium (Figure 3B), where the formation of a fungal mat appeared to promote optimal contact between extracellular enzymes and the azo dye. While the disappearance of color and the transition from blue to orange suggest active biodegradation through the cleavage of azo bonds (—N=N—), it is important to note that this was not experimentally quantified through direct enzymatic assays. According to the literature, such a color shift is consistent with the transformation of the azo dye into intermediate metabolites, like phenolic compounds or aromatic amines, prior to full mineralization. (de Almeida AP et al., 2021)

To bridge the gap between solid-state success and the challenges of liquid systems, we propose the "immobilized myco-raft" as a novel industrial perspective. This approach would use modified sawdust as a floating scaffold to concentrate azo dyes around the biomass through natural adsorption while providing the high oxygen levels required for aerobic oxidative degradation. Furthermore, the lignin content in the sawdust would naturally stimulate the production of the necessary LiP and MnP enzymes. While this model offers a "green" and economical alternative to chemical oxidation by avoiding toxic secondary sludge, its implementation remains a theoretical solution that requires further experimental validation (Aksu, 2005; Zhuo & Fan, 2021; Przysaś et al., 2018).

CONCLUSION

To summarize, this study evaluated the potential of *Phanerochaete chrysosporium* as a biological tool for the degradation of azo dyes released by the textile industry. This project assessed the relevance of mycoremediation as a sustainable alternative to conventional wastewater treatment methods.

The results demonstrated that *Phanerochaete chrysosporium* was able to induce visible decolorization of the azo dye Unicert Blue 05601-J, particularly under solid culture conditions. The formation of clear decolorization halos on PDA medium indicated an effective enzymatic activity and confirmed the ability of this fungus to act on complex dye structures. Notably, at Day 3, *P. chrysosporium* demonstrated a significantly superior decolorization diameter of 5.9 cm

compared to 2.6 cm for *P. ostreatus* (Mann-Whitney test, $p = 0.034$), confirming its superior efficacy.

In contrast, experiments conducted in liquid medium showed more variable and less reproducible results, with fluctuations in absorbance values. These variations are likely related to turbidity effects, measurement limitations, and non-optimized ratios between dye concentration and fungal biomass.

Solid media appeared more suitable for assessing azo dye degradation under the conditions tested, while liquid systems would require further optimization to improve reproducibility and reliability.

Overall, this work provides a proof of concept for the use of *Phanerochaete chrysosporium* in the mycoremediation of azo dyes. Future studies could focus on optimizing culture parameters, improving experimental designs, and developing systems closer to real industrial or environmental applications. This project also illustrates the scientific value of student-led research in exploring innovative and sustainable solutions to environmental pollution.

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CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

REFERENCES

1. Berradi, M., Hsissou, R., Khudhair, M., Assouag, M., Cherkaoui, O., El Bachiri, A., & Elharfi, A. (2019). Textile finishing dyes and their impact on aquatic environs. *Heliyon*, 5, e02711. <https://doi.org/10.1016/j.heliyon.2019.e02711>
2. Lellis, B., Fávaro-Polonio, C. Z., Pamphile, J. A., & Polonio, J. C. (2019). Effects of textile dyes on health and

- the environment and bioremediation potential of living organisms. *Biotechnology Research and Innovation*, 3(2), 275–290. <https://doi.org/10.1016/j.biori.2019.09.001>
3. European Parliament & Council of the European Union. (2000). Directive 2000/60/EC establishing a framework for Community action in the field of water policy. *Official Journal of the European Communities*, L 327, 1–73. <http://data.europa.eu/eli/dir/2000/60/oj>
 4. European Parliament & Council of the European Union. (2006). Regulation (EC) No 1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH). *Official Journal of the European Union*, L 396, 1–849. <https://reachonline.eu/reach/en/annex-xvii.html>
 5. Kersten, P., & Cullen, D. (2007). Extracellular oxidative systems of the lignin-degrading Basidiomycete *Phanerochaete chrysosporium*. *Fungal Genetics and Biology*, 44(2), 77–87. <https://doi.org/10.1016/j.fgb.2006.07.007>
 6. Singh, R. L., Singh, P. K., & Singh, R. P. (2015). Enzymatic decolorization and degradation of azo dyes – A review. *International Biodeterioration & Biodegradation*, 104, 21–31. <https://doi.org/10.1016/j.ibiod.2015.04.027>
 7. Herath, I. S., Udayanga, D., Jayasanka, D. J., & Hewawasam, C. (2024). Textile dye decolorization by white rot fungi – A review. *Bioresource Technology Reports*, 25, 101687. <https://doi.org/10.1016/j.biteb.2023.101687>
 8. Antošová, Z., Herkommerová, K., Pichová, I., & Sychrová, H. (2018). Efficient secretion of three fungal laccases from *Saccharomyces cerevisiae* and their potential for decolorization of textile industry effluent – A comparative study. *Biotechnology Progress*, 34(1), 69–80. <https://doi.org/10.1002/btpr.2559>
 9. Rajhans, G., Barik, A., Sen, S., & Raut, S. (2021). Degradation of dyes by fungi: An insight into mycoremediation. *BioTechnologia*, 102(4), 445–455. <https://doi.org/10.5114/bta.2021.111109>
 10. de Almeida, A. P., Macrae, A., Ribeiro, B. D., & do Nascimento, R. P. (2021). Decolorization and detoxification of different azo dyes by *Phanerochaete chrysosporium* ME-446 under submerged fermentation. *Brazilian Journal of Microbiology*, 52(2), 727–738. <https://doi.org/10.1007/s42770-021-00458-7>
 11. Aksu, Z. (2005). Application of biosorption for the removal of organic pollutants: A review. *Process Biochemistry*, 40(3–4), 997–1026. <https://doi.org/10.1016/j.procbio.2004.04.008>
 12. Zhuo, R., & Fan, F. (2021). A comprehensive insight into the application of white rot fungi and their lignocellulolytic enzymes in the removal of organic pollutants. *Science of The Total Environment*, 778, 146132. <https://doi.org/10.1016/j.scitotenv.2021.146132>
 13. Przysaś, W., Zabłocka-Godlewska, E., & Grabińska-Sota, E. (2018). Efficiency of decolorization of different dyes using fungal biomass immobilized on different solid supports. *Brazilian Journal of Microbiology*, 49(2), 285–295. <https://doi.org/10.1016/j.bjm.2017.06.010>

PET Mycoremediation: an Exploratory Study Using Enzymes from *Phanerochaete chrysosporium*

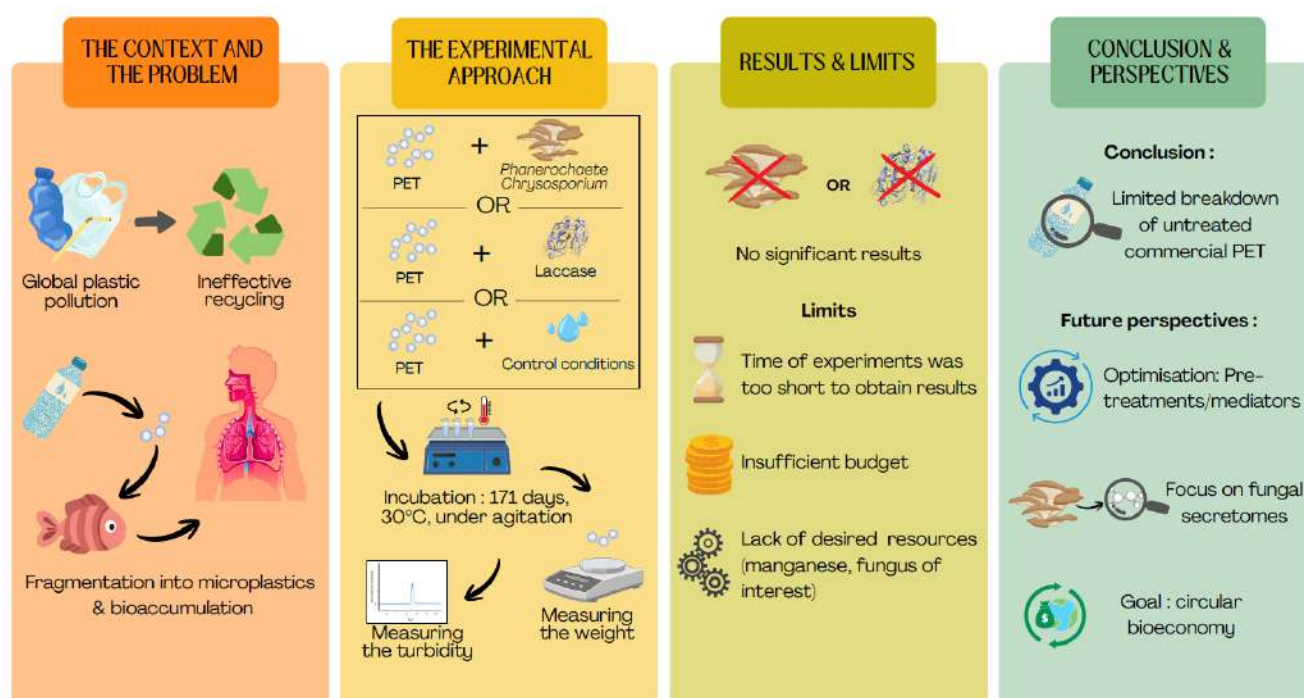
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VISUAL ABSTRACT



PET Mycoremediation: an Exploratory Study Using Enzymes from *Phanerochaete chrysosporium*

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Abstract – The global accumulation of polyethylene terephthalate has created a critical environmental crisis, with microplastics now infiltrating human tissues and global food chains. Traditional mechanical and chemical recycling methods remain energy-intensive and largely inefficient for contaminated waste. Mycoremediation offers a promising biological alternative; however, current fungal degradation rates remain too slow for industrial application. This study investigates the degradation potential of the white-rot fungus *Phanerochaete chrysosporium* and the oxidative enzyme laccase on PET fragments sourced from commercial beverage bottles. By analyzing fungal growth dynamics and enzymatic interactions over a 171-day period, we evaluated the effectiveness of these biological systems through mass loss and turbidity variations. Our findings aim to optimize the use of fungal secretomes to enhance polyethylene terephthalate breakdown, providing a more scalable and cost-effective solution for plastic waste management and bridging the gap between laboratory research and circular bio-economy requirements. We have tested our theories through laboratory experiments which led us to optimize ideas for potential next trials.

Index Terms – Biodegradation, Fungal enzymes, Laccase, Mycoremediation, *Phanerochaete chrysosporium*, PET, PET degradation, Plastic pollution.

INTRODUCTION

Global plastic production has reached alarming levels, now exceeding 400 million tons per year. Among these synthetic polymers, polyethylene terephthalate (PET) is the most dominant, with annual production exceeding 50 million tons¹. While its durability, lightweight nature, and versatility make it ideal for packaging and textiles, these same properties lead to critical environmental persistence. It is estimated that a single PET bottle can take up to 48 years to degrade under ambient conditions². Projections suggest that without

intervention, global plastic waste accumulation could reach 12,000 million tons by 2050³.

The environmental persistence of PET is not merely a logistical challenge; it poses a direct threat to living organisms. Through weathering and mechanical stress, PET debris fragments into microplastics, which contaminate both terrestrial and aquatic ecosystems. These particles act as "Trojan horses," carrying toxic additives and heavy metals into the food chain via bioaccumulation⁴. In wildlife, ingestion leads to internal physical damage, chronic inflammation, and reproductive toxicity. Furthermore, the detection of microplastics in human blood and cardiac tissues highlights a systemic health crisis, as these particles bypass natural biological barriers, potentially causing long-term cellular damage and metabolic disruption⁵.

Current waste management strategies, such as incineration and landfilling, are resource-intensive and often release toxic secondary pollutants. Mechanical recycling remains inefficient for contaminated or composite plastics, which cannot be easily reprocessed². This crisis is particularly acute in regions like sub-Saharan Africa, where infrastructure is often lacking⁶. Consequently, there is an urgent need for mycoremediation, the use of fungi to degrade or sequester environmental contaminants.

Fungi possess a unique "enzymatic arsenal," including laccases and peroxidases, capable of breaking down recalcitrant polymers. In this context, the white-rot fungus *Phanerochaete chrysosporium* stands out as a primary candidate for PET degradation. Its selection is justified by its specialized ligninolytic system, which evolved to decompose lignin, a complex natural polymer that shares structural similarities with the aromatic backbone of PET (Figure 1). Recent studies have confirmed that *P. chrysosporium* can effectively colonize PET surfaces, inducing structural erosion and chemical cleavage of ester bonds through the action of high-redox potential oxidative enzymes⁷. This degradation

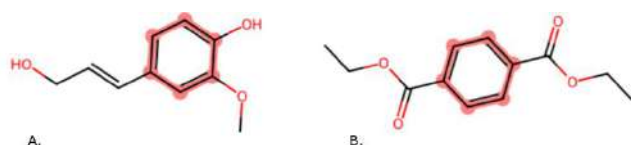


Figure 1: Structural similarities between lignin (A) and PET (B) (the aromatic circle are highlighted in red, and the oxygen are in red). Lignin and PET share aromatic structures that confer high stability but become vulnerable after oxidative activation. This similarity explains their susceptibility to radical-based degradation by oxidoreductases such as laccase and manganese peroxidase.

process typically begins with hydroperoxidation, increasing the polymer's hydrophilicity and making it more susceptible to further enzymatic attack⁸.

Despite the identification of over 400 plastic-degrading microorganisms, a significant knowledge gap remains regarding the optimization of fungal consortia for industrial-scale PET mineralization. Most current biological approaches achieve degradation rates that are too slow for commercial viability, and the synergy between specific fungal enzymes remains poorly understood.

This study aimed to determine whether the strategic combination of specific fungal secretomes could improve PET degradation compared to approaches using a single strain or isolated enzymes. By systematically evaluating appropriate fungal cultures, enzymatic treatments, and abiotic controls over an extended incubation period, we sought to better understand the interactions between biological systems and PET substrates under controlled laboratory conditions. Such an approach contributes to the development of optimized biological degradation strategies and can support the design of scalable enzymatic platforms. Ultimately, this work lays

the foundation for future biofactory models and advances efforts toward sustainable plastic waste management within a circular bioeconomy.

MATERIALS AND METHODS

EXPERIMENTAL OBJECTIVE AND BIOLOGICAL MATERIAL

This study aimed to evaluate the biodegradation potential of polyethylene terephthalate (PET) using two biological approaches: the application of oxidative enzymes (laccase) and the use of the white-rot filamentous fungus *Phanerochaete chrysosporium*.

Fungal Preparation *P. chrysosporium* (Cirm.fungi) was initially cultivated during 2 weeks at 30°C on a solid medium PDA plate. To prepare the inoculum, a suspension was created in sterile water. The concentration of the mother strain and subsequent dilutions were precisely determined using a Malassez counting chamber at multiple intervals (Day 0, 1, 4, and the end of the study) to monitor growth dynamics (6.63×10^6 spores/mL), days 2 & 3 were not measured because we needed the expertise of our referent who was not available.

Enzymatic Solution Laccase (Merk) was prepared in a sterile culture medium (3.47×10^2 U/mL).

PET Preparation, PET was sourced from commercial beverage bottles (e.g., Coca-Cola). This material was chosen for its high crystallinity and environmental relevance. The bottles were cleaned, and fragments were mechanically punched into uniform pellets with a size of ~6mm, refer to Figure 3, to ensure consistent surface area. All PET samples were lightly decontaminated by using 70% ethanol followed

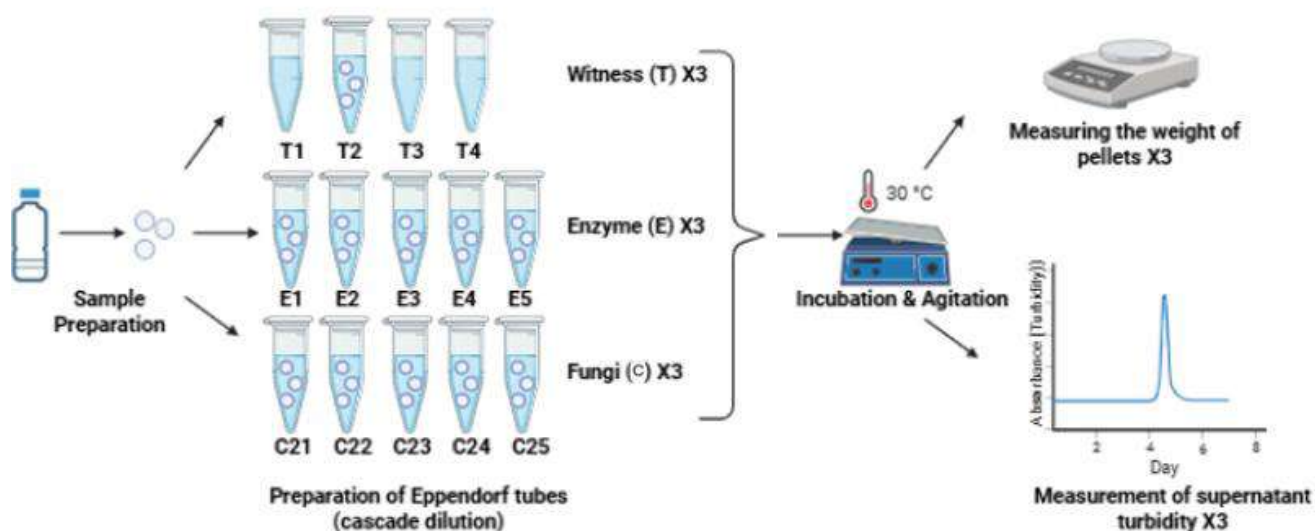


Figure 2: Simplified schematic of the experimental protocol. Schematic created with BioRender.com. PET pellets were prepared and distributed into Eppendorf tubes under different conditions: controls (T), enzyme treatments (E), and fungal treatments (C). Samples were incubated at 30 °C under agitation at 125 rpm and remained in these conditions during the entire project time frame. Supernatant turbidity and dry pellet weight were measured to evaluate PET degradation. Each tube contained $n = 3$ PET pellets. Each condition was performed in duplicate (two independent tubes per treatment).

by rinsing with sterile water and air drying near a Beseen burner to maintain aseptic condition.

Other Consumable Microtube nest (2mL), universal graduated tip (1000UL), serological pipette (5ML), BD Plastipak syringe (20mL), 96-well plate, size M and S gloves, pH paper 0 to 14, ethyl acetate (Supbiotech), plastic cup (Sigma-Aldrich)

EXPERIMENTAL DESIGN AND MONITORING

All procedures were conducted under sterile conditions. The experimental procedure is illustrated in Figure 2.

Experimental Groups and Controls: To isolate the effects of the biological agents, a set of controls was established (total of 30 Eppendorf tubes of 1,5mL. 2 tubes per condition (duplicata)): T1 contains the culture medium only (System sterility control); T2 contains the medium with PET pellets (Physical degradation control); Tr3 contains the medium and the Laccase enzyme (Enzymatic stability control); T4 contains the medium and the fungus (Fungal growth control without plastic).

The experimental groups consisted of PET pellets exposed to two distinct biological treatments, each tested at five concentrations obtained through serial dilution. First, the fungal treatment was applied at the following concentrations: C21 ($6,63 \times 10^6$ spores/mL), C22 ($2,31 \times 10^6$ spores/mL), C23 ($4,30 \times 10^5$ spores/mL), C24 ($1,00 \times 10^5$ spores/mL), and C25 ($2,00 \times 10^4$ spores/mL). Secondly, the laccase enzymatic treatment was tested following the same dilution protocol, yielding concentrations: E1 ($3,47 \times 10^2$ U/mL), E2 ($7,49 \times 10^{-3}$ U/mL), E3 ($1,50 \times 10^{-3}$ U/mL), E4 ($2,99 \times 10^{-4}$ U/mL), and E5 ($5,99 \times 10^{-5}$ U/mL).

Incubation was maintained at 30 °C, the optimal temperature for both *P. chrysosporium* growth and laccase enzymatic (Sigma-Aldrich product information sheet) stability with continuous agitation at 125 rpm.

ANALYTICAL METHODS AND OBSERVATIONS

The degradation process was monitored through a combination of physical, optical, and microscopic methods. Indeed, fungal morphology was documented through periodic photography of the Petri dishes. Observations were recorded in a standardized table focusing on shape, color, opacity, and texture (fuzziness). Additionally, wet mount preparations were examined under a optic microscope to observe the hyphal structure and ensure the purity of the strain. As an indicator of fungal proliferation or byproduct of PET degradation release, absorbance was measured at 600nm (as commonly used for turbidity experiments) using a microplate reader (350 µL samples). Measurements were taken at six

time points: Days 0, 1, 2, 3, 4, and 171. Throughout this paper, time points are labeled using 'J' (from the French *Jour*), where

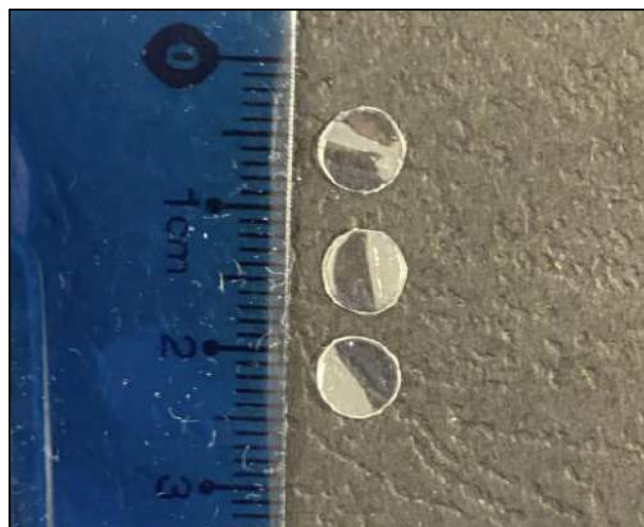


Figure 3: Photograph of PET pellets used in the experiments.
Photograph of the PET pellets used in this study.

J1 corresponds to Day 1. PET pellets were rinsed thoroughly to remove biofilm, dried, and weighed using a precision balance to detect any significant mass loss at days 1, 2, 3, 4, and 171 of culture. We waited several days to analyze the results and assess whether there had been any significant deterioration, as five days of experimentation were not sufficient to study a deterioration phenomenon. After the experimental phase conducted in the laboratory, we also devoted the internship period to this study, then, upon our return, we continued the analysis and interpretation of the experiments to have a longitudinal data (as recommended in the literature ~90 days minimum).

BIostatistical ANALYSIS

Data processing and visualization were performed using RStudio (version 4.4.3). The workflow included data cleaning, filtering of outliers, and temporal conversion of measurements.

To evaluate the significance of the results, a one-way ANOVA was applied to each system (Groups C, E, and T) to compare mean absorbance and mass values across the incubation period. The results were considered significant at $p < 0.05$. If the ANOVA indicated significant differences, a Tukey Honestly Significant Difference (HSD) test was performed to identify specific days where degradation or growth was statistically relevant. Final data were presented in trend graphs incorporating p-values and replicates to provide a clear overview of the efficiency of each biological system.

RESULTS

RESULTS CONTROLS

To validate the experimental conditions and isolate the effects of the biological agents, four control groups were monitored:

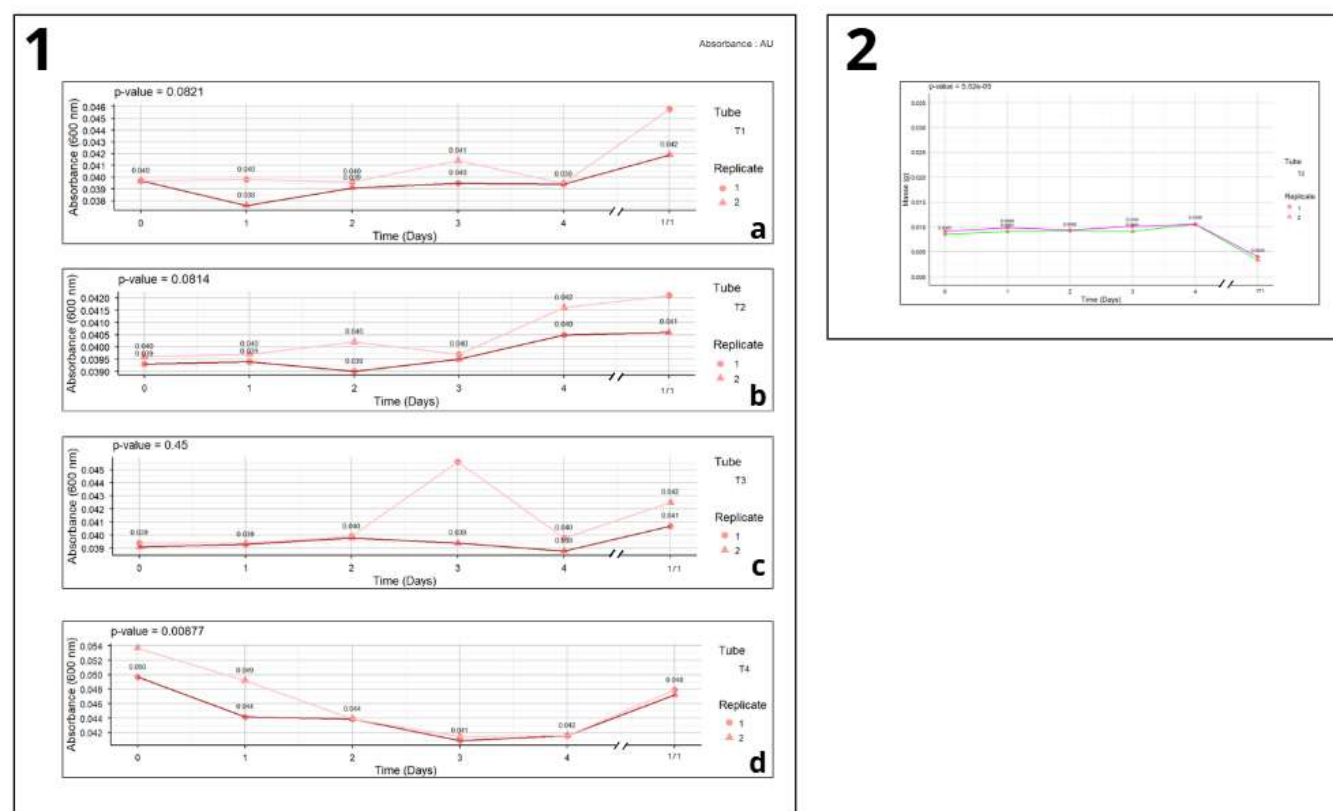


Figure 4: Temporal evolution of absorbance in control conditions (T1–T4) and plastic pellet mass in condition T2. Absorbance was measured at different time points for two replicates under the following control conditions: A: (T1) medium only; B: (T2) medium + plastic; C: (T3) medium + laccase; and D: (T4) medium + *Phanerochaete chrysosporium* (without plastic). Plastic pellet mass (g) was measured over time for condition T2 (medium + plastic). Note: For each representation, the top curve joins the bigger values for each day during the time frame, contrary to the lower curve that joins the lower values.

T1 (Sterility), T2 (Abiotic/Physical), T3 (Enzymatic stability), and T4 (Fungal viability). The monitoring period included an initial kinetic tracking (Days 0 to 4) and a final endpoint measurement at Day 171.

The first system sterility control, designated as T1, consisted of culture medium only to confirm the absence of contamination throughout the 171-day study. As shown in Figure 4.1.a, absorbance remained low and stable between Day 0 and Day 4 (approximately 0.040). A slight, non-significant increase was observed at the final endpoint (Day 171), reaching 0.046 for Replicate 1 and 0.042 for Replicate 2. The statistical analysis (p-value = 0.0821) indicates no significant variation, confirming that the medium remained free of external contamination throughout the experiment.

The second control, designated as T2, was established to assess physical degradation by monitoring a mixture of culture medium and PET pellets. This control assessed whether the PET pellets passively released optically active substances or degraded abiotically. The absorbance profile shown in Figure 4.1.b, remained stable during the initial phase (Day 0–4) around 0.039–0.040. At Day 171, a marginal increase was recorded (approximately 0.042). With a p-value of 0.0814, these variations are not statistically significant,

confirming that the mere presence of PET in the medium does not interfere with optical density readings in the absence of biological activity.

The third control, designated as T3, consisted of culture medium supplemented with laccase to monitor the stability of the enzymatic solution. As illustrated in Figure 4.1.c, the absorbance showed minor fluctuations during the first 4 days, with a transient spike in Replicate 1 at Day 3 (0.045) which returned to baseline. By Day 171, values remained consistent with the initial levels (approximately 0.041–0.042). The high p-value (p-value = 0.45) confirms that the enzymatic solution remained optically stable and did not undergo significant precipitation or degradation over the long term.

The fourth control, designated as T4, served as a positive control to validate the viability of *Phanerochaete chrysosporium* when grown in the culture medium without the stress of plastic exposure. Unlike the abiotic controls, Figure 4.1.d displays a significant evolution (p-value = 0.00877). Initial absorbance was high (0.050–0.054), reflecting the density of the inoculum. A decrease was observed during the initial phase (Day 0 to Day 3), likely corresponding to the lag phase. However, a clear recovery was recorded at Day 171, with absorbance rising back to

approximately 0.048. This confirms that the fungus remained viable and capable of growth in the standard medium over the 6-month duration.

The T2 control was also evaluated from a gravimetric perspective to assess the physical stability of PET pellets in the abiotic culture medium. As shown in Figure 4.2, corresponding to the T2 mass figure, the weight remained relatively stable or slightly increased due to water absorption during the initial phase (Day 0 to Day 4), peaking at 0.0106 g. However, a highly significant drop was recorded at the final endpoint (Day 171), with mass falling to 0.0040 g (p -value = 5.52×10^{-5}). This unexpected mass loss in the control group suggests the presence of experimental artifacts, such as pellet fragmentation during the final washing steps or hydrolysis over the extended incubation period, which must be considered when interpreting the biodegradation results of the test groups.

RESULTS FUNGI

The evolution of optical density (OD) at 600 nm was monitored from Day 1 (J1) to Day 171 (J171) with a break between day 4 and day 171, for five initial fungal concentrations (C21 to C25). All assays were performed in duplicate.

Monitoring of Fungi Activity (Absorbance at 600 nm)

As illustrated in Figure 5.1.a (C21: 6.63×10^6 spores/mL), both replicates began at a similar absorbance of 0.282 on Day 1. Replicate 1 maintained a relatively stable density, with a slight growth peak at J3 (0.298) followed by a gradual decline to 0.090 at J171. Conversely, replicate 2 exhibited a sharp drop as early as J2 (0.070), attempted a recovery at J4 (0.206), and slightly decrease to 0.168 by J171 (p -value = 0.202).

According to Figure 5.1.b (C22: 2.31×10^6 spores/mL), an initial disparity was noted at J0 (0.075 for R1 vs. 0.045 for R2). However, absorbance values converged by J2 and remained consistent until the final measurement, ending at 0.049 for R1 and 0.044 for R2 at J171 (p -value = 0.537).

In Figure 5.1.c (C23: 4.30×10^5 spores/mL), the group followed a trend similar to Figure 5.1.b. At J1, R2 was at 0.078 and R1 at 0.044. Throughout the observation period, R1's absorbance remained stable around 0.040 and showed a little increase at J4 around 0.048 (± 0.04). R2 showed a drastic decrease at J2 to 0.040, and remained stable (± 0.04) (p -value = 0.456).

As demonstrated in Figure 5.1.d (C24: 1.00×10^5 spores/mL), both replicates shared an identical absorbance of 0.039 at J1. Values remained stable around 0.040 (± 0.02), except at J171, where R1 increased to 0.044 while R2 decreased to 0.038 (p -value = 0.962).

Finally, in Figure 5.1.e (C25: 2.00×10^4 spores/mL) the replicates remained stable and nearly identical from J1 to J4

at approximately 0.039. By J171, R2 remained at 0.040, while R1 increased to 0.070 (p -value = 0.459).

Evaluation of Plastic Degradation in the Presence of the fungi (PET Pellet Mass)

The weight of the PET pellets was recorded from J0 (initial weight before contact) to J171.

As illustrated in Figure 5.2.a (C21: 6.63×10^6 U/mL), both replicates started with an initial mass of 0.091g. They followed a synchronized trend with a variance of approximately ± 0.015 g. At J171, an increase in mass was observed for both, reaching 0.0255g for R1 and 0.0140g for R2 (p -value = 0.144).

According to Figure 5.2.b (C22: 2.31×10^6 spores/mL), the initial weights at J0 were 0.0102g (R1) and 0.0091g (R2). Evolution remained stable during the first four days (± 0.0010 to ± 0.0015 g). At J171, a significant mass increase was noted, reaching 0.0176g for R1 and 0.0140g for R2 (p -value = 0.0191).

As shown in Figure 5.2.c (C23: 4.30×10^5 spores/mL), both replicates began with an initial mass of 0.0091g. Stability was maintained from J1 (0.0096g) to J4 (0.0098g). At J171, masses increased to 0.0190g (R2) and 0.0180g (R1), reflecting a highly significant statistical difference (p -value = 8.34×10^{-6}).

In Figure 5.2.d (C24: 1.00×10^5 spores/mL), initial mass was 0.0090g for both replicates. A stable trend was observed from J0 to J4, followed by an increase at J171 to 0.030g (R2) and 0.0160g (R1) (p -value = 0.0653).

Finally, Figure 5.2.e (C25: 2.00×10^4 spores/mL) reveals a starting mass of 0.0085g. While weights remained relatively stable around 0.097g between J0 and J4 (± 0.001 g), a sharp decrease was observed at J171, with both replicates dropping to 0.001g (p -value = 0.000216).

Interpretation of growth and mass variations from fungus' figures

The monitoring of absorbance at 600 nm demonstrates that *P. chrysosporium* did not undergo a sustainable exponential growth phase across the tested concentrations (C21 to C25). Optical density values remained generally stationary or declined, indicating a progressive senescence phase. Statistical analyses confirm this observation, with predominantly non-significant p -values (e.g., p -value = 0.202 for C21 and p -value = 0.962 for C24), suggesting that the observed fluctuations are attributable to experimental variability rather than a structured biological response. This stagnation implies that the culture medium, and specifically the PET, did not provide a sufficiently bioavailable carbon source to support fungal metabolism (Figure 5.1).

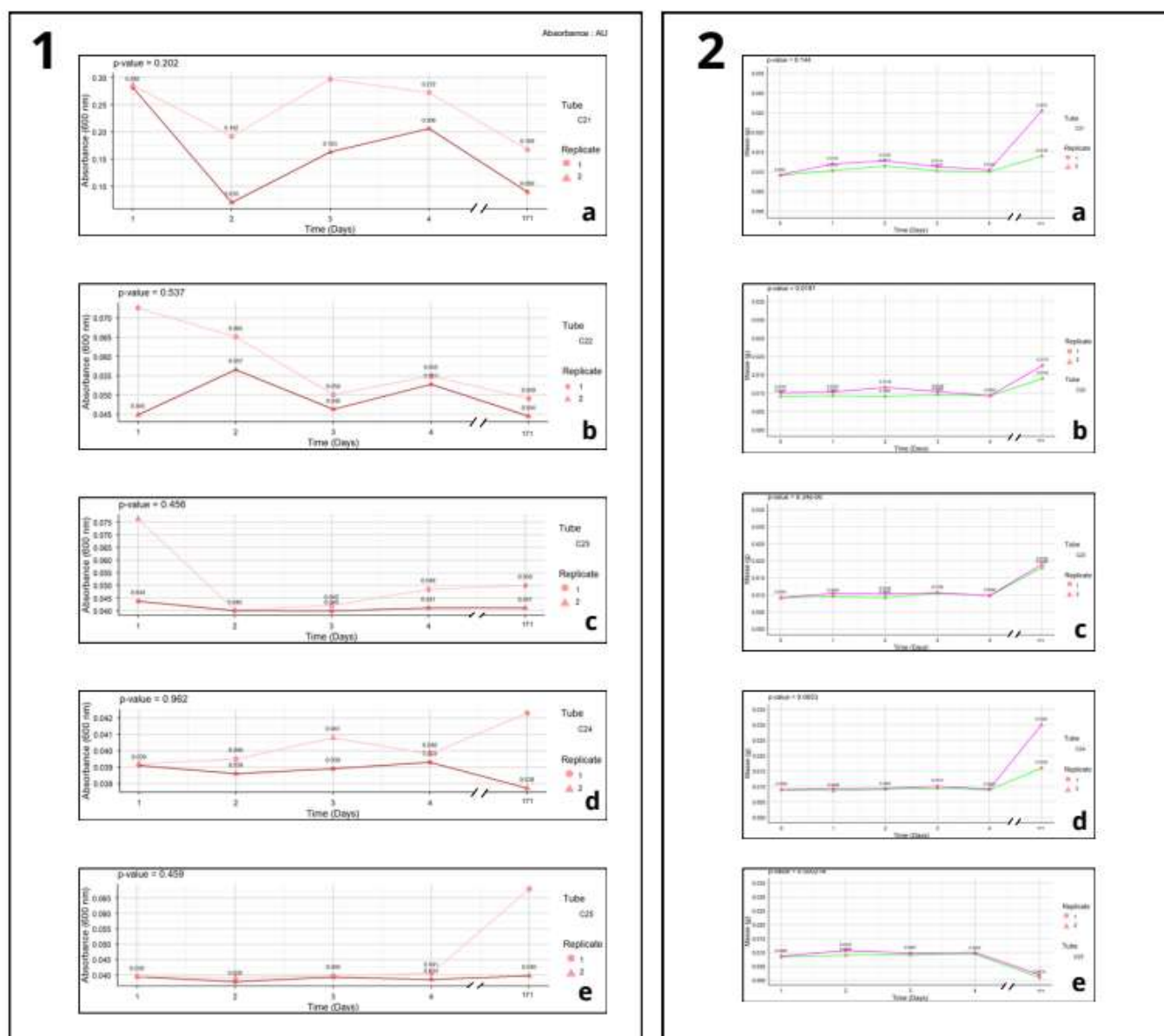


Figure 5: Evolution of absorbance (1) and plastic pellet mass (2) over time in the presence of *Phanerochaete chrysosporium*. Absorbance and mass (g) of plastic pellets were measured at different time points for two replicates in a degradation solution containing *P. chrysosporium*. Panels A–E correspond to tubes C21 (6.63×10^6 U/mL), C22 (2.31×10^6 spores/mL), C23 (4.30×10^5 spores/mL), C24 (1.00×10^5 spores/mL), and C25 (2.00×10^4 spores/mL), respectively. A non-parametric ANOVA was applied. When $p\text{-value} < 0.05$, significant heterogeneity was detected, Tukey post-hoc tests were carried out. Note: For each representation, the top curve joins the bigger values for each day during the time frame, contrary to the lower curve that joins the lower values.

However, the biological control group (T4), which contained fungus without plastic, exhibited a recovery in absorbance at Day 171 ($p\text{-value} = 0.00877$). This confirms that *Phanerochaete chrysosporium* remained viable in the culture medium over the six-month period. Therefore, the lack of exponential growth in the test groups (C21–C25) cannot be attributed to intrinsic fungal death or medium toxicity, but rather to the inability of the strain to metabolize the PET as a primary carbon source.

Concurrently, as shown in Figure 5.2, gravimetric analysis highlighted a counter-intuitive phenomenon for concentrations C21 through C24: an increase in final mass at Day 171. Although some increases were statistically significant (C22: $p\text{-value} = 0.0191$ and C23: $p\text{-value} < 0.001$), they do not reflect mineralization. Instead, this mass gain suggests the formation of a tenacious fungal biofilm on the polymer surface or the absorption of water and solutes into the plastic's microporosities⁹. The isolated case of condition C25, which presented a drastic mass loss ($p\text{-value} < 0.001$), must be interpreted with caution. Crucially, the abiotic control

group (T2) also displayed a significant mass decrease at Day 171 (p -value < 0.0001), dropping from an initial approximately 0.009 g to 0.0040 g. Since no biological agent was present in T2, this loss suggests the occurrence of abiotic fragmentation or experimental artifacts (e.g., loss of pellet fragments during the final washing and drying steps). Consequently, the mass loss observed in C25 is likely a reflection of these methodological limitations rather than genuine enzymatic mineralization.

RESULTS ENZYME

Monitoring of Enzymatic Activity (Absorbance at 600 nm)

The evolution of absorbance at 600 nm was monitored from Day 1 (J1) to Day 171 (J171) for five enzymatic conditions (E1 to E5). All assays were performed in duplicate.

As shown in Figure 6.1.a (E1: 3.47×10^2 U/mL), both replicates exhibited similar initial absorbance values at J1 (approximately 0.109). Absorbance increased progressively from J1 to J4 and continued to rise over time, reaching about 0.420 at J171 for both replicates. Statistical analysis revealed a significant difference between the measurements (p -value = 0.00203).

In Figure 6.1.b (E2: 7.49×10^{-3} U/mL), the two replicates also started with comparable absorbance values around 0.050 at J1. Absorbance increased in both replicates until J4, reaching a maximum of 0.223 for replicate 1 and 0.3306 for replicate 2. Subsequently, a decrease was observed at J171, with values of 0.173 and 0.2544, respectively. This variation between replicates was statistically significant (p -value = 0.00726).

According to Figure 6.1.c (E3: 1.50×10^{-3} U/mL), initial absorbance values were similar between replicates at J1 (around 0.042). Replicate 1 showed a slight increase from J1 to J3, followed by a decrease at J4, and ultimately reached 0.0655 at J171. In contrast, replicate 2 exhibited a moderate increase up to J2, then slight fluctuations before decreasing to 0.0487 at J171. No statistically significant difference was observed between replicates (p -value = 0.0732).

As illustrated in Figure 6.1.d (E4: 2.99×10^{-4} U/mL), both replicates displayed low initial absorbance values at J1 (0.0451 and 0.0399). Absorbance remained globally stable from J1 to J4, with only minor fluctuations observed during this period. At J171, the values remained close to those measured at the beginning of the experiment. Statistical analysis did not reveal any significant difference (p -value = 0.621).

Finally, Figure 6.1.e (E5: 5.99×10^{-5} U/mL) shows that both replicates exhibited low and comparable absorbance values from J1 onward (around 0.040). Slight fluctuations were observed between J1 and J4, without any clear increasing trend. At J171, absorbance values remained close to baseline, reaching 0.0418 and 0.0475, respectively. No statistically significant difference was detected (p -value = 0.349).

Evaluation of Plastic Degradation in the Presence of Enzymes (PET Pellet Mass)

The weight of the PET pellets was recorded from J0 (initial weight before contact) to J171. Only the mean pellet mass measured for each enzymatic condition was considered.

As shown in Figure 6.2.a (E1: 3.47×10^2 U/mL), the mean mass of the plastic pellets at J0 was 0.0100 g and 0.0087 g for the two replicates. During the first days of incubation, only minor variations were observed, and mean mass values remained relatively stable from J1 to J4. At J171, an increase in the mean pellet mass was recorded compared to the initial measurement, indicating no detectable plastic mass loss over time. The comparison between J0 and J171 did not reveal any statistically significant difference (p -value > 0.05).

In Figure 6.2.b (E2: 7.49×10^{-3} U/mL), initial measurements indicated pellet masses of 0.010 g and 0.009 g for the two replicates. For the first replicate, mass values remained relatively stable during the early incubation period before increasing to 0.0280 g at J171. In contrast, the second replicate exhibited stable mass values throughout the experiment, remaining close to its initial value at J171. Statistical analysis did not reveal any significant difference between measurements (p -value = 0.448).

According to Figure 6.2.c (E3: 1.50×10^{-3} U/mL), pellet mass at J1 was 0.009 g for replicate 1 and 0.010 g for replicate 2. Replicate 1 remained stable during the early incubation period before exhibiting an increase at J171, reaching 0.026 g. In contrast, replicate 2 displayed stable mass values throughout the experiment, remaining close to its initial value at J171. Statistical analysis did not reveal any statistically significant difference (p -value = 0.591).

As illustrated in Figure 6.2.d (E4: 2.99×10^{-4} U/mL), pellet mass at J1 was 0.102 g for replicate 1 and 0.009 g for replicate 2. Replicate 2 remained stable throughout the experiment, whereas replicate 1 showed a decrease, reaching 0.0053 g at J171. However, statistical analysis did not reveal any statistically significant difference (p -value = 0.536).

Finally, Figure 6.2.e (E5: 5.99×10^{-5} U/mL) shows that pellet mass at J1 was 0.089 g for both replicates. Mass values remained stable up to J4 before increasing at J171, reaching 0.019 g for both replicates. Statistical analysis revealed a statistically significant difference between measurements (p -value = 1.74×10^{-5}).

Interpretation of growth and mass variations from enzyme's figures

The monitoring of absorbance at 600 nm reveals distinct behaviors depending on the enzyme concentration. In the high-concentration conditions (E1 and E2), a significant evolution in optical density was observed. Specifically,

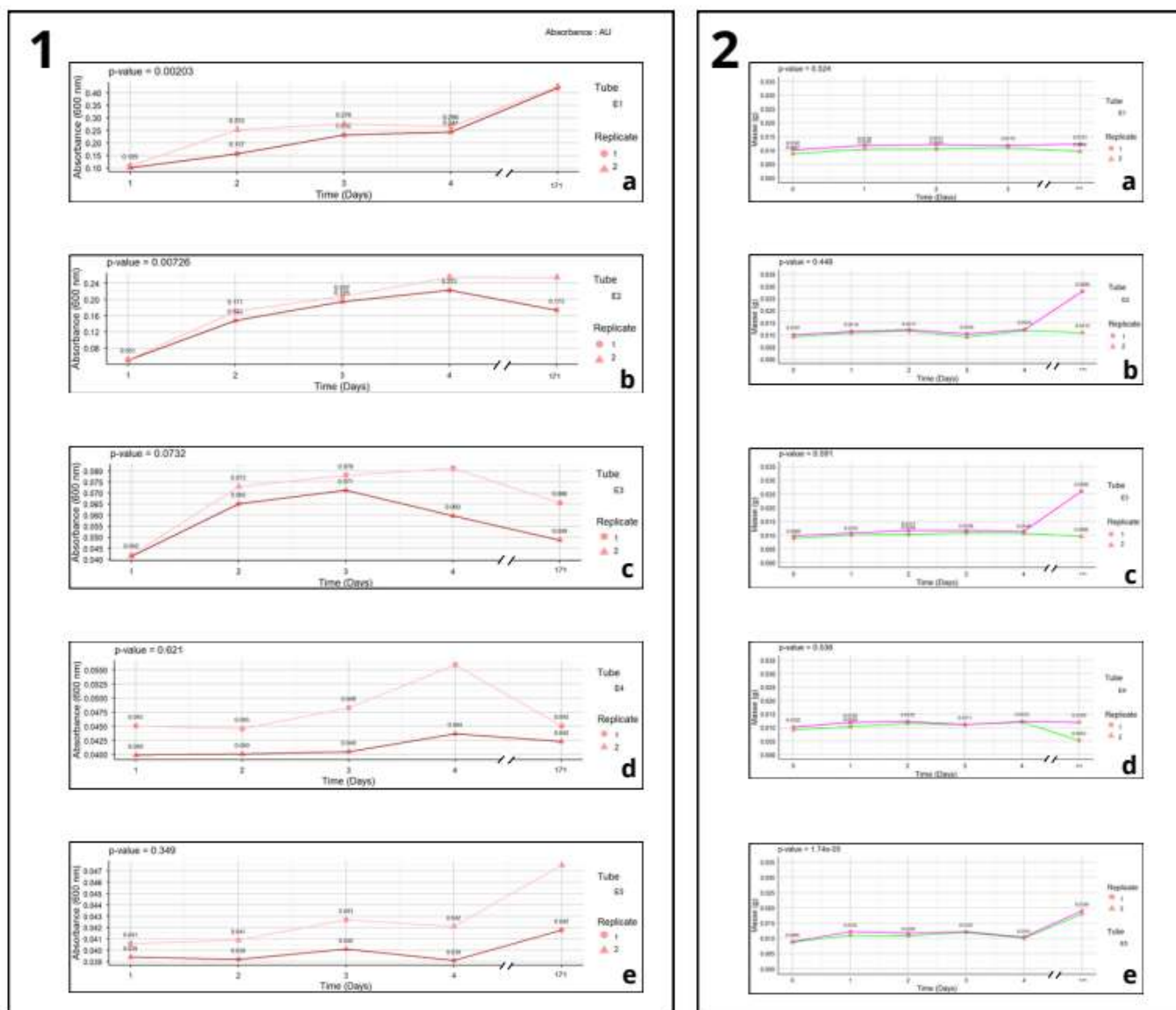


Figure 6: Evolution of absorbance (1) and plastic pellet mass (2) over time in the presence of laccase. Absorbance and mass (g) were measured at different time points for two replicates in a degradation solution containing laccase. Panels A–E correspond to tubes E1 (3.47×10^2 U/mL), E2 (7.49×10^{-3} U/mL), E3 (1.50×10^{-3} U/mL), E4 (2.99×10^{-4} U/mL), and E5 (5.99×10^{-5} U/mL), respectively. A non-parametric ANOVA was applied. When p -value < 0.05 , significant heterogeneity was detected, Tukey post-hoc tests were carried out. Note: For each representation, the top curve joins the bigger values for each day during the time frame, contrary to the lower curve that joins the lower values.

condition E1 (3.47×10^2 U/mL) showed a continuous increase in absorbance, reaching 0.420 at Day 171 (p -value = 0.00203). Similarly, E2 exhibited a rise followed by a decline. Since the T1 control (Sterility) confirmed the absence of microbial contamination (stable absorbance, p -value = 0.0821) and the T3 control (Enzymatic stability) showed that the enzyme alone in the medium remains stable (p -value = 0.45), the increased turbidity in E1 and E2 is likely attributable to a physical interaction between the high-concentration enzyme and the PET surface. This suggests protein precipitation or aggregation induced by the polymer, rather than biological activity or contamination.

Conversely, for the lower concentrations (E3 to E5), absorbance values remained low and generally stationary throughout the 171 days, mirroring the baseline values of the sterility control. The lack of significant variation (e.g., E4: p -value = 0.621) indicates that at these dilutions, the enzyme remained in solution without forming significant aggregates, but also without generating optically active degradation products.

Gravimetric analysis of the PET pellets reveals a lack of genuine enzymatic degradation, characterized by two contrasting non-biological phenomena: mass gain and abiotic fragmentation.

For the highest enzymatic concentration (E1), a mass increase was recorded at Day 171 compared to the initial weight. While not statistically significant (p -value > 0.05), this trend aligns with the turbidity results observed in Figure 6.1.A. This "negative degradation" suggests the adsorption of the enzyme onto the plastic surface (protein coating) or the absorption of water into the polymer matrix, masking any potential mass loss¹⁰.

In conditions E4 and E5, mass loss or significant fluctuations were observed (e.g., E5: p -value $= 1.74 \times 10^{-5}$). However, these results must be interpreted with caution. Crucially, the Abiotic Control (T2) also displayed a highly significant mass decrease at Day 171 (p -value $= 5.52 \times 10^{-5}$), dropping from approximately 0.010 g to 0.0040 g. Since T2 contained no biological agent, this loss is indisputably the result of abiotic factors, such as pellet fragmentation during the extended incubation or loss of material during the final washing and drying steps. Consequently, the statistically significant differences observed in the test groups (particularly E5) are likely a reflection of these methodological limitations and experimental artifacts rather than catalytic hydrolysis of the PET by the laccase.

DISCUSSION

The cross-analysis of fungal growth and PET pellet mass variation over a 171-day period reveals no significant degradation of the polymer, despite certain statistical variations observed., which may in part reflect the very small sample size ($n = 2$) that limits statistical power and makes p -values highly sensitive to additional replicates. As the aim of this study is to document the methodological limitations encountered, these results provide crucial insights into the recalcitrance of PET under this specific mycoremediation protocol using *Phanerochaete chrysosporium* (Figure 5).

Similarly, the enzymatic assays involving commercial laccase did not demonstrate effective depolymerization of the PET pellets. While high enzymatic concentrations (E1: 3.47×10^2 U/mL) induced a significant increase in turbidity (p -value $= 0.00203$) and a slight mass gain, these phenomena are interpreted as protein adsorption or aggregation on the polymer surface rather than hydrolysis¹⁰. Lower concentrations (E3–E5) showed no significant deviation from the abiotic controls (T1, T2). The significant mass loss observed in the lowest concentration group (E5) mirrors the abiotic fragmentation recorded in the physical control (T2), confirming that commercial laccase alone, without specific mediators, was insufficient to initiate backbone cleavage of the semi-crystalline PET.

TECHNICAL LIMITATIONS AND EXPERIMENTAL CONSTRAINTS

The absence of degradation can be attributed to several critical technical limitations encountered during the study. Primarily, the experimental design was constrained by the inability to perform in-house enzyme extraction or to culture

the specific fungal strains initially targeted. Due to regulatory restrictions on importing specific biological species into France, we were unable to utilize high-potency strains cited in literature, such as *Neopestalopsis Phangnang*¹¹, necessitating the substitution with *P. chrysosporium*. Furthermore, reliance on commercial laccases rather than fresh, experimentally induced secretomes may have resulted in lower enzymatic activity or a lack of the specific mediators required for PET hydrolysis.

Moreover, the scope of the enzymatic study was severely restricted by supply chain logistics. Initially, the experimental design included the formulation of an enzymatic cocktail combining Laccase with Manganese Peroxidase (MnP). Literature suggests that the synergistic action of these ligninolytic enzymes significantly enhances the oxidation potential, allowing for more effective attack on recalcitrant carbon backbones compared to mono-enzymatic treatments¹². However, the MnP shipment was not received within the allotted experimental timeframe, compelling us to proceed solely with the laccase assay, potentially limiting the oxidative power necessary to destabilize the PET surface.

Analytical limitations also hindered the precise assessment of degradation. The reliance on gravimetry (mass balance) proved insufficient for detecting micro-degradation events, as evidenced by the unexpected mass loss in the abiotic control (T2). This confirms that simple weighing lacks the sensitivity to distinguish between actual biodegradation and mechanical disintegration or handling errors over long incubation periods. Additionally, despite the strict sterile conditions maintained in the laminar flow hood, the extended incubation period of 171 days introduced a risk of contamination, which may have competed with the target fungus for nutrients in the control media. Time constraints also prevented the implementation of a secondary screening phase to re-adapt the surviving fungal colonies to higher stress environments. Finally, we have to account for the interference of abiotic factors. These are essentially the non-living physical drivers in the environment, things like mechanical agitation from the shaker, temperature fluctuations, spontaneous hydrolysis, or even minor handling inconsistencies. The issue is that these factors tend to parasitize the results. For instance, the constant physical motion required to keep the culture aerated can cause micro-fragmentation of the plastic that mimics biological degradation.

The experimental design and analytical depth of this study were strictly defined by significant resource constraints, specifically a limited operating budget (€200) and a restricted timeframe for active manipulation (one week). These limitations directly impacted the protocol, specifically, the inability to perform in-house genetic sequencing or source alternative isolates compelled us to rely exclusively on a standard commercially available *P. chrysosporium*. Also, the budget precluded the use of advanced surface analysis techniques such as Scanning Electron Microscopy (SEM)¹³ or

Fourier Transform Infrared Spectroscopy (FTIR)¹⁴. Consequently, we relied solely on gravimetry, which proved insufficient for distinguishing between biofilm accumulation and polymer mass loss at the micro-scale. Additionally, the restricted manipulation time prevented the implementation of a "pre-adaptation" phase, where fungi are progressively exposed to plastic as a sole carbon source to induce specific enzymatic pathways before the main experiment.

COMPARATIVE ANALYSIS WITH LITERATURE

Our results stand in contrast to successful mycoremediation studies where specific pre-conditions were met. Research suggests that while *P. chrysosporium* is highly effective on lignin due to its peroxidase system, it may struggle to hydrolyze the ester bonds of semi-crystalline thermoplastics like PET without adding specific mediators^{15, 16}. Furthermore, despite maintaining our cultures under agitation to ensure adequate oxygenation, a condition often cited as critical for white-rot fungi, degradation did not occur. This suggests that the limiting factor in our experiment was not oxygen availability, but rather the biochemical incompatibility between the fungal secretome and the untreated PET surface.

THE VALUE OF NEGATIVE RESULTS

While positive results are traditionally prioritized, reporting negative findings is equally critical for scientific advancement. Our tests showed that *Phanerochaete chrysosporium*, under these specific laboratory conditions, was not effective at degrading PET. We present these results for three main reasons.

First, to prevent redundancy. By documenting that this specific experimental setup yields null results, we allow other researchers to avoid repeating the exact same path and immediately focus resources on alternative variables, such as pH, temperature, or the pre-treatment of the plastic or on a completely different fungal strain.

Second, we want to promote transparency in environmental biotechnology. Addressing pollution challenges is an urgent matter and requires a comprehensive understanding of biodegradation potential. Publishing inconclusive data prevents the loss of knowledge associated with unpublished trials and helps build a more complete map of biotechnological possibilities.

Finally, this contributes to the reality of the scientific process. Science is rarely linear, negative results are essential data points that narrow the scope of investigation. This study serves as a necessary baseline, enabling future research to refine experimental questions and accelerate the identification of effective solutions.

METHODOLOGICAL INSIGHTS AND FUTURE OPTIMIZATION

To overcome these barriers in future iterations, prioritizing the physicochemical alteration of the plastic is essential. Since

agitation and oxygenation were sufficient in our protocol yet failed to induce degradation, the primary obstacle remains the high crystallinity and hydrophobicity of commercial beverage bottles. Future studies, even those with limited budgets, should incorporate a low-cost pretreatment step (such as prolonged UV exposure or thermal oxidation) to increase surface roughness¹⁵. Additionally, investing in a pre-culture phase to acclimatize the fungus to the plastic substrate could trigger the expression of dormant metabolic pathways, potentially yielding better results than direct inoculation.

CONCLUSION

This work investigated the feasibility of PET biotransformation with a fungal system and oxidative enzymatic treatment extended prolonged incubation conditions. Beyond evaluating degradation performance, the study primarily aimed to assess the practical constraints and experimental limitations associated with implementing biological approaches on untreated commercial PET.

The results highlight the intrinsic resistance of semi-crystalline PET to biological attack in the absence of prior physicochemical modification or optimized enzymatic synergy. Rather than demonstrating polymer degradation, the data highlights the complexity of distinguishing true depolymerization from surface interactions, biomass accumulation, or abiotic alterations when relying on conventional laboratory measurements. These observations reinforce the need for more sensitive analytical techniques and refined methodological designs to study the long-term biodegradation of plastics.

Importantly, this study clarifies critical parameters that must be considered to improve future research, including substrate pretreatment, adaptive fungal conditioning, and the integration of complementary oxidative enzymes. By identifying these experimental obstacles, the current work establishes a methodological foundation that can guide more targeted and effective research strategies.

Ultimately, advancing biological solutions for plastic waste management will require not only optimized biocatalysts, but also carefully designed experimental frameworks capable of overcoming the physicochemical stability of synthetic polymers. The knowledge gained here contributes to shaping this framework and supports the continued development of sustainable biotechnological approaches from a circular bioeconomy perspective.

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AUTHORS' CONTRIBUTIONS

M.K. and L.S. wrote the Abstract, the Introduction section and the Conclusion section of the article. P.F., M.K. and L.S. wrote Materials and Methods section. D.Y. and P.F. wrote the Results (fungi and control section) and Discussion. J.D. and S.N. wrote the Results (enzyme section). Y.M. did the figures and the statistical tests, she was assisted by J.D. and S.N.

CONFLICTS OF INTEREST

The authors declared no potential conflicts of interest with respect to the research, authorship, and publication of this article.

REFERENCES

- Kornberger, D., Paatsch, T., Schmidt, M. & Salat, U. New combined absorption/¹H NMR method for qualitative and quantitative analysis of PET degradation products. *Environ. Sci. Pollut. Res.* **31**, 20689–20697 (2024).
- Zrimec, J., Kokina, M., Jonasson, S., Zorrilla, F. & Zelezniak, A. Plastic-Degrading Potential across the Global Microbiome Correlates with Recent Pollution Trends. *mBio* **12**, e0215521 (2021).
- Ekanayaka, A. H. *et al.* A Review of the Fungi That Degrade Plastic. *J. Fungi Basel Switz.* **8**, 772 (2022).
- Ali, M., Xu, C. & Li, K. Micro- and Nanoplastics as Emerging Threats to Both Terrestrial and Aquatic Animals: A Comprehensive Review. *Vet. Sci.* **12**, (2025).
- Liu, F., Wang, T., Liu, X.-H., Xu, N. & Pan, X.-L. Efficient biodegradation and upcycling of polyethylene terephthalate mediated by cell-factories. *Front. Microbiol.* **16**, 1599470 (2025).
- Yeboaa, C., Tetteh, E. K., Chollom, M. N. & Rathilal, S. Sustainable Solutions for Plastic Waste Mitigation in Sub-Saharan Africa: Challenges and Future Perspectives Review. *Polymers* **17**, 1521 (2025).
- Ekanayaka, A. H. *et al.* Linking the Metabolic Activity of Plastic-Degrading Fungi to Their Taxonomy and Evolution. *J. Fungi* **11**, 378 (2025).
- Oliveira, J. *et al.* Marine Environmental Plastic Pollution: Mitigation by Microorganism Degradation and Recycling Valorization. *Front. Mar. Sci.* **7**, (2020).
- Biofilm formation and its influences on the properties of microplastics as affected by exposure time and depth in the seawater - ScienceDirect. <https://www.sciencedirect.com/science/article/abs/pii/S0048969720327546>.
- Badino, S. F., Bååth, J. A., Borch, K., Jensen, K. & Westh, P. Adsorption of enzymes with hydrolytic activity on polyethylene terephthalate. *Enzyme Microb. Technol.* **152**, 109937 (2021).
- Khruengsai, S., Sripahco, T. & Pripdeevech, P. Microbial degradation of low-density polyethylene by *Neopestalotiopsis phangngaensis*. *J. Gen. Appl. Microbiol.* **68**, 287–294 (2023).
- Iiyoshi, Y., Tsutsumi, Y. & Nishida, T. Polyethylene degradation by lignin-degrading fungi and manganese peroxidase. *J. Wood Sci.* **44**, 222–229 (1998).
- Mittal, A., Soni, R. K., Dutt, K. & Singh, S. Scanning electron microscopic study of hazardous waste flakes of polyethylene terephthalate (PET) by aminolysis and ammonolysis. *J. Hazard. Mater.* **178**, 390–396 (2010).
- Chen, Z., Hay, J. N. & Jenkins, M. J. FTIR spectroscopic analysis of poly(ethylene terephthalate) on crystallization. *Eur. Polym. J.* **48**, 1586–1610 (2012).
- Wei, R. & Zimmermann, W. Microbial enzymes for the recycling of recalcitrant petroleum-based plastics: how far are we? *Microb. Biotechnol.* **10**, 1308–1322 (2017).
- Plastics: Environmental and Biotechnological Perspectives on Microbial Degradation | Applied and Environmental Microbiology. <https://journals.asm.org/doi/10.1128/aem.01095-19>.



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